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Overcoming Distribution Shifts and Imbalance Challenges in Representation Learning for Deep Regression Models

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Abstract

Deep learning has shown remarkable potential for industrial applications, particularly in predictive maintenance and condition monitoring. A key challenge in this domain is prognostics, specifically predicting the Remaining Useful Life (RUL) of assets — a continuous value prediction problem. However, two critical challenges limit the effective deployment of deep learning models for RUL prediction in real-world industrial settings. First, the high variability of operating conditions and system configurations (such as different units of a fleet) leads to significant distribution shifts between training and deployment, resulting in substantial degradation of model performance. Second, in safety-critical systems, preventive maintenance practices prevent assets from reaching highly degraded states, resulting in severely imbalanced datasets that predominantly reflect early degradation states.

While these challenges are common to both classification and regression tasks, they are more pronounced in regression tasks such as RUL prediction, as regression models are more sensitive to variations in input data and feature scaling, making generalization under domain shifts and data imbalance particularly challenging. Although domain adaptation methods have shown promise in mitigating distribution shifts, they have been primarily developed for classification tasks, with relatively limited research addressing the distinct characteristics of regression tasks.

This dissertation addresses these real-world industrial challenges and proposes a comprehensive framework that effectively handles domain shifts and data imbalance in deep regression models. The thesis makes four main contributions.

Firstly, we develop a phase-aware distribution alignment method that leverages expert knowledge to individually align overrepresented and underrepresented operational phases in the data. By aligning the marginal distributions of each operational phase separately across domains, this method improves RUL prediction accuracy and generalization across different operating conditions.

Secondly, we introduce regression-specific adaptation methods based on closed-form solutions and uncertainty estimation. This contribution consists of two distinct methods. We propose aligning the inverse Gram matrices to preserve feature relationships and scales during the alignment, aiming to align the features from the regressor's perspective.

Additionally, we incorporate uncertainty quantification to guide the adaptation process, providing confidence measures alongside predictions. This serves two purposes: to enrich the embedding space of regression models (which in computer vision tasks tends to be lower-rank than in classification) and to provide uncertainty estimates, not just point estimates, which is important for real-world deployment.
Thirdly, we propose an in-context learning approach to address the challenge of imbalanced data distributions, particularly in learning from minority regions. We theoretically study the impact of data density in this form of learning. This method enables models to generalize better in data-sparse regions without requiring additional training, by conditioning on representative examples during inference.
Finally, we present a source-free open-set adaptation strategy that tackles the problem of newly presented unknown classes in the target domain and aims to adapt the model without requiring access to source domain data. By generating synthetic data points and adapting the classifier to accommodate these unknown classes, we aim to learn their semantics directly instead of discarding them. This method facilitates adaptation in scenarios where certain failure trajectories are unobserved in the source domain data, and the RUL prediction is represented as a discrete task.
Through extensive validation on both industrial datasets and computer vision benchmarks, we demonstrate that our framework significantly improves regression model performance under real-world constraints. The practical impact of this work has already been demonstrated through a patent submission for predicting health states in plasma spray gun technology. By addressing fundamental challenges in current approaches, this framework enables more reliable deployment of regression models across various industrial applications beyond prognostics, while reducing the need for extensive labeled data and adapting to new operating conditions with minimal supervision.

Résumé

L'apprentissage profond a démontré un potentiel remarquable pour les applications industrielles, notamment pour la maintenance prédictive et la surveillance de l'état des équipements. Un défi majeur dans ce domaine est le pronostic, en particulier la prédiction de la durée de vie résiduelle (RUL) des actifs — un problème de prédiction de valeurs continues. Cependant, deux défis critiques limitent le déploiement efficace des modèles d'apprentissage profond pour la prédiction de la RUL en environnements industriels réels. Premièrement, la forte variabilité des conditions d'exploitation et des configurations systèmes (comme les différentes unités d'une flotte) entraîne des décalages de distribution significatifs entre l'entraînement et le déploiement, ce qui dégrade substantiellement les performances des modèles. Deuxièmement, dans les systèmes à sécurité critique, les pratiques de maintenance préventive empêchent les équipements d'atteindre des états de dégradation avancée, ce qui se traduit par des ensembles de données fortement déséquilibrés qui reflètent principalement les états de dégradation précoce.

Bien que ces défis soient communs aux tâches de classification et de régression, ils sont plus prononcés dans les tâches de régression comme la prédiction de la RUL, car les modèles de régression sont plus sensibles aux variations des données d'entrée et au redimensionnement des caractéristiques, rendant la généralisation particulièrement difficile en présence de décalages de domaine et de données déséquilibrées. Bien que les méthodes d'adaptation de domaine aient montré des résultats prometteurs pour atténuer les décalages de distribution, elles ont été principalement développées pour les tâches de classification, avec une recherche relativement limitée abordant les caractéristiques distinctes des tâches de régression.

Cette thèse aborde ces défis industriels concrets et propose un framework complet qui gère efficacement les décalages de domaine et les déséquilibres de données dans les modèles de régression profonde. Cette thèse apporte quatre contributions principales.

Premièrement, nous développons une méthode d'alignement de distribution tenant compte des phases opérationnelles, qui exploite l'expertise du domaine pour aligner individuellement les phases opérationnelles surreprésentées et sous-représentées dans les données. En alignant séparément les distributions marginales de chaque phase opérationnelle entre les domaines, cette méthode améliore la précision de la prédiction de la RUL et la généralisation à travers différentes conditions d'exploitation.

Deuxièmement, nous introduisons des méthodes d'adaptation spécifiques à la régression basées sur des solutions analytiques et l'estimation de l'incertitude. Cette contribution comprend deux méthodes distinctes. Nous proposons d'aligner les matrices de Gram-inverses pour préserver les relations entre caractéristiques et leur échelle pendant l'alignement, visant à aligner les caractéristiques du point de vue du régresseur. De plus, nous intégrons la quantification de l'incertitude pour guider le processus d'adaptation, fournissant des mesures de confiance avec les prédictions. Cela sert deux objectifs : enrichir l'espace de représentation des modèles de régression (qui tend à être de rang inférieur à celui des tâches de classification en vision par ordinateur) et fournir des estimations d'incertitude, pas seulement des estimations ponctuelles, ce qui est crucial pour le déploiement en conditions réelles.
Troisièmement, nous proposons une approche d'apprentissage contextuel pour répondre au défi des distributions de données déséquilibrées, particulièrement dans l'apprentissage à partir des régions minoritaires. Nous étudions théoriquement l'impact de la densité des données dans cette forme d'apprentissage. Cette méthode permet aux modèles de mieux généraliser dans les régions à faible densité de données sans nécessiter d'entraînement supplémentaire, en se conditionnant sur des exemples représentatifs pendant l'inférence.
Enfin, nous présentons une stratégie d'adaptation sans source en ensemble ouvert qui aborde le problème des nouvelles classes inconnues dans le domaine cible et vise à adapter le modèle sans nécessiter l'accès aux données du domaine source. En générant des points de données synthétiques et en adaptant le classifieur pour prendre en compte ces classes inconnues, nous cherchons à apprendre directement leur sémantique plutôt que de les écarter. Cette méthode facilite l'adaptation dans les scénarios où certaines trajectoires de défaillance ne sont pas observées dans les données du domaine source, et où la prédiction de la RUL est représentée comme une tâche discrète.
À travers une validation approfondie sur des jeux de données industriels et de vision par ordinateur, nous démontrons que notre framework améliore de manière significative la performance des modèles de régression sous des contraintes réelles. L'impact pratique de ce travail a déjà été démontré par le dépôt d'un brevet pour la prédiction des états de santé dans les technologies de projection plasma. En abordant les défis fondamentaux des approches actuelles, ce framework permet un déploiement plus fiable des modèles de régression dans diverses applications industrielles au-delà du pronostic, tout en réduisant le besoin de données étiquetées extensives et en s'adaptant à de nouvelles conditions d'exploitation avec une supervision minimale.

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1 Introduction

1.1 Motivation

In deep learning, regression problems play a fundamental role in tasks where precise, continuous-value predictions are critical. Unlike classification tasks, where models assign discrete labels to inputs, regression aims to predict quantitative values. This approach finds application not only in computer vision tasks such as head-pose estimation (T.-Y. Yang et al., 2019), facial landmark detection (H. Li et al., 2022), and depth estimation (Godard et al., 2019), but also in medical imaging for tasks like forensic age estimation from hand CT/MRI (M. Zhang, 2022) and cell detection in histology (S. He et al., 2021). In addition, regression models are widely used in industrial and engineering applications, such as product quality prediction (L. Ren et al., 2020), surrogate modeling for tasks like car drag coefficient prediction (Song et al., 2023), condition monitoring (Serin et al., 2020), and Remaining Useful Life (RUL) prediction (Arias Chao et al., 2021).

Remaining Useful Life prediction aims to estimate the remaining operational time during which an asset is expected to perform its intended function reliably before reaching a point where maintenance or replacement becomes necessary, typically before the asset reaches the end of its life. RUL prediction is a key element of Prognostics and Health Management (PHM), which seeks to optimize the health and performance of complex systems by minimizing downtime, reducing maintenance costs, and extending the operational lifespan of assets. With advancements in sensor technologies, collecting extensive health monitoring data has become feasible, enabling the development of advanced, data-driven approaches for PHM (R. Zhao et al., 2019). These approaches have shown promise in tasks like fault detection, diagnosis, and RUL prediction (Fink et al., 2020). However, their effectiveness heavily relies on the assumption that the underlying data distribution remains consistent over time, which is often not the case due to domain shifts. Similar challenges with distribution shifts have been observed across industrial applications, such as in customized production where variations in order specifications lead to significant changes in operational data distributions (Senoner et al., 2023).

Domain shift refers to the situation where the statistical properties of the data-generating process change between the training and testing phases, leading to a distribution discrepancy that can negatively impact the performance of the trained models. Real-world applications are often influenced by environmental conditions, operational variability, and other factors, leading to inevitable domain gaps between datasets. In the context of PHM, there are three main types of domain gaps: (1) between varying environmental and operating conditions (Y. Pan et al., 2019), (2) between different units of a fleet (Michau & Fink, 2021), and (3) between simulations and real data (Gao et al., 2021; K. Yu et al., 2021; J. Zhao & Li, 2024). The distribution shift in the data caused by the high variability of operating conditions and system configurations can pose significant challenges for PHM and can result in a performance drop.

These domain shifts pose particular challenges for regression tasks, where the objective is to predict continuous outputs, such as RUL estimation. In regression tasks, especially those involving precise predictions like RUL, the challenges are more pronounced compared to classification problems. While classification models predict discrete categories (clusters), regression aims to predict continuous outputs that are more sensitive to variations in input data and feature scaling (X. Chen et al., 2021). Even slight changes in operating conditions can lead to substantial prediction errors, making generalization under domain shift particularly challenging for regression tasks.

In addition to challenges related to representation learning, RUL prediction also faces issues of data scarcity and imbalanced datasets (Zio, 2022). In many industrial settings, particularly safety-critical applications such as aircraft engines and large electrical transformers, allowing assets to reach complete failure is not acceptable due to the significant risks involved. Consequently, these systems are often taken out of operation preventively, before reaching highly degraded states (close-to-failure). As a result, condition monitoring data typically reflects the early stages of degradation more frequently than the critical degradation states, creating an inherent imbalance in the available training data. The limited representation of severe degradation states poses a challenge for developing accurate RUL prediction models that must generalize effectively across varying health conditions, despite limited data from the critical states that are crucial for reliable predictions.

Unsupervised domain adaptation is a promising approach for addressing the challenges posed by the distribution shift in industrial applications (Singhal et al., 2023). It aims to transfer knowledge from a labeled source domain to a related, but unlabeled target domain by aligning the source and target distributions (Wilson & Cook, 2020b). These methods can be particularly valuable in industrial settings involving multiple operating conditions. They can improve the generalizability of models to new and evolving conditions without the need for further annotation of the data. Finally, domain adaptation can provide the necessary tools for effectively detecting faults early, predicting the health state of equipment, and optimizing maintenance schedules by estimating the RUL of equipment in industrial applications.

While many domain adaptation (DA) methods have been developed and evaluated for classification and segmentation problems, most are not directly transferable to DA regression (X. Chen et al., 2021). Despite the importance of adapting models for regression problems, only a limited number of research studies have addressed this task. Although some methods proposed for classification can also be applied to regression, they often fail to address the fundamental differences between discrete and continuous predictions. These challenges stem from several key characteristics of regression problems. First, deep neural networks have been found to be less robust to feature scaling, as reported by X. Chen et al., 2021. Therefore, aligning the distributions of deep representations can potentially modify the feature scale, thereby impeding domain adaptation regression. Furthermore, regression models need to learn to map inputs to a continuous output space while preserving the underlying relationships between features and target values. The domain shift in the target domain makes maintaining these relationships particularly challenging, as small variations in the input space can lead to significant errors in the predicted values. To mitigate these issues, existing UDA methods focusing on regression have typically relied on some labeled data (Singh & Chakraborty, 2020).

In addition to the challenges of adapting to domain shifts, the continuous nature of regression tasks presents unique challenges when dealing with imbalanced data distributions (Y. Yang et al., 2021). Unlike classification tasks where class imbalance is clearly defined, regression imbalance appears as varying densities across the continuous label space. Approaches to address this imbalance generally fall into two main categories. The first category involves density-based sample weighting strategies (Steininger et al., 2021), which balance the influence of samples by assigning higher weights to less frequent values. This adjustment aims to provide better representation across the label spectrum by counteracting biases in label frequency. The second approach focuses on preserving local and global relationships in the continuous label space by introducing additional learning objectives, such as ranking or contrastive losses (Y. Gong et al., 2022). However, these methods, which rely on updating model weights through gradient descent, often struggle to generalize effectively in regions with limited training data. This limitation is particularly evident in the tail regions of the distribution, where models tend to default toward predicting values closer to the mean of the more densely populated regions of the training data (J. Ren et al., 2022).

1.2 Aim and Scope

The aim of this research is to develop a comprehensive framework for regression tasks in industrial applications, specifically addressing the challenges of domain shift and imbalanced data distributions in real-world settings where data constraints and limited supervision. Our framework consists of four interconnected modules, each addressing a distinct challenge associated with domain shifts and imbalanced regression tasks : (1) The first module addresses distribution alignment across operational phases, leveraging expert knowledge to balance between underrepresented and over-represented phases and improve RUL predictions; (2) The second module develops regression-specific domain adaptation techniques, exploiting closed-form solutions and incorporating uncertainty estimation to guide the adaption; (3) The third module addresses imbalanced data distributions through in-context learning, improving model generalization in data-sparse regions; (4) Finally, the fourth module introduces source-free open-set adaptation, allowing the framework to handle missing labels and learn unknown class semantics without access to source domain data. In addition to predicting RUL, understanding the health state of assets — which can also assumed to be discrete — plays a critical role in safety-critical applications, where extremely degraded health states may not be represented in the available data. This module bridges classification and regression by providing prior knowledge of health states, enabling better prediction in scenarios with limited degradation data.

By integrating these modules, our proposed framework aims to bridge the gap between theoretical advances in domain adaptation and imbalanced regression, and practical industrial applications. The scope of this work includes the development, implementation, and evaluation of these methods on real-world industrial datasets, particularly for RUL prediction and related PHM tasks, and computer vision benchmarks to demonstrate broader applicability and transferability.

1.3 Background & Research Gap

1.3.1 Unsupervised Domain Adaptation

Unsupervised Domain Adaptation (UDA) aims to transfer knowledge from a labeled source domain to an unlabeled target domain, improving model generalization in scenarios where labeled data is scarce or unavailable in the target domain. Three primary approaches have emerged within domain adaptation research: distribution alignment, adversarial learning, and self-training.

Distribution alignment methods focus on minimizing discrepancies between source and target domain distributions by aligning statistical measures. Common techniques include Maximum Mean Discrepancy (MMD) (Long et al., 2015), correlation alignment (B. Sun & Saenko, 2016), and optimal transport (Damodaran et al., 2018), which align statistical moments between domains to improve generalization (Carlucci et al., 2017; Chang et al., 2022; Long et al., 2017). Adversarial learning has shown strong performance by employing a discriminator to encourage the feature extractor to learn domain-invariant representations (Ganin & Lempitsky, 2015; Long et al., 2018; Rangwani et al., 2022; Tang & Jia, 2020; Tzeng et al., 2017). This setup effectively aligns source and target domains at the feature level. Self-training methods further support adaptation by iteratively refining pseudo-labels for target domain samples, guiding the model with gradually updated pseudo-labels to improve alignment (Y. Luo et al., 2020; Melas-Kyriazi & Manrai, 2021; F. Pan et al., 2020; Q. Wang, Dai, et al., 2021b).

Recent advances in prompt learning and Large Language Models (LLMs) have opened new research directions in UDA, enabling more flexible and generalizable adaptation strategies, such as open-vocabulary and taxonomy mapping for example. These approaches extend core UDA techniques in vision tasks through the rich semantic understanding provided by LLMs (S. Bai et al., 2024; Ge et al., 2022, 2023a; Lai et al., 2024; Sohn et al., 2023).

While domain adaptation has achieved notable success in classification tasks, many real-world applications, such as RUL prediction, require regression models that predict continuous outcomes. This shift from classification to regression presents unique challenges, including sensitivity to feature scale and the assumption of constant variance across predictions, both of which may not hold under domain shifts (X. Chen et al., 2021). Early work by Cortes and Mohri, 2011b laid the theoretical foundation for unsupervised domain adaptation in regression (UDAR), expanding on classification-oriented analyses (such as Ben-David et al., 2006). However, recent approaches to UDAR often focus on learning a shared feature extractor to align distributions between source and target domains (X. Chen et al., 2021; J. Wu et al., 2022b). This approach implicitly assumes that reducing feature discrepancy will naturally enable the learning of a shared regressor across domains. This focus on feature alignment overlooks challenges specific to the regression model itself.

Furthermore, reducing prediction uncertainty has shown promise in domain adaptation settings, particularly by addressing target domain samples with high entropy. Entropy minimization has been effectively used in classification tasks, where source samples typically exhibit lower entropy than target samples due to domain shift (Prabhu et al., 2021; Saito et al., 2019; Westfechtel et al., 2024), and source-free domain adaptation methods (such as proposed in Roy et al., 2022) apply uncertainty weighting to target samples. Roy et al., 2022 proposed to use information maximization and last-layer Laplace approximations to estimate prediction uncertainty, an approach effective in classification. However, adapting such methods to regression poses a unique challenge: in regression tasks, only a single value per sample providing no direct measure of certainty.

Incorporating uncertainty quantification (UQ) into regression could help evaluate distribution shifts by observing differences in uncertainty levels. Properly calibrated UQ could then guide the adaptation process by prioritizing samples with higher confidence and reducing reliance on uncertain points, facilitating more effective alignment in regression.

1.3.2 Unsupervised Domain Adaptation in Prognostics and Health Management

In recent years, domain adaptation has gained attention in PHM applications (W. Li, Huang, et al., 2022), particularly for *fault diagnostics*. Three primary types of domain gaps exist in the PHM context: between varying operating conditions (Y. Pan et al., 2019; Z. Zhang et al., 2020), across different units within a fleet (Michau & Fink, 2021), and between simulated and real-world data (Gao et al., 2021; Q. Wang, Taal, & Fink, 2021; B. Yang et al., 2019; K. Yu et al., 2021).

Fault Diagnostics

Most research in PHM has addressed domain shifts between discrete operating conditions (Q. Wang et al., 2019b). Classic domain adaptation techniques, such as distribution alignment (X. Li et al., 2019; W. Lu et al., 2016) and adversarial alignment (H. Bao et al., 2021; B. Zhang et al., 2018), have shown effectiveness in this context (Q. Wang et al., 2019b). Discrepancy-based adaptation methods, employing criteria like Maximum Classification Discrepancy (H. Bao et al., 2021) and Maximum Mean Discrepancy (T. Han et al., 2021; Tong et al., 2018), have also been widely explored for aligning feature distributions across different operating conditions.

Deep domain adversarial frameworks such as DANN (Jiao et al., 2020; W. Li, Yang, et al., 2022; S. Liu et al., 2023; J.-F. Yang et al., 2024) have been developed for fault diagnosis under varying conditions. Recent extensions include methods that use conditional discriminators to provide a more refined alignment between domains (X. Yu et al., 2020; Q. Zhang et al., 2021). While these methods have successfully promoted domain-invariant feature learning, they primarily focus on classification tasks, overlooking the unique

challenges of continuous predictions and feature scaling that arise in regression-based tasks like RUL prediction.

Beyond the closed set scenarios, PHM applications often face scenarios where fault classes may be only partially observed or entirely missing in either domain. This has led to the development of specialized approaches for partial and open-set domain adaptation (Q. Wang, Michau, & Fink, 2021). To address these challenges, several GAN-based domain adaptation methods were proposed (S.-W. Huang et al., 2018; X. Xie et al., 2020). For instance, a Wasserstein GAN framework was proposed to support the controlled generation of distinct fault signatures (Rombach et al., 2023), while a cycle GAN was proposed to map data across multiple source domains to manage cases of incomplete source data (Q. Wang, Xu, et al., 2024).

Remaining Useful Life Prediction

Standard domain adaptation approaches have been successfully applied in RUL prediction tasks to facilitate the transfer of knowledge across different operating conditions and different units. Several adversarial methods have been developed to improve RUL predictions by aligning source and target domains (da Costa et al., 2020). For example, da Costa et al., 2020 used adversarial training with Long Short-Term Memory models to extract domain-invariant features. Moreover, a Wasserstein distance-based weighted domain adversarial neural network (WD-WDANN) was proposed by Hu et al., 2022 to adapt RUL models across operating conditions by weighting source samples based on their similarity to the target domain. A discriminative-based adaptive framework was proposed in Xiong et al., 2023 that dynamically selects models according to failure modes.

Alternative domain adaptation paradigms, such as distribution alignment through MMD, have also shown promising results in RUL prediction (Zhu et al., 2020; Zhuang et al., 2021). For example, a digital twin-driven model combines graph domain adaptation with Laplacian MMD to handle non-Euclidean data, allowing feature transferability in bearing degradation data (Cui et al., 2024). Similarly, a residual convolutional network was proposed that incorporates MMD with an attenuation penalty to avoid negative transfer, facilitating cross-machine predictions (X. Lu et al., 2024). A memory-based network for RUL prediction was also proposed that combines MMD with domain adversarial learning to align temporal and spatial features across operating conditions, providing domain invariance in the feature space (J. Li et al., 2024). While such methods showed improvement by incorporating both adversarial training and distribution alignment, they primarily focus on general alignment strategies without addressing the specific requirements of regression tasks. This highlights the need for approaches tailored for regression that preserve feature scaling and maintain continuous relationships between inputs and predictions across domains while handling the additional complexity of temporal dependencies.

Moreover, existing approaches to domain adaptation typically apply adaptation techniques across entire operational cycles, assuming uniform feature distributions across all phases. This assumption overlooks the unique dynamics of each operational phase, especially the overrepresentation of certain phases, such as steady-state operations in industrial settings. Overrepresented phases can bias the model's predictions toward these frequently observed phases, potentially limiting generalization across all phases of operation. Expert knowledge could be leveraged to automatically detect or segment different phases, helping the model adapt more effectively to varied operational conditions.

1.3.3 Open-Set Domain Adaptation

Open-set and partial domain adaptation techniques have been widely explored across various domains, particularly in fault diagnostics within industrial applications (W. Zhang et al., 2021b; Y. Zhang, Ren, et al., 2023). Fault diagnostics typically involves discrete fault types, with the key challenge being the accurate classification of fault categories under domain shifts. However, not all fault types are typically observed, leading to scenarios where several fault classes may be missing in the source or target domain. Open-set domain adaptation (OSDA) techniques tackle the challenge of adapting to new, previously unseen fault types in the target domain (Z. Chen et al., 2023; W. Zhang et al., 2021a). In contrast, partial domain adaptation (PDA) addresses scenarios where the target domain includes only a subset of the classes available in the source domain (X. Li & Zhang, 2020; Q. Wang, Michau, & Fink, 2021; Y. Zhang, Ji, et al., 2023). Both approaches aim to mitigate negative transfer by filtering irrelevant source classes or target classes during adaptation.

While OSDA and PDA have been extensively studied in fault diagnostics, their application in prognostics remains underexplored. Prognostics differs fundamentally from diagnostics in its objective, which involves predicting continuous degradation trajectories rather than classifying discrete fault types. These degradation trajectories can also be represented through discrete health states (Ramasso et al., 2012), providing a classification-like perspective for health assessment. However, critical degradation states — particularly those close to failure — are often absent or underrepresented in the source domain due to operational and safety constraints.

Various OSDA methods developed in computer vision offer potential solutions to these challenges (Dong, Chatzi, & Fink, 2024; G. Li et al., 2021; Panareda Busto & Gall, 2017). These include methods that assign target domain samples to source categories while identifying and excluding unrelated samples (Panareda Busto & Gall, 2017), and adversarial techniques that create distinctions between known and unknown samples (Jang et al., 2022; Saito, Yamamoto, et al., 2018). More sophisticated approaches like Separate to Adapt (H. Liu et al., 2019) progressively separate unknown and known class samples using a coarse-to-fine weighting mechanism, while Rotation-based Open Set

(Bucci et al., 2020) leverages self-supervised tasks for unknown class detection. Recent advances include projecting features to hyperspherical latent spaces (M. Jing et al., 2021) and addressing semantic-level bias through causal alignment (W. Li et al., 2023).

However, these methods typically require access to source domain data during adaptation, which is often restricted in industrial settings due to privacy and proprietary constraints. This limitation has led to the emergence of Source-Free Open-Set Domain Adaptation (SF-OSDA) where models must adapt to both known and unknown classes using only a pre-trained source model and unlabeled target data (Liang et al., 2020; Roy et al., 2022). This scenario is particularly relevant in PHM applications, where both data privacy and emerging fault patterns are common concerns.

Several source-free methods have been extended to handle the presence of unknown classes in the target domain. For instance, SHOT (Liang et al., 2020) was adapted to identify unknown samples through entropy thresholding, while AaD (S. Yang et al., 2022) extended its clustering-based approach to separate unknown classes. U-SFAN (Roy et al., 2022) incorporated uncertainty estimation to detect samples from unknown classes. However, these adaptations primarily focus on known class semantics and discarding the unknown classes. Recent Universal Domain Adaptation approaches (Liang et al., 2021; Qu et al., 2023, 2024) attempt to handle both domain shifts and label set differences more systematically, using techniques like one-vs-all clustering and subspace decomposition to identify common and private target classes. Progressive Graph Learning (Y. Luo et al., 2023) similarly decomposes the target hypothesis space into shared and unknown subspaces. However, these methods either require specific source model training to incorporate unknown classes (Kundu, Venkat, M V, & Babu, 2020; Kundu, Venkat, Revanur, Babu, et al., 2020) or rely on fixed thresholds to distinguish known from unknown classes, making predictions sensitive to threshold selection.

Nevertheless, these approaches do not fully exploit the semantic information of the unknown samples. In industrial settings, particularly for health state assessment, unknown classes often represent critical degradation states not present in the source domain. Rather than simply filtering these states out, exploiting their semantic structure requires extending the classifier's ability to account for unknown classes. Increasing the number of classes predicted by the classifier would enable it to learn and directly differentiate between known and unknown classes, facilitating more efficient adaptation, even in the presence of a label shift.

1.3.4 Imbalanced Regression

Imbalanced regression poses unique challenges distinct from imbalanced classification due to the continuity in the label space. Unlike classification, where categories are discrete, regression labels are continuous, preserving ordinal relationships between samples through

their relative ranks and distances in the label space (Y. Yang et al., 2021). One approach to address this imbalance is to discretize the label space, applying methods from imbalanced classification. However, this approach requires a careful definition of an acceptable error threshold for discretization (J. Ren et al., 2022), which can limit its effectiveness and loss of the valuable ordinal information inherent in regression tasks.

Several strategies have emerged to better manage these label and feature relationships in regression tasks. Label Distribution Smoothing (LDS) (Y. Yang et al., 2021) and DenseLoss (Steininger et al., 2021) use kernel density estimation to smooth the label distribution, aiming to refine data reweighting and mitigate imbalance by estimating the underlying label density. Feature Distribution Smoothing (Y. Yang et al., 2021) similarly smooths the feature space by aligning feature statistics with nearby labels, improving generalization in imbalanced settings.

Beyond smoothing methods, some approaches introduce additional regularization tasks to strengthen feature representations, leveraging local and global relationships within the data (Y. Gong et al., 2022; Keramati et al., 2024; Z. Wang & Wang, 2024). For example, RankSim (Y. Gong et al., 2022) employs ranking similarity regularization, while contrastive learning techniques have been adapted for regression, as in CONR (Keramati et al., 2024), which enhances feature continuity in regression tasks. A recent method introduced by Z. Wang and Wang, 2024 combines deep evidential regression (Amini et al., 2020) with variational distribution learning, weighting data probabilistically by computing the latent distribution among data with similar target labels.

However, these approaches rely heavily on in-weight learning, which can lead to overfitting in minority regions under long-tailed settings, as pointed out by J. Ren et al., 2022. The use of Mean Square Error loss in imbalanced regression can exacerbate this bias toward majority regions. Recent work suggests that alternative paradigms, such as in-context learning, offer a promising solution by enabling models to adapt to new regions without requiring gradient updates (Chan et al., 2022). Unlike in-weight learning, in-context learning leverages diverse pre-training to adapt to various tasks using only a few context examples, potentially circumventing the limitations of gradient-based optimization.

1.4 Proposed Framework

In industrial applications, particularly in predictive maintenance and prognostics, regression tasks are essential for assessing asset conditions. RUL prediction, for instance, requires continuous-value predictions while also accounting for discrete health states that characterize degradation patterns. However, real-world scenarios often present challenges like domain shifts and imbalanced label distributions. While domain adaptation techniques have shown success in classification tasks, their application to regression problems remains limited. Furthermore, practical settings rarely meet assumptions such as balanced datasets and abundant labeled source data.

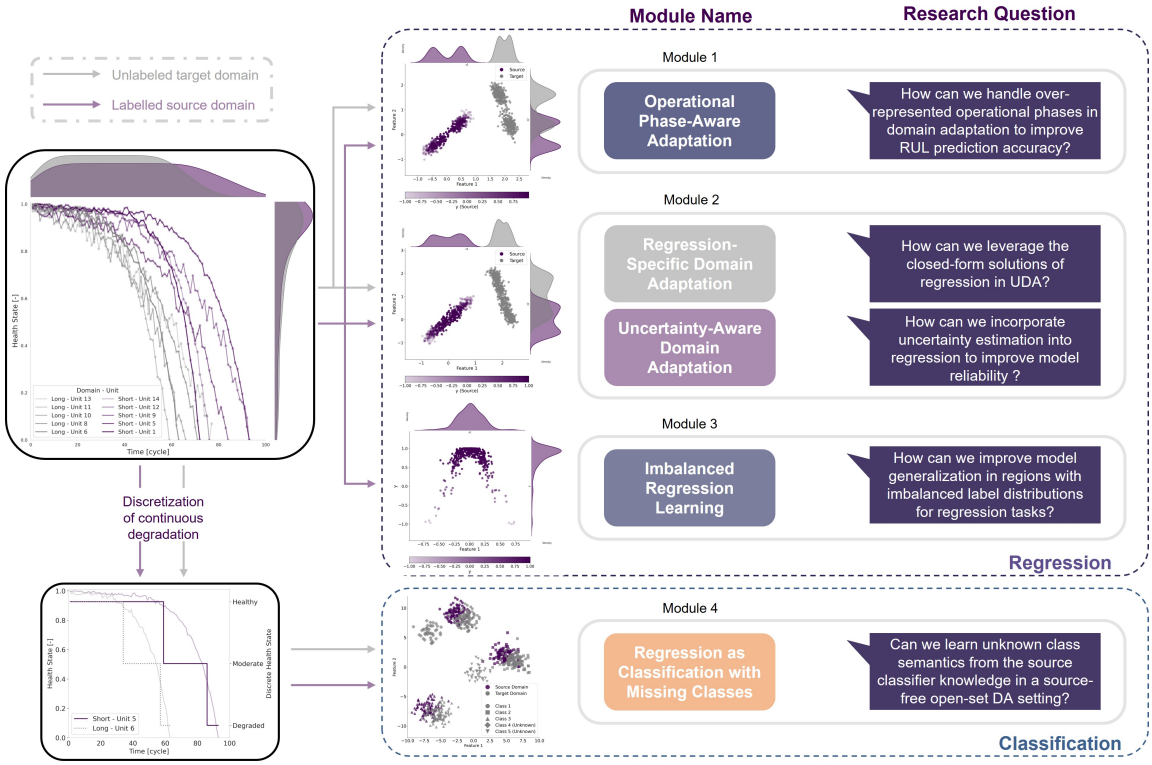


Figure 1.1: Overview of the Proposed PHM Framework for Domain Adaptation with Modules Addressing Real-world Challenges in Regression Tasks.

This thesis introduces a novel framework that addresses both domain adaptation and imbalanced regression. The proposed framework consists of four interconnected modules, each targeting specific challenges: (1) phase-aware adaptation that leverages expert knowledge to balance underrepresented and overrepresented operational phases, (2) regression-specific domain adaptation methods that preserve feature relationships and incorporate uncertainty estimation, (3) in-context learning strategies for generalization in minority regions of imbalanced distributions, and (4) adapting to open-set, source-free scenarios in which the target domain may contain new or unknown states of degradation. In practice, due to the lack of complete degradation trajectories and constraints on source

data, health states are often unknown or missing in the source domain. The source-free open-set module specifically addresses this need to incorporate unlabeled health states and adapt without relying on source data.

This combination of modules enables the framework to tackle domain adaptation and data imbalance as distinct yet interconnected challenges. The proposed methods aim to generalize effectively across different domains and data distributions, making them well-suited for real-world prognostic tasks where domain shifts, label imbalance, and label shift are frequent. An overview of the proposed framework is shown in Figure 1.1.

1.4.1 Addressed Challenges

The proposed framework is designed to address several critical challenges commonly encountered in industrial applications:

1. Industrial machinery typically operates in ordered sequences of operating phases (e.g., transient, steady-state), each impacting system degradation differently. Certain phases, like steady-state operation, may be over-represented in datasets, potentially biasing models. The proposed framework addresses this by **separating and aligning data across distinct operational phases** ensuring the model can adapt to these shifts without assuming uniform conditions.

2. While domain adaptation has been widely studied for classification, regression tasks present unique challenges due to their continuous nature and sensitivity to feature scale. Traditional approaches focus primarily on feature alignment, overlooking regression-specific challenges in the adaptation process. The proposed framework addresses this by **developing regression-specific adaptation methods** that preserve feature relationships and scales across domains.

3. Most existing machine learning approaches in prognostics, including domain adaptation methods, focus on providing point estimates without quantifying the uncertainty in their predictions. This limitation has been one of the reasons limiting the broad adoption of data-driven prognostics in industry, where reliable uncertainty quantification is required for risk assessment and decision-making. The framework addresses this by **incorporating uncertainty-guided domain adaptation**, which not only provides confidence measures alongside predictions but also leverages uncertainty estimation to guide the adaptation process itself.

4. Industrial datasets, particularly in PHM, are often imbalanced. For instance, data from initial or normal operation states may vastly outnumber observations from severe degradation states or critical failure events, as these highly degraded states are seldom observed due to preventive maintenance. The framework accounts for

this by explicitly **targeting imbalanced data distributions**, ensuring that the model can generalize even when data from failure states is scarce.

5. In many cases, the source domain may not contain data for all operational or degradation states present in the target domain. This is particularly relevant in the context of health assessment, where not all possible states of degradation may be represented in the source domain due to incomplete degradation trajectories. The framework addresses these limitations by **enabling adaptation to novel classes in a source-free setting**.

1.4.2 Modules

The **Proposed Framework for Domain Adaptation and Imbalanced Regression** consists of four interconnected modules designed to address key challenges in prognostics for industrial applications. The phase-aware distribution alignment module serves as a foundation, addressing the unique challenges of RUL prediction by explicitly considering distinct operational phases during adaptation. Building on this, a regression-specific domain adaptation module introduces two novel approaches: aligning inverse Gram matrices to preserve feature relationships and scales across domains, and incorporating uncertainty estimation to guide the adaptation process. To address the challenges of imbalanced data distributions common in industrial settings, an in-context learning module enables effective generalization in minority regions without requiring gradient updates. Furthermore, to handle scenarios where certain degradation states are missing in the source domain and source data is not available during adaptation, a source-free open-set adaptation module allows the framework to learn and incorporate novel class semantics during adaptation. An overview of the proposed framework is shown in Figure 1.1. The modules with their corresponding methods are:

- **Operating-phase Aware Distribution Alignment** This module leverages expert knowledge to align specific operational phases independently, focusing on phases with distinct degradation dynamics, such as the take-off, cruise, and landing phases. In RUL prediction, variations in operating phases result in differing impacts on system degradation. Domain adaptation methods for PHM often overlook these phase-specific distinctions, potentially causing misalignment due to the over- or under-representation of certain operating phases. This misalignment, particularly when phases like the cruise phase dominate, can lead to biased model predictions.

  To address this, we propose an approach in which separate discriminators are applied to each operational phase rather than relying on a single-domain discriminator, as commonly seen in methods such as Domain-Adversarial Neural Networks. By aligning the marginal distributions of each operational phase independently in the source and target domains, our approach ensures phase-specific adaptation. This method accounts for each unique operational profile phase, allowing for either

discrete assignments to specific regimes or probabilistic assignments that account for transition phases. As a result, the model achieves better generalization across units within a fleet, even when varying operating conditions induce domain shifts.

- **Regression-Specific Domain Adaptation** This module introduces two novel approaches for domain adaptation in regression by leveraging the inverse Gram matrix and uncertainty-guided alignment to address challenges unique to regression under domain shifts.
  - **Inverse Gram Matrix Alignment:** From a regressor's perspective and inspired by the least squares solution, we propose aligning the inverse Gram matrices of source and target features to preserve both feature relationships and feature scale between domains. Unlike prior methods, which overlook the importance of feature scaling, aligning the inverse Gram matrix ensures consistency in the range of predictions across domains while preserving relationships between features that are critical in regression for both source and target domains.
  - **Uncertainty-Guided Alignment:** We observe that regression models in domain adaptation tend to learn low-rank representations, which can lead to a loss of information in the feature space. This issue arises due to the constant variance assumption in Mean Squared Error loss, which is often violated in cases of domain shift. To address this, we propose leveraging an uncertainty estimation framework that captures variance shifts across samples. By incorporating uncertainty estimation, the model is guided to focus on reliable, high-confidence predictions that are more likely to transfer effectively across domains.

- **Imbalanced Regression Learning** Industrial datasets often exhibit significant imbalance, with an overrepresentation of low-severity or minor degradation states and only a few samples representing critical degradation states. We propose exploring alternative learning strategies through an in-context learning approach, leveraging priors obtained from meta-training across diverse regression tasks to improve robustness in underrepresented regions.

  Unlike gradient-based approaches, which tend to overfit in sparse regions, our in-context learning method adapts by conditioning on specific, representative examples from minority regions. Through our experiments, we discovered a counterintuitive result: *providing the model with more extensive context can reduce performance, as it biases the model toward majority regions*, particularly at the expense of minority region predictions. This finding emphasizes the need for a localized context.

- **Source-Free Open-Set Adaptation Module** In many industrial scenarios, certain degradation states may not have been observed in the previous run-to-failure trajectories, especially in safety-critical environments where assets are maintained

before severe degradation can occur. This absence of data in critical regions poses a challenge for regression-based models, as they rely on continuous predictions across all states. To address this, we propose transitioning to a classification approach to handle unknown degradation states in a source-free open-set setting. This shift allows the model to classify and differentiate between known and unknown degradation categories rather than making continuous predictions in regions with no available data.

In this module, we propose to use the classifier differently from previous methods. Traditional approaches often rely on high-entropy scores to discard data points considered as belonging to unknown categories, overlooking the opportunity to learn the underlying semantics of these points. To address this limitation, we propose generating new data points optimized to exhibit high entropy, allowing the model to learn new boundaries for a target classifier to accommodate unknown classes in the target domain. This enables the model to learn the semantics of these unknown categories, making it especially useful in environments where complete class information is unavailable or impractical to obtain.

1.5 Contributions

This cumulative thesis incorporates five key contributions that address critical challenges in distribution shifts and imbalance within representation learning for deep regression models. These works focus specifically on targeting domain adaptation for regression, uncertainty estimation, and learning under imbalanced conditions in both industrial and computer vision contexts. Together, these papers form the foundation of the thesis and are summarized below.

1.5.1 Domain Adaptation with Operational Phase Alignment for Prognostics

Most domain adaptation methods treat the entire operational cycle as a single distribution, overlooking both the distinct characteristics and inherent imbalances of different operating phases (e.g., take-off, cruise) that impact asset degradation and performance in different ways. In Chapter 2, we propose a novel domain adaptation framework for remaining useful life prediction that explicitly considers these operational phases in the alignment process. This framework introduces two approaches for aligning each operational phase separately between domains, significantly improving prediction accuracy in complex industrial environments with variable operating conditions.

Specific Contributions:

- Introduction of an operational-phase-aware domain adaptation approach for RUL prediction, that aligns the marginal distributions of each operational phase separately across domains.
- Introduction of two novel alignment strategies:
  - A hard assignment approach (OPS-DANN hard) that uses dedicated domain discriminators for each operational phase to ensure precise phase-specific distribution alignment.
  - A soft assignment approach (OPS-DANN soft) that handles uncertain phase transitions by probabilistically assigning samples to multiple phase-specific discriminators.
- Comprehensive empirical validation on the N-CMAPSS turbofan engine dataset, demonstrating that phase-specific alignment significantly improves RUL prediction accuracy across different flight profiles (short, medium, and long-haul) compared to state-of-the-art domain adaptation methods.

1.5.2 Unsupervised Domain Adaptation Regression by Aligning Inverse Gram Matrices

Existing unsupervised domain adaptation approaches often treat regression in the same way as classification, overlooking critical feature-scale differences. Chapter 3 proposes a novel approach leveraging closed-form solutions, specifically aligning the inverse Gram matrix in regression tasks to ensure robustness to domain shifts.

Specific Contributions:

- Proposal of a regression-specific domain adaptation method that aligns feature representations based on the inverse Gram matrix.
- Analytical demonstration of the limitations of traditional feature alignment for regression due to sensitivity to feature scale.
- Experiments on real-world and synthetic datasets to validate the efficacy of this approach for domain adaptation in regression.

1.5.3 Uncertainty-Guided Domain Adaptation for Regression

This work addresses a critical gap in domain adaptation literature by introducing a novel uncertainty-guided framework specifically designed for regression tasks. While classification models inherently produce uncertainties via predicted probabilities, regression models typically output single-point estimates without associated confidence measures. Chapter 4 addresses this by introducing an uncertainty-guided framework for regression-based domain adaptation, leveraging uncertainty estimates to regularize and guide the feature alignment process.

Specific Contributions:

- We provide a theoretical foundation linking domain adaptation gaps and mutual information loss under MSE, showing the benefits of incorporating uncertainty to mitigate this loss during domain adaptation,
- We design an uncertainty-aware domain adaptation method for regression, incorporating evidential learning to estimate prediction confidence.
- We leverage uncertainty estimates as a regularization mechanism to guide the adaptation process, improving model robustness under domain shifts by focusing alignment on more confident samples.
- Empirical validation on datasets from computer vision and prognostics, showing that uncertainty-guided adaptation significantly improves regression accuracy under domain shifts.

1.5.4 In-Context Learning for Imbalanced Regression

In imbalanced regression, where data is skewed toward certain outcomes, traditional gradient-based learning approaches often lead to overfitting. Chapter 5 explores the application of in-context learning as a solution, enabling the model to make adaptive predictions in data-sparse regions without parameter updates.

Specific Contributions:

- We propose IM-Context, a novel paradigm shift towards in-context learning for addressing imbalanced regression, showing that models can effectively generalize in minority regions without parameter updates, overcoming the limitations of gradient-based weight learning methods that tend to overfit in underrepresented regions.
- We provide a theoretical analysis of the expected prediction error in the context of imbalanced regression, revealing that increasing context size can counter-intuitively degrade performance in minority regions. We establish error bounds showing distinct behaviors between majority and minority regions, with the bound widening in sparse regions as more context samples are added.
- We propose a localized context selection strategy, enabling the model to condition on semantically relevant samples.
- Experimental demonstration that in-context learning outperforms state-of-the-art in-weight learning methods, particularly in regions of high imbalance.

1.5.5 Source-Free Open-Set Domain Adaptation for Regression as Classification

Most domain adaptation methods assume identical label spaces across domains, limiting their applicability when the target domain includes classes absent from the source domain. Additionally, privacy constraints often restrict access to source data during adaptation. Chapter 6 addresses this limitation by proposing a source-free open-set domain adaptation approach that generates synthetic data points for both known and unknown classes. This enables the model to adapt while explicitly learning discriminative features for target-private unknown classes while operating in a source-free setting.

Specific Contributions:

- We propose a simple yet effective framework for source-free open-set domain adaptation that explicitly learns the semantics of unknown classes, rather than treating them as outliers to be filtered out.

- We propose a method that integrates a novel target classifier with additional decision boundaries, achieved by optimizing synthetic feature points to exhibit low entropy for known classes and high entropy for unknown classes, enabling direct classification and direct semantic learning of unknown target classes.
- We demonstrate that with these additional decision boundaries, any off-the-shelf source-free closed-set domain adaptation method using self-training can be effectively leveraged for open-set scenarios, simplifying the open-set adaptation problem.

1.6 Relevance to Science and Economy

This thesis addresses fundamental limitations in domain adaptation and data imbalance for regression tasks — a paradigm less studied in the literature in comparison to classification and image segmentation, making these methods more practical and robust under the constraints typical of industrial settings. The methodologies developed in this thesis hold practical relevance; they are designed to reduce the need for extensive labeled data and adapt to new operating conditions with minimal supervision.

While the thesis primarily focuses on prognostics applications, the proposed methodologies are broadly applicable to other domains, such as demand forecasting for power system optimization, manufacturing quality control, and performance optimization. In many of these applications, obtaining sufficient labeled data is not only costly or time-consuming but often impossible; for example, in prognostics, failures cannot be artificially produced, making it difficult to obtain sufficient labeled failure data.

The operational-phase-aware domain adaptation framework, for instance, directly addresses variability in operating conditions across distinct phases — such as differences in bearing stress between low torque and peak load phases — that heavily influence asset wear. By aligning these phases separately, the method provides more reliable predictions without requiring extensive labeled data for each operational condition, supporting effective adaptation to industrial environments with over- or under-represented phases under domain shift.

For managing domain shifts, uncertainty quantification plays a critical role in industrial applications where models are deployed across diverse operating conditions. This is particularly relevant in sectors like railways, where different rolling stock sub-fleets operate under varying environmental and operating conditions, or in power generation, where turbines may experience different load profiles and environmental factors. Even within a single operation, units may exhibit significant variations due to differing configurations or external conditions. The uncertainty-guided method adapts model alignment by leveraging predicted confidence levels, providing a practical mechanism to monitor and respond to these shifts in real-time. By quantifying prediction uncertainty, the method serves as a safeguard, identifying when a model is operating within its validated domain

(in-domain) versus when it may produce unreliable predictions (out-of-domain). This capability is increasingly important given current regulatory frameworks, such as the EU AI Act, which require system reliability assessments for AI applications.

To address the challenge of data imbalance, the in-context learning approach enables streamlined deployment of machine learning models in environments where models cannot be frequently retrained offline. Industrial installations are often isolated from the internet due to safety regulations, making it challenging to update models regularly. Traditional in-weight learning requires repeated offline retraining, which in turn necessitates on-site updates and version management. In-context learning alleviates this complexity by allowing a single model to be deployed across various machines, using locally collected data as contextual input. This data can be incorporated automatically, providing self-sustaining improvements in model predictions by the end of each lifecycle, without needing frequent offline retraining.

This research has been practically applied in collaboration with Oerlikon through a *patent submitted* for a system designed to predict health states and estimate remaining useful life for spray plasma gun technology using raw voltage data, underscoring its direct impact on maintenance costs and asset reliability.

Beyond prognostics, these methods offer broad applicability. In engineering optimization, surrogate models are crucial for reducing the computational burden of expensive simulations. The ability to learn from low-density regions (few high-quality designs) while managing domain shifts across different design spaces can significantly enhance efficiency. Similarly, these methods offer value in other fields where regression tasks face similar challenges: in medical prognosis, where patient data may differ across hospitals or populations; in energy demand forecasting, where consumption patterns shift across regions or seasons; and in environmental monitoring, where sensor distributions and conditions vary across locations. In all these applications, the methods developed in this thesis provide a framework for addressing both distribution shifts and data imbalance, enabling more reliable deployment of regression models in real-world applications.

1.7 Publications

The following works were published or submitted during my doctoral study:

- **Nejjar, Ismail**, Jean Meunier-Pion, Gaëtan Frusque, and Olga Fink. "DG-Mix: Domain Generalization for Anomalous Sound Detection Based on Self-Supervised Learning." *DCASE*, 2022.
- **Nejjar, Ismail**, Qin Wang, and Olga Fink. "DARE-GRAM: Unsupervised Domain Adaptation Regression by Aligning Inverse Gram Matrices." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- Dong, Hao, **Ismail Nejjar**, Han Sun, Eleni Chatzi, and Olga Fink. "SimMMDG: A Simple and Effective Framework for Multi-modal Domain Generalization." *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- **Nejjar, Ismail**, Fabian Geissmann, Mengjie Zhao, Cess Taal, and Olga Fink. "Domain Adaptation via Alignment of Operation Profile for Remaining Useful Lifetime Prediction." *Reliability Engineering & System Safety (RESS)*, 2024.
- Frusque, Gaëtan, **Ismail Nejjar**, Majid Nabavi, and Olga Fink. "Semi-Supervised Health Index Monitoring with Feature Generation and Fusion." *IEEE Transactions on Reliability*, 2024.
- **Nejjar, Ismail**, Faez Ahmed, and Olga Fink. "IM-Context: In-Context Learning for Imbalanced Regression Tasks." *Transactions on Machine Learning Research (TMLR)*, 2024.
- **Nejjar, Ismail**, Gaetan Frusque, Florent Forest, and Olga Fink. "Uncertainty-Guided Alignment for Unsupervised Domain Adaptation in Regression.", 2024.
- **Nejjar, Ismail**, Hao Dong, and Olga Fink. "Recall and Refine: A Simple but Effective Source-free Open-set Domain Adaptation Framework." , 2024.
- Niresi, Keivan Faghih, **Ismail Nejjar**, and Olga Fink. "Efficient Unsupervised Domain Adaptation Regression for Spatial-Temporal Air Quality Sensor Fusion.", 2024.
- Sun, Han, Rui Gong, **Ismail Nejjar**, and Olga Fink. "DynAlign: Unsupervised Dynamic Taxonomy Alignment for Cross-Domain Segmentation." *International Conference on Learning Representations (ICLR)*, 2025.

2 Domain Adaptation via Alignment of Operation Profile for Remaining Useful Lifetime Prediction

This chapter corresponds to the published article[I]:
Nejjar, Ismail, Fabian Geissmann, Mengjie Zhao, Cess Taal, and Olga Fink "Domain adaptation via alignment of operation profile for Remaining Useful Lifetime prediction." *Reliability Engineering & System Safety*, 2024

Abstract: Effective Prognostics and Health Management (PHM) relies on accurate prediction of the Remaining Useful Life (RUL). Data-driven RUL prediction techniques rely heavily on the representativeness of the available time-to-failure trajectories. Therefore, these methods may not perform well when applied to data from new units of a fleet that follow different operating conditions than those they were trained on. This is also known as domain shifts. Domain adaptation (DA) methods aim to address the domain shift problem by extracting domain invariant features. However, DA methods do not distinguish between the different phases of operation, such as steady states or transient phases. This can result in misalignment due to under- or over-representation of different operation phases. This paper proposes two novel DA approaches for RUL prediction based on an adversarial domain adaptation framework that considers the different phases of the operation profiles separately. The proposed methodologies align the marginal distributions of each phase of the operation profile in the source domain with its counterpart in the target domain. The effectiveness of the proposed methods is evaluated using the New Commercial Modular Aero-Propulsion System (N-CMAPSS) dataset, where sub-fleets of turbofan engines operating in one of the three different flight classes (short, medium, and long) are treated as separate domains. The experimental results show that the proposed methods improve the accuracy of RUL predictions compared to current state-of-the-art DA methods.

[I]Please note, this is the author's version of the manuscript published in Reliability Engineering & System Safety. Changes resulting from the publishing process, namely editing, corrections, final formatting for printed or online publication, and other modifications resulting from quality control procedures may have been subsequently added.

2.1 Introduction

Remaining Useful Life (RUL) prediction, also referred to as prognostics, is a key element of Prognostics and Health Management (PHM). Prognostics aim to estimate how long a system can maintain its specified functionality before reaching its end of life (Si et al., 2011). Accurate RUL predictions enable scheduling maintenance from the perspective of operational and resource availability, avoid costly downtime, and prevent critical failures. In recent years, technological advancements and decreasing sensor costs have led to an increase in the amount of data collected from industrial assets (R. Zhao et al., 2019). This increased data availability allows taking advantage of advanced data-driven approaches such as Deep Neural Networks (DNNs), which have recently shown great potential in PHM applications such as fault detection and diagnosis (Fink et al., 2020).

Traditional supervised machine learning methods assume that, on the one hand, a representative labeled dataset is available and, on the other hand, that the training and test datasets stem from similar distributions. Unfortunately, in real-world scenarios, both assumptions do not hold for many applications (Rombach et al., 2021). While these challenges are also present in fault detection and diagnostics, they are particularly pronounced in prognostics. On the one hand, collecting a sufficiently representative dataset of run-time failure trajectories is often impossible due to the potentially catastrophic consequences of such an event in reality. On the other hand, deep learning models still face the challenge of domain shift due to the wide variety of operating conditions and limited training samples. These two challenges considerably impact the generalization ability of a model trained on a specific dataset and applied to a different but related dataset.

Unsupervised Domain Adaptation (UDA) has been shown to be effective in addressing the domain gap problem by adapting a model that has been trained on one domain (the source) to a different, unlabeled domain (the target). Previous research on domain adaptation has largely focused on classification tasks (Wilson & Cook, 2020b), and therefore, UDA has been extensively applied to fault diagnostics tasks (C. Li et al., 2020; Q. Wang et al., 2019b). Recently, several research studies have also proposed applying UDA to prognostics tasks. These approaches typically involve aligning the feature distributions between the source and target domains through adversarial training (da Costa et al., 2020) or reducing the disparity between feature distributions (Cheng et al., 2021; Zhuang et al., 2021). Although adversarial training has been widely used in prognostics, it has not taken into account the distinct phases of cyclical operation profiles, such as those found in flights, for example, take-off, cruise and landing phases. A distinct marginal distribution defines each phase of the operation profile, and adversarial adaptation methods may fail to capture such multi-modal structures in condition monitoring data. Therefore, the alignment of the source and target domains using DANN (Ganin & Lempitsky, 2015) assumes that the marginal distributions of each phase of the operation profile are also aligned. However, it may lead to misalignment of the phase of the operation profile due

to under- or over-representation since the different operation phases may have different durations in the different operation profiles, for example.

To improve the ability to adapt between different domains, we propose utilizing the common sequence of operation profile phases which each system undergoes. We take advantage of the prior knowledge of the operating cycles since the phases always occur in the same order, share specific characteristics, and their existence and sequence are invariant across all domains. For instance, in the aviation domain, a flight can be divided into several phases, that a flight follows, such as take-off, cruise, and descent. The flight phases differ in terms of, e.g., altitude, duration, speed, etc., and have distinct marginal distributions. Within the same flight class, e.g., the cruise phase will have similar characteristics. However, between different flight classes, the characteristics of the flight phases will be more different. Each phase has a different impact on engine stress and degradation and consequently also on the RUL. By incorporating the operation profile into domain adaptation, we aim to improve the target RUL prediction in the case of a domain shift.

In this paper, we consider the scenario where time-to-failure trajectories for complex systems such as aircraft jet engines and power plants are available within one fleet but not for units from a different fleet. To address this challenge, we define domains as sub-fleets of these systems that are operated under similar conditions. Our focus is on the domain adaptation problem, where we aim to transfer knowledge from a labeled source domain to an unlabeled target domain. The units within the same fleet may have a heterogeneous composition of operating conditions during their respective missions. These operating conditions can include differences in operating speed, machine load, working temperature, and environmental noise. These differences can lead to diverse distributions of marginal characteristics, making it difficult to adapt models to new domains. Previous research has shown that when the feature distribution is multimodal, adapting only the feature representation can be challenging for adversarial networks (Arjovsky & Bottou, 2017; Long et al., 2018).

To tackle this issue, we propose to replace the single-domain discriminator from DANN (Ganin et al., 2016) method with a discriminator for each phase of the operation profile. This approach is inspired by the success of multi-task learning (Zamir et al., 2018), and auxiliary tasks (Q. Wang, Dai, et al., 2021a) in facilitating adaptation between domains. The proposed methodologies align the marginal distributions of each phase of the operation profile in the source domain with its counterpart in the target domain. This work investigates in depth the learning of invariant features by adversarial learning for the alignment of sub-fleets (units operated in a similar way), taking into account their operation profiles. Two methods are proposed to deal with the above scenarios: 1) each part of the operational profiles can be assigned to a specific regime, or 2) the parts of the operational profile are smoothly assigned to all regimes with a defined probability, which is particularly useful for transition phases.

We evaluate the performance of the proposed algorithm on the NASA New Commercial Modular Aero-Propulsion System Simulation (N-CMAPSS) turbofan degradation dataset (Arias Chao et al., 2021). The dataset contains three flight classes: short (S), medium (M), and long (L), with a total of 15 units, with five units in each flight class. The proposed domain adaptation methods were evaluated on three adaptation tasks of increasing difficulty. The models were applied between sub-fleets, each consisting of five units, flying in a specific flight class. The adaptation scenarios consist of transferring from medium to long flights ($M \rightarrow L$), from short to medium flights ($S \rightarrow M$), and finally from short to long flights ($S \rightarrow L$). The adaptation scenarios always involve the adaptation from a flight class where run-to-failure trajectories are available to a flight class without any labels. The results of the three tasks demonstrate that the proposed algorithm, which takes into account the operation profile, can improve over the traditional DANN (Ganin et al., 2016) and outperform other comparative benchmark methods.

2.2 Related Work

2.2.1 Unsupervised Domain Adaptation

Unsupervised domain adaptation (UDA) (S. J. Pan & Yang, 2009) aims to improve the performance of a model on a target domain in the presence of a domain shift between the labeled source domain and unlabeled target domain. Several UDA methods have been proposed to align the feature distributions between the two domains during training using either discrepancy losses or adversarial training.

Prior works mainly relied on distribution alignment. While Deep Adaptation Networks (DAN) (Long et al., 2015) minimize the Maximum Mean Discrepancy (MMD) over domain-specific layers, Joint Adaptation networks (Long et al., 2017) align the joint distributions of domain-specific layers across different domains based on a Joint Maximum Mean Discrepancy (JMMD). In contrast, adversarial-based methods strive to obtain domain-invariant representations through adversarial training. For instance, Domain-Adversarial Training of Neural Networks (DANN) (Ganin et al., 2016) aims to create a domain-invariant representation by using a domain discriminator. To achieve this, the model is trained with a reversal gradient layer to make the feature space indistinguishable for different domains. Maximum Classification Discrepancy (MCD) (Saito, Watanabe, et al., 2018) is another adversarial-based method that aims to reduce domain divergence and align the distribution of a target domain by considering task-specific decision boundaries through the use of task-specific classifiers in an adversarial training approach. Other approaches focused on the distribution of features for both domains across the batch normalization layers. Such approaches include Adaptive Batch Normalization (AdaBN) (Y. Li et al., 2016b) and Automatic Domain Alignment Layers (AutoDial) (Carlucci et al., 2017), which aligns the distributions via modified batch normalization layers.

Most DA approaches have been developed for classification tasks, and domain adaptation for regression has received little attention. While classification approaches aim at generating decision boundaries to separate data into different classes, regression methods, on the other hand, aim at predicting continuous numerical outputs with a defined ordinal relationship. Early works on domain adaptation for regression problems introduced a weighting scheme that assigns a weight according to the similarity between the training and test samples (Cortes & Mohri, 2011b; Mansour et al., 2009a). Recent works in the deep representation learning regime proposed different strategies to close the domain gap. For example, Representation Subspace Distance (X. Chen et al., 2021) explores the Riemannian geometry of the Grassmann manifold to reduce the domain gap using the orthogonal bases representation of the subspace as opposed to instance representations. More recently, taking inspiration from the closed-form solution of least-square problems, DARE-GRAM (Nejjar, Wang, & Fink, 2023b) proposed a strategy centered on aligning the inverse Gram matrices of source and target features, with the aim of mitigating this issue.

Many of the mentioned methods for classification can be naturally extended to regression problems. Nevertheless, when dealing with complicated regression problems, there are still no straightforward solutions to the fundamental problem of unsupervised domain adaptation for regression.

2.2.2 Domain Adaptation applied to PHM

In recent years, domain adaptation has been increasingly applied to PHM applications (W. Li, Huang, et al., 2022), particularly in the area of fault diagnostics classification. Three main domain gaps exist in the PHM context: between varying operating conditions (Y. Pan et al., 2019; Z. Zhang et al., 2020), between different units of a fleet (Michau & Fink, 2021), and between simulations and real data (Gao et al., 2021; Q. Wang, Taal, & Fink, 2021; B. Yang et al., 2019; K. Yu et al., 2021).

Most research in this area has focused on the transfer between discrete operating conditions. Classic domain adaptation methods, such as distribution alignment (X. Li et al., 2019; W. Lu et al., 2016) and adversarial alignment (H. Bao et al., 2021; B. Zhang et al., 2018), have been widely used in this context and have been found to be beneficial (Q. Wang et al., 2019b). Proposed approaches include discrepancy-based domain adaptation, using various criteria such as Maximum Classification Discrepancy (MCD) (H. Bao et al., 2021) and Maximum Mean Discrepancy (MMD) (T. Han et al., 2021; Y. Li et al., 2021; Tong et al., 2018). Deep domain adversarial frameworks such as DANN (Jiao et al., 2020; W. Li, Yang, et al., 2022), (S. Liu et al., 2023) have been developed and applied for fault diagnosis of machines under varying working conditions. Some works have sought to improve upon adversarial alignment through the use of conditional discriminators (X. Yu et al., 2020; Q. Zhang et al., 2021).

Remaining useful life predictions have received increasing attention. (Y. Li et al., 2023) proposes a novel RUL prediction approach that combines knowledge of the sensor relationship and deep learning models. Recently, (Xiong et al., 2023) proposed an adaptive framework that can select the correct trained model according to failure modes. DSCN-DTAM was proposed in (Cheng et al., 2022) and employs multiple regularization strategies and a double transferable attention mechanism to improve feature transferability in the presence of shift. Furthermore, transfer learning has also been recently increasingly applied to RUL prediction to increase the transferability of the learned feature,(Cheng et al., 2022; Ding et al., 2021; A. Zhang et al., 2018). For example, a transfer learning algorithm based on Bidirectional Long Short-Term Memory (BLSTM) recurrent neural networks was proposed in (A. Zhang et al., 2018) for RUL estimation. The algorithm fine-tunes a model trained on a large amount of data from a source task with a small amount of data from a target task, usually a different but related task. The results showed that transfer learning is effective, except when transferring from a dataset of multiple operating conditions to a dataset of a single operating condition, leading to negative transfer learning. In addition to cases where labels are available for the target, also unsupervised domain adaptation approaches have been applied for RUL prediction. Different adversarial methods have been proposed to align the source and target domains and improve the performance of RUL prediction (da Costa et al., 2020). For example, (da Costa et al., 2020) used adversarial training to extract domain-invariant features from a Long Short-Term Memory (LSTM) model. Recent work (Hu et al., 2022) introduced a novel Wasserstein distance-based weighted domain adversarial neural network (WD-WDANN) for RUL prediction under different operating conditions by measuring the similarity of source samples to the target domain to determine the sample quality. Multiple-kernel maximum mean discrepancies (MK-MMD) were proposed and shown to be more robust than single-kernel methods (Zhu et al., 2020; Zhuang et al., 2021) and help to minimize the distribution discrepancy between different failure behaviors in the feature space.

Previous approaches proposed for RUL prediction in the context of PHM have mainly focused on applying domain adaptation on the entire operating cycle without considering the distinct phases of the operation profile for each domain. Such an approach aligning source and target domains using domain invariant feature learning implicitly assumes that the marginal distributions of different operating conditions are also aligned. We propose replacing the single-domain discriminator with multiple-domain discriminators for each operating condition. In this way, we ensure that the different marginal distributions are aligned with their counterparts in the other domain.

2.3 Methodology

2.3.1 Terminology

Before delving into the details of the proposed algorithm, it is important to define some key terms that will be used throughout the text. In this study, we make use of the terms *operation profile* and *operation condition*.

An operation condition: refers to the specific state or set of conditions under which a system is operating and can be thought of as different domains.

An operation profile: is a detailed and specific way of describing how industrial assets are operated and are controlled under different conditions while sharing similar characteristics in the sequences in which the operating conditions occur. An operation profile is typically composed of **distinct and discrete phases**, which are characterized by definite control and condition parameter ranges defining the states of the system that are retained for specific periods of time and are characterized by a unique marginal distribution of its characteristics.

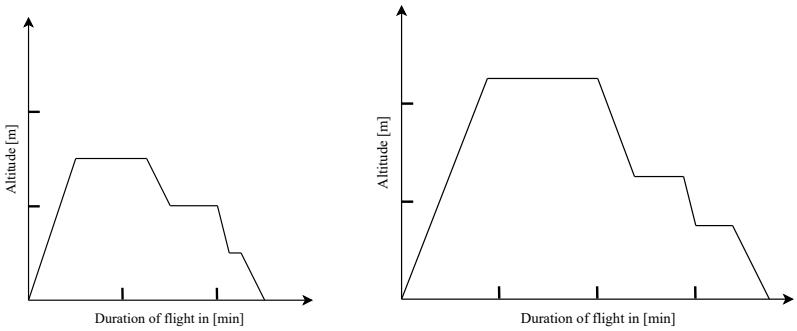
Together, the operation profile (with different phases) and operation conditions provide a comprehensive understanding of how industrial assets function and perform over time.

2.3.2 General Problem Definition

In this paper, we study the unsupervised domain adaptation problem in the context of regression. During training, we are given access to a set of n_s labeled samples from a source domain $D_s = \{(x_s^i, y_s^i)\}_{i=1}^{n_s}$ and n_t unlabeled samples from a target domain $D_t = \{(x_t^i)\}_{i=1}^{n_t}$. Each observation $x^i \in \mathcal{X} \subset \mathbb{R}^{K \times T}$ is a multivariate time sequence of K raw measurements of length T, where T is the length of the observation window. Finally, given a sample i from the set of all trajectories in the source domain, $y^i \in \mathcal{Y} \subset \mathbb{R}$ denotes the Remaining Useful Lifetime (RUL) value of the observed sequence x^i. Labels of target data are not available during training. The goal of the task is to train a model using labeled set D_s and unlabeled set D_t multivariate time-series data of aircraft engines operated under different operating conditions.

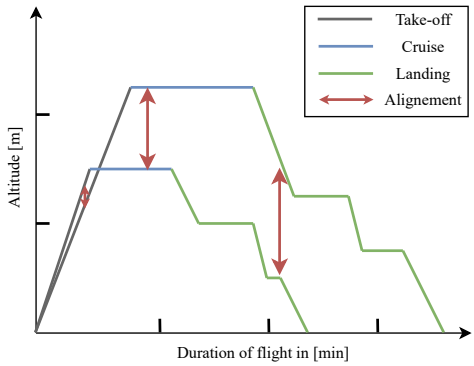
2.3.3 Operation profile

Industrial assets may operate under different operating conditions but share similar phases of the operation profile. For example, in aviation, a flight is usually composed of different phases, such as take-off, cruise, and descent, requiring different thrust levels to propel the aircraft forward (Fig.2.1), thus applying different types and levels of stress on the jet engine.



(a) Operating Condition: Short flight Altitude profile

(b) Operating Condition: Long flight Altitude profile



(c) Altitude level at the different phases of the flight Take-off, cruise, and Landing.

Figure 2.1: The figure 2.1a and 2.1b show two flights that were operated differently and have different profiles. Figure 2.1c illustrates the presence of similar phases in both flights, which were used for the alignment process.

More generally, the operation profile can be determined by discretizing a particular measured or control parameter based on domain knowledge or using an unsupervised clustering algorithm on the control parameters or multivariate observations. The final objective is to obtain a set of n_p discrete and common phases of the operation profile for each of the two domains.

This research assumes that a finite number of discrete operation profile labels can either be provided based on domain knowledge or identified using the measured or control parameters. We argue that taking into account the different phases of the operation profile can improve the alignment process.

The following section presents how these operation profile labels can be exploited to extend upon the existing DANN methods to support the alignment process.

2.3.4 Domain-Adversarial Neural Networks

As the operating conditions of the source and target sub-fleets differ, a model trained on the source data would be able to easily distinguish the feature vectors of the target domain from the source domain due to the domain shift, leading to poor performance when applied to the target domain. Adversarial distribution alignment methods aim to address this problem by ensuring that the feature extractor is unbiased with respect to the characteristics of the source and target domains.

DANN (Ganin & Lempitsky, 2015) is a broadly applied domain adaptation method that aligns the distributions of source and target features by adding a domain discriminator and introducing adversarial training.

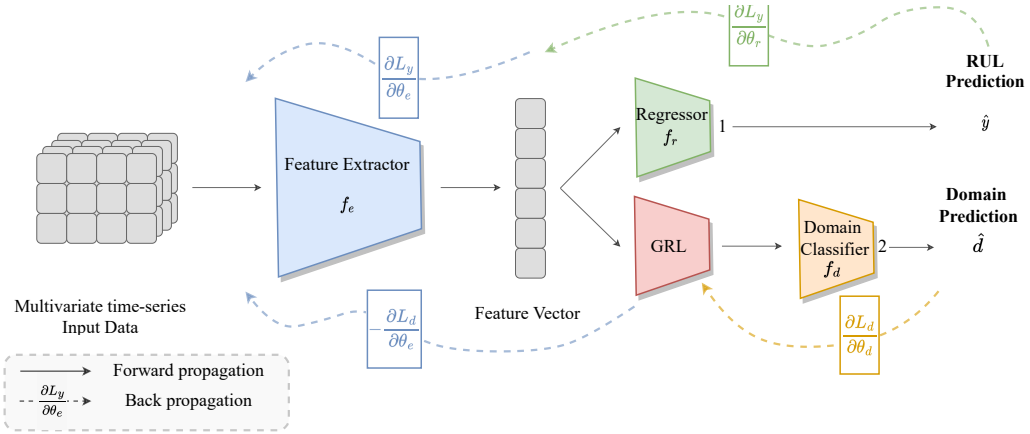


Figure 2.2: Standard architecture of DANN approaches. The gradient Reversal Layer (GLR) ensures that the feature distributions over the two domains are indistinguishable for the domain classifier

As illustrated in Figure 2.2, the DANN model is composed of a feature extractor f_e, a regressor f_r, and a domain classifier f_d parameterized by $\theta_e, \theta_r, \theta_d$ respectively. The feature extractor f_e takes the input data x and learns a feature representation denoted by $f_e(x)$. The regressor predicts the RUL label for each input sample $\hat{y} = f_r(f_e(x))$. The RMSE loss is used for minimizing the error between the true and the predicted RUL on the source data:

$$\mathcal{L}_{y,s}(\theta_e, \theta_r) = \sqrt{(\hat{y}_s - y_s)^2} \tag{2.1}$$

The second part of the model includes a domain discriminator, which is responsible for aligning the two domains. The domain classifier predicts the domain label for each input sample $\hat{d} = f_d(f_e(x))$, where $d \in \{0, 1\}$ is a domain label assigned to each training example to indicate its origin. The domain classifier uses the binary cross entropy loss function shown in Equation 2.2.

$$\mathcal{L}_d(\theta_e, \theta_d) = -\Big(d \log(\hat{d}) + (1-d) \log(1-\hat{d})\Big) \tag{2.2}$$

While the domain discriminator weights θ_d are trained to minimize the binary cross-entropy loss, the feature extractor weights θ_e are updated to maximize the binary cross-entropy loss. The feature extractor learns to extract domain-invariant features, rendering the domain discriminators incapable of predicting the true domain label. These two contradicting objectives are trained in an adversarial procedure utilizing a gradient reversal layer (GRL). GRL is an operation that acts as an identity function during the forward pass and reverses the sign of the gradient during the backward pass. The GRL allows us to implement this optimization problem in practice easily (Ganin & Lempitsky, 2015).

Finally, the regular stochastic gradient solvers (SGD) can be adjusted to optimize the model as suggested by Equations 2.3 and 2.4.

$$(\hat{\theta}_e, \hat{\theta}_r) = \arg\min_{\theta_e, \theta_r} \mathcal{L}(\theta_e, \theta_r, \hat{\theta}_d) \tag{2.3}$$

$$\hat{\theta}_d = \arg\max_{\theta_d} \mathcal{L}(\hat{\theta}_e, \hat{\theta}_r, \theta_d) \tag{2.4}$$

2.3.5 Operation Profile-Specific Alignment

In this research, we propose the Operation Profile-Specific (OPS) alignment framework, which aims to align the marginal distributions of the specific phases of the operation profile between different domains. These phases are often overlooked by other domain-invariant feature-learning techniques. Our approach extends the alignment component of DANN to consider each operating phase individually when aligning the source and target domains. In previous research, it has been demonstrated that when the feature distribution is multimodal, adapting only the feature representation can be challenging for adversarial networks (Arjovsky & Bottou, 2017; Long et al., 2018). Therefore, to improve DANN's performance in these scenarios, we chose to use the DANN method, as it has a strong track record in domain adaptation tasks and has previously been successful in regression tasks for PHM (da Costa et al., 2020). We propose two different approaches for operation profile-specific alignment:

- Hard assignment : OPS-DANN (hard)
- Soft assignment: OPS-DANN (soft)

For hard assignment, each sample is strictly assigned to its corresponding phase within the operational profile. In contrast, soft assignment involves assigning samples to multiple discriminators, which can be advantageous when operation phases are uncertain. This approach proves particularly useful during transitional periods between phases, where samples could be associated with more than one operation phase.

OPS-DANN (hard)

Our first proposed extension to implement OPS alignment involves using dedicated domain discriminators for each operation profile. A hard assignment is made for each sample to its respective phase of the operation profile. The goal of OPS alignment is to separately align the distinct marginal distributions of each regime of the operation profile across the two domains.

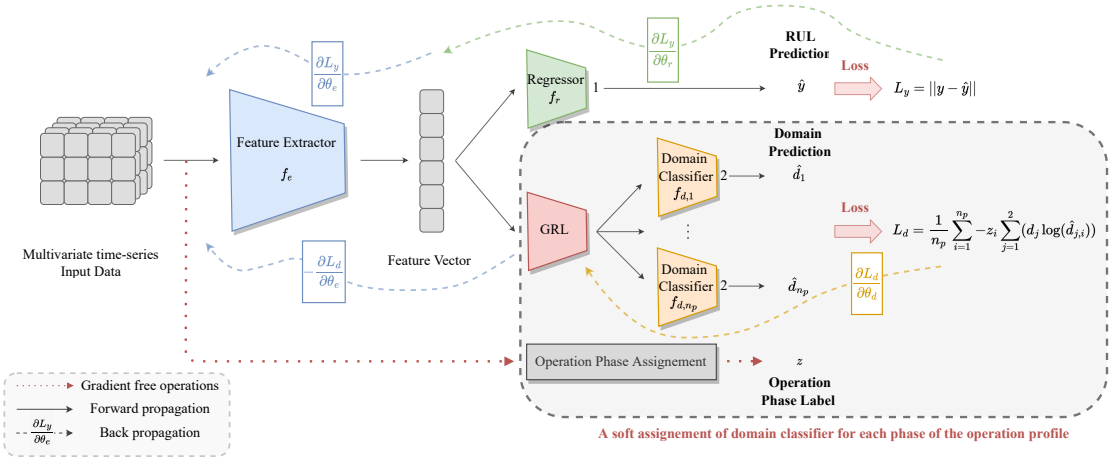


Figure 2.3: Proposed OPS-DANN (hard) approach. The source and target samples of each operation profile are aligned with a dedicated domain discriminator.

This can be achieved by replacing the single domain discriminator task that discriminates between the source and the target domain with as many individual discriminators as there are discrete operating phases. Each individual domain discriminator still aims to differentiate between the source and the target domain. However, it only does so for samples belonging to one operating phase. This ensures that the marginal distribution of each operating phase is aligned with its counterpart in the other domain. The proposed framework is visualized in Figure 2.3.

The domain discriminators are separated for each n_p operating phases. Each domain discriminator is parameterized with its own set of weights θ_{d_j} with $j \in \{1, \ldots, n_p\}$ and aims to predict the domain label $\hat{d}_{j,i} = f_{d_j}(f_e(x))$ for a sample that is assigned to the i-th domain discriminator. The operating phase label z^i assigns each sample to one of the n_p discriminators. The binary cross entropy loss function shown in Equation 2.5 is utilized to determine the loss of each domain discriminator, where z^i is a binary variable indicating whether a sample belongs to operating phase i or zero otherwise. The new domain discriminator loss is presented in Equation 2.5:

$$\mathcal{L}_{d,i}(\theta_e, \theta_d) = -z_i \left(d \log(\hat{d}) + (1 - d) \log(1 - \hat{d}) \right) \tag{2.5}$$

OPS-DANN (soft)

A soft assignment of samples to multiple discriminators can be useful when the operation phase labels are not certain. This can happen, for example, when a system is transitioning between two phases, and samples can be considered to belong to both operation phases. Instead of using a hard assignment of each sample to a single phase of the operation profile, we propose using a probabilistic assignment (soft assignment). To achieve this, we add an additional classifier after the feature extractor to classify the operation phase. This classifier is trained in a supervised manner using the available operating phase labels and outputs a probability distribution over the n_p phases. The soft assignment allows samples to be assigned to multiple domain discriminators when the operating phase classifier is uncertain. The predicted probabilities are then used to weigh each sample's contribution to each of the n_p domain discriminators. This model is depicted in Figure 2.4.

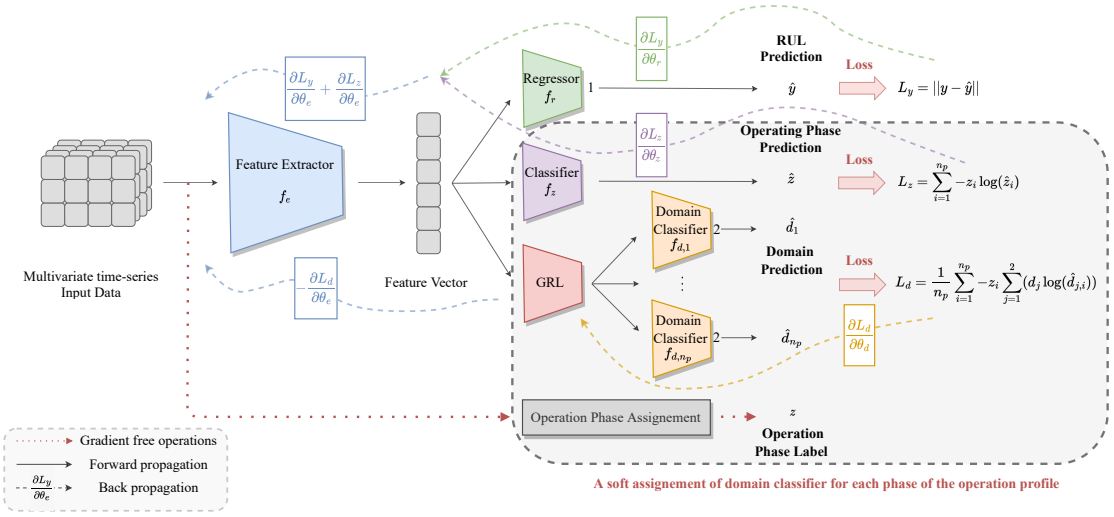


Figure 2.4: Proposed OPS-DANN (soft) approach. An additional operation profile classifier is added to allow for the assignment of each sample to multiple domain discriminators depending on its probability of belonging to the respective operating regime.

For OPS-DANN (soft), the previously introduced equations change slightly due to the additional operating phase classifier parameterized by θ_z. For each source and target sample, it predicts a probability of a sample belonging to each of the n_p operating phases $\hat{z} = f_z(f_e(x))$. The prediction $\hat{z}$ can be written as a n_p-dimensional vector $\hat{z} = [\hat{z}_1, \ldots, \hat{z}_{n_p}]$. The cross-entropy loss function compares the model's predictions to the true operating phase labels.

$$\mathcal{L}_z^i(\theta_f, \theta_z) = -z^i \log(\hat{z}^i) \tag{2.6}$$

The final loss for the OPS-DANN (soft) can be written as :

$$\mathcal{L}_{d,i}(\theta_e, \theta_d) = -\hat{z}_i \left(d \log(\hat{d}) + (1-d) \log(1-\hat{d}) \right) \tag{2.7}$$

2.4 Case Study

2.4.1 N-CMAPSS Dataset

We evaluate our proposed methods for Domain adaptation on the new Commercial Modular Aero-Propulsion System Simulation (N-CMAPSS) dataset, which contains run-to-failure trajectories of large turbofan engines (Arias Chao et al., 2021). We compare the performance to previously proposed DA approaches. The dataset was created using NASA's high-fidelity simulation model (Frederick et al., 2007), which allows the simulation of flight data over a wide range of flight conditions. Concretely, the flight data covers take-off, cruise, and descend flight conditions corresponding to different commercial flight routes. Each engine unit is assigned to one of the three flight classes based on the duration of the individual flights. Short flights are defined as flights that last between one and three hours, medium flights are between three and five hours long, and long flights last longer than five hours.

The degradation behavior of each engine unit is modeled as the combination of three contributors: an initial degradation, a normal degradation, and an abnormal degradation due to a fault. In the first phase, the engine degrades due to the normal degradation until the onset of a fault, whereby different fault types can occur. *Normal degradation* is modeled linearly. Once a fault is initiated, the engine enters an *abnormal degradation* phase until it ultimately reaches its end-of-life (EOL). During the abnormal degradation phase, the engine health decays exponentially. Figure 2.5 shows sample degradation trajectories of six engine units.

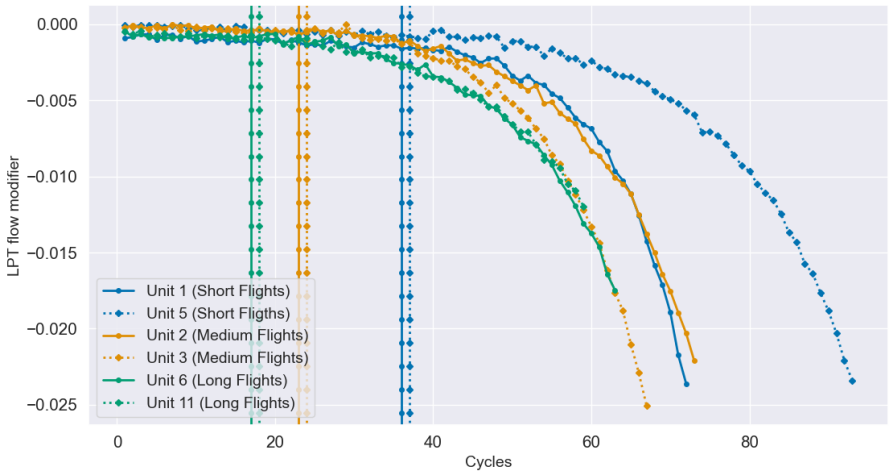


Figure 2.5: Example of degradation trajectories of engine units over time for engines from different domains. Vertical lines denote the fault initialization of each engine unit.

The degradation trajectories are given as a multivariate time series of the sensor readings, containing measurements from 14 sensors from the turbofan engine, as well as scenario descriptors containing 4 parameters characterizing the operating condition of the flight,

Table 2.1: Number of units and samples of the three considered domains.

Domain Flights	**Short (S)**	**Medium (M)**	**Long (L)**
Number of units	5	5	5
Number of samples	104k	202k	362k

also referred to as scenario descriptors. The signals and the scenario descriptors are sampled at a frequency of 1 Hz. The different sensors include various temperature and pressure measurements, fan speeds, and fuel flow. The scenario descriptors include the altitude, speed, throttle-resolved angle, and total temperature at the fan inlet.

Each turbofan engine consists of five main components: the fan, the low and high-pressure compressor, and the low-and high-pressure turbine. During the simulated lifetime of each unit, one or a combination of several sub-components fails, leading to the onset of one of seven possible fault types. Fault severity increases over time. The N-CMAPSS dataset introduced several improvements compared to the popular original CMAPSS dataset (Saxena et al., 2008), frequently used by researchers working on RUL-prediction tasks (Chao et al., 2022; da Costa et al., 2020; Fan et al., 2020; X. Li et al., 2018). The N-CMAPSS differs with respect to two main aspects. First, it considers actual flight conditions recorded on board of a commercial aircraft. Secondly, it extends the modeling of degradation by linking the degradation process to its operational history. The RUL prediction aims to indicate how many flights (i.e., cycles) of a particular engine (unit) are left before its EOL.

The N-CMAPSS dataset (Arias Chao et al., 2021) contains eight datasets, each with different fault types and flight classes. In this study, we consider the different flight classes as distinct domains and focus on dataset three (DS03) of the N-CMAPSS dataset, which is characterized by a single fault type (HPT-efficiency failure combined with LPT flow and efficiency failure) and equally distributed flight classes. We define the sub-fleets operated under different flight classes as a domain. The subset used for this paper contains an equal number of five units per flight class. Despite the identical number of units, the total number of samples of the three domains varies considerably due to the different lengths of the flights, as shown in Table 2.1.

In aviation, engines are subject to high stress during take-off. Therefore, in this dataset, short-haul flights have more take-offs and landings compared to long-haul flights, resulting in a different type of degradation. In addition, the sensor measurements of long-haul flights are significantly different from those of short-haul flights, as they reach higher altitudes and faster speeds, as can be seen in Figure 2.6.

Given the three domains, there are six possible domain adaptation tasks, of which only the three presumably most difficult tasks are considered in this research. The three tasks

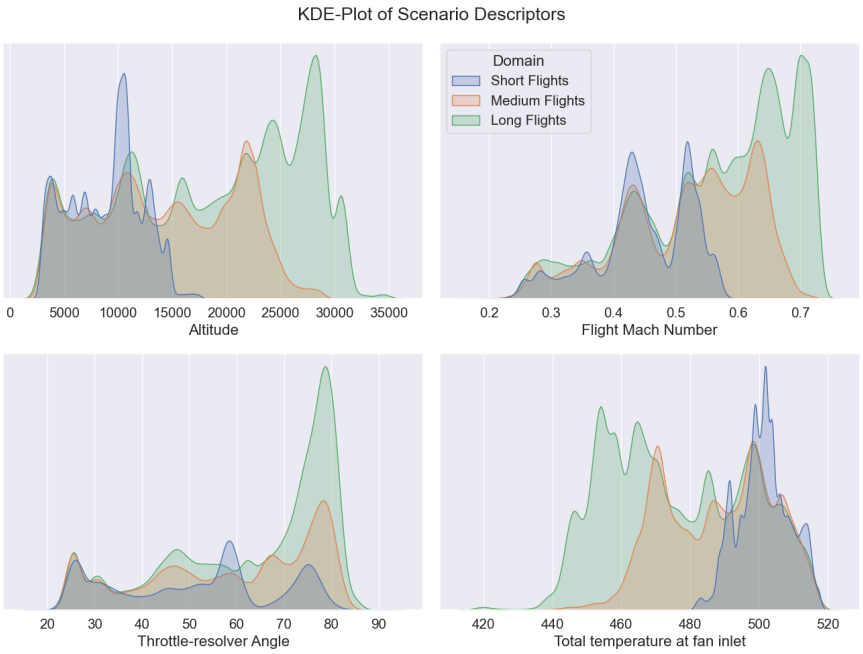


Figure 2.6: The kernel density estimate (KDE) shows the probability distribution of the four scenario descriptors for each domain separately. In three out of four cases, it can be seen that the long-haul flights span the widest range of feature values, while the short-haul flights cover a noticeably smaller part.

comprise the transfer from a sub-fleet of short-range flights to the sub-fleets of mid-range and long-range flights, as well as from mid-range to long-range flights. These three tasks are particularly difficult because the marginal distribution of the characteristics of shorter flights does not cover those of longer flights, as visualized using the four scenario descriptors in Figure 2.6. The corresponding considered adaptation tasks are abbreviated as $S \rightarrow M$ for short to medium flight length adaptation tasks by $S \rightarrow L$ and $M \rightarrow L$ for the other two tasks, respectively.

In general, it seems intuitive to subdivide a flight into take-off, cruise, and landing operating conditions. However, in this work, we found a better suitable division of the operation profile: in ascending, steady, and descending operating conditions. This division allows a more fine-grained separation and is more closely related to how the airplane is operated. For example, the operating conditions of an airplane that changes its flight altitude during a flight would only be assigned to the cruise condition with the more intuitive partitioning approach. In contrast, the partitioning proposed in this research allows separating the two steady flight sequences from the descending one in between (see Figure 2.7).

Ascending, steady, and descending flight conditions can be identified using the first-order derivative of the altitude measurement. To that end, the change in altitude

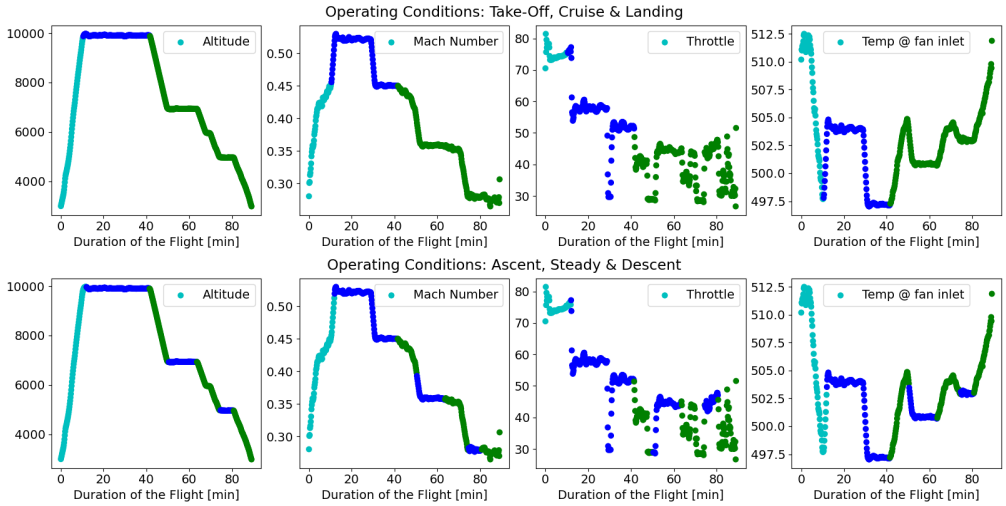


Figure 2.7: Direct comparison of two possible separations of the operating conditions. Ascent and take-off are colored in turquoise, respectively; cruise and steady flight in dark blue; and descent and landing in green. The short periods of steady flying are differentiated from the rest using the change in altitude.

between neighboring sampling points was considered and grouped into the three operating conditions using a threshold value. The threshold was experimentally set to $T = 0.5\frac{ft}{s}$. The Operation Phase label z can be defined based on the following inequalities:

$$\mathrm{z} = \begin{cases} \text{Ascending} & \text{if } \frac{x_{alt}^{i+1} - x_{alt}^{i}}{\Delta t} \geq T \\ \text{Descending} & \frac{x_{alt}^{i+1} - x_{alt}^{i}}{\Delta t} \leq -T \\ \text{Steady} & \text{otherwise} \end{cases} \tag{2.8}$$

A median filter with a length of 51 was applied to smoothen the predictions.

The key idea behind the proposed OPS alignment is to consider each operation phase separately during the alignment to ensure that its marginal distribution is matched with its counterpart in the other domain. Figure 2.8 shows the marginal distributions of three exemplary sensors for the short, medium, and long flight domains. Most of the other sensor values follow similar patterns.

2.4.2 Preprocessing

Before a fault initiates, the normal degradation process is slow, and it can be assumed that the RUL is very large. Thus, for this work, we only start predicting the RUL after the onset of the fault. This results in a two-step process in which fault detection is performed first, and the RUL prediction is only initiated after the fault has been detected. In this paper, we are only focusing on the second step and assume that the fault onset

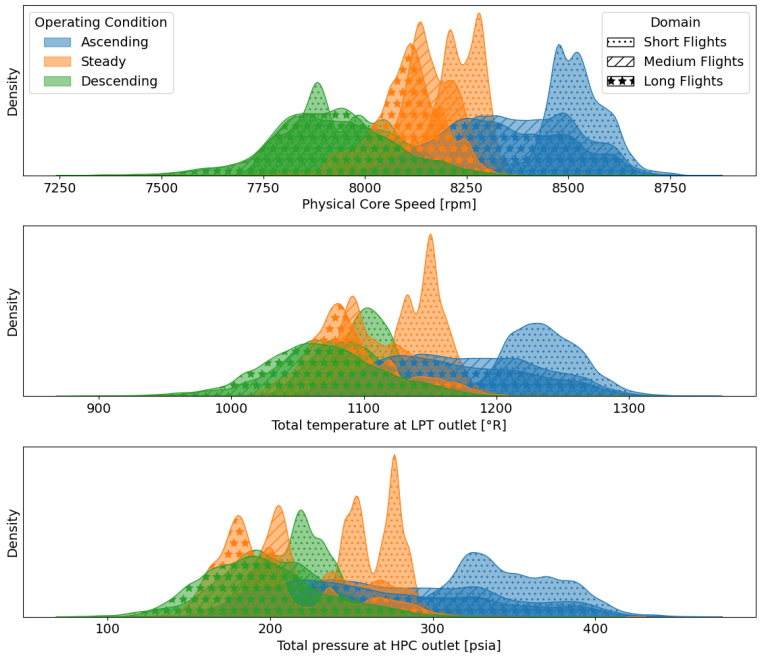


Figure 2.8: Kernel density estimate for marginal distributions of each operation profile for the three domains. Three exemplary sensor values were selected: Physical Core speed, Total Temperature at the LPT outlet, and total pressure at the HPC outlet.

detection is given. Therefore, the RUL prediction task aims to predict the remaining cycles after a fault has been initiated and detected.

Sensor measurements and scenario descriptors are taken as inputs to ensure a realistic usage scenario, resulting in 18 sensor values. Other quantities, such as virtual sensors and model health parameters, which require internal parameters of a simulator, are not considered in this research.

The dataset was downsampled by a factor of ten using a Chebyshev filter with order eight, which applies an anti-aliasing filter before the down-sampling process. This ensured the same 0.1 Hz sampling frequency that was used in (Chao et al., 2022).

Each signal measurement was scaled to a range $[-1, 1]$ using min-max normalization. The scaler was fitted on the source domain data for each adaptation task and subsequently applied to the target domain data.

The RUL was normalized for each engine unit to decrease from one to zero **between the point in time when the fault occurs and the end of life**. To that end, all samples belonging to one engine unit were divided by the maximum number of cycles of the respective unit.

2.4.3 Model Architecture

In this research, we compare the proposed methodology to other commonly used DA approaches. All of the DA methods use the same basic architecture to ensure a fair comparison. In particular, we used a one-dimensional-convolutional neural network (1DCNN) with the architecture inspired by that proposed in (Chao et al., 2022). The feature extractor consists of three 1D-convolutional layers (L=3). The first two layers have ten channels each, while the third layer condenses the feature representation into a single channel. Filters of size ten are used along with a stride of one, and zero-padding is added to ensure that inputs keep the same length when passing through the network. The network uses *ReLU* as the activation function. The feature extractor has a total of 3k trainable model parameters.

The RUL-regressor contains two fully connected layers (L=2). The first one takes the flattened output from the feature extractor and passes it through a fully-connected layer with 50 neurons. Then, after another *ReLU* activation function, the last layer predicts a single output value, the RUL, which is subsequently normalized to a range between 0 and 1 using a *Sigmoid* activation function. In total, there are approximately 3k parameters in the RUL-regressor.

The domain discriminator has a similar architecture as the RUL regressor but contains an additional layer. A first fully-connected layer with 50 neurons is followed by a second one with 30 neurons. The last layer is also fully-connected and ends with a single output neuron. Again, a *Sigmoid* activation function is applied to the final output, and *ReLU* activation functions are used in between fully connected layers. The number of layers and neurons is loosely inspired by an earlier work, which also used a DANN architecture on a similar dataset (da Costa et al., 2020). The resulting domain discriminator has roughly 4k parameters.

OPS-DANN (hard): This model utilizes three domain discriminators, as there are three different phases for the operation profile in the dataset used in this work. The architecture of each discriminator is identical to the one described above, leading to a total number of 18k parameters.

OPS-DANN (soft): Similarly to the OCS-DANN (hard) version, this model utilizes three distinct domain discriminators. Additionally, it uses an operating phase classifier, which is added after the feature extractor. This classifier uses the same architecture as the domain discriminator, except for the last activation function. A $Softmax$ activation function is used in the model instead of a $Sigmoid$ activation function, and this is because there are multiple classes that the model needs to predict and not just two classes since $Softmax$ activation is designed to handle multi-class classification. OPS-DANN (soft) has the largest number of trainable model parameters, with a total of 22k.

Multi-Class OPS-DANN: We propose to explore the impact of the domain alignment of the operation phases in an ablation analysis. Compared to the architecture introduced above, this model only requires changes in the architecture of the domain discriminator of DANN. The domain discriminator predicts the domain of each sample and its operating phases. As there are three operating conditions, the discriminator requires six output neurons, one for each domain and operating phase pair. Consequently, the $Sigmoid$ function must be exchanged by the $Softmax$ activation to handle the multi-class output. This model has slightly more parameters than the original DANN, with roughly 10k parameters.

2.4.4 Comparison Methods

To fairly evaluate the performance of our proposed methods, we first establish a baseline using a feature extractor with a regressor that is trained only on the source data. We then compare the results of our Operation Profile-specific Domain Adaptation Network (OPS-DANN) and its variants to established domain adaptation techniques such as AdaBN, MK-MMD, and DANN. These methods have been previously applied to similar tasks in the field of prognostics and health management and have been shown to have strong performance, as reported in previous research studies such as (Q. Wang et al., 2019b).
MK-MMD: This DA method is a domain-invariant feature learning technique. However, instead of using adversarial training, it aims to minimize a divergence measure between the two domains. To that end, the feature representation f, found by a feature extractor, is used to compute the MMD measure as shown in Equation 2.9. Multiplications of the feature transformation $\phi(\cdot)$ can be readily computed by taking advantage of the kernel trick as demonstrated in Equation 2.10. For the multiple kernel version of MMD, K Gaussian kernels with different bandwidth parameters γ are combined to enhance model performance by adding them together, as shown in Equation 2.11.

$$MMD(F_S, F_T) = \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(f_S^i) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(f_T^j) \right\|^2 \tag{2.9}$$

$$k(f_S^i, f_T^j) = \langle \phi(f_S^i), \phi(f_T^j) \rangle \tag{2.10}$$

$$k(f_S^i, f_T^j) = \sum_{k=1}^{K} k_k(f_S^i, f_T^j) \tag{2.11}$$

For this work, five kernels were selected, similar to previous applications as reported in (Cheng et al., 2021). The bandwidths parameter was set to 0.01, 0.1, 1, 10, and 100.

Table 2.2: Range of the hyperparameter search for all considered DA-methods.The optimal hyperparameters found on the $S \rightarrow L$ adaptation task are emphasized.

Model Name	Learning Rate	Momentum	Tradeoff Alignment Loss	Others
Baseline	{**0.01**, 0.05, 0.25}	{0.5, **0.9**}	-	-
AdaBN	{0.01, 0.05, **0.25**}	{0.5, **0.9**}	-	-
MK-MMD	{0.01, **0.05**, 0.25}	{**0.5**, 0.9}	{0.003, **0.01**, 0.03}	$\{2, 5, \mathbf{10}\}^1$
DANN	{0.01, **0.05**, 0.25}	{**0.5**, 0.9}	{0.1, **0.3**, 1}	-
OPS-DANN Hard	{**0.05**}	{**0.5**}	{0.03, 0.1, 0.3, **1**, 3}	-
OPS-DANN Soft	{**0.05**}	{**0.5**}	{0.1, **1**, 3}	$\{0.03, \mathbf{0.1}, 0.3\}^2$
Multi-class OPS-DANN	{**0.05**}	{**0.5**}	{0.1, **0.3**, 1}	-

AdaBN: Unlike the other baseline methods introduced above, AdaBN (Y. Li et al., 2016a) is not a domain-invariant feature learning technique. Instead, it aims to replace the normalization statistics of batch-norm layers computed on the source domain with those of the target domain. Notably, the target domain data is only used to update the normalization statistics, while all the other model parameters are trained using source data only. A necessary condition to utilize AdaBN is an architecture with batch-norm layers; consequently, the feature extractor's standard architecture had to be adapted. A 1D-batch-norm layer was placed following each of the three convolutional layers.

In summary, except for AdaBN, which uses additional batch-norm layers, all other baseline methods use the same architecture for the feature extractor and the RUL regressor to ensure a fair comparison.

2.4.5 Training Procedure

Before training, all models were initialized using Xavier normal initialization (Glorot & Bengio, 2010). All the data from each domain are considered during training for each adaptation task, and no test-training split is used.

In this research, several consecutive measurement points are combined into one input sample as input to the model using sequences of length 50 from the multivariate time series with a step size of one.

Model updates were performed using batch gradient descent with batches of size 256. One batch of source and target domain data was processed for each training step.

All DA methods were trained for the same number of epochs for each of the adaptation tasks to ensure a fair comparison. For the $S \rightarrow L$ and $M \rightarrow L$ adaptation tasks, the models were trained for 15 epochs. However, in the $S \rightarrow M$ adaptation task, the models were trained for 25 epochs due to the lower number of model updates per epoch.

The baseline was trained for 40 epochs in case the source domain was short flights S, and for 20 epochs in case the source domain was medium flights M.

All DA methods based on DANN additionally required the definition of the reverse gradient factor ρ. For this work, the same update rule for ρ is used as the one proposed in the original DANN (Ganin et al., 2016), which gradually updates ρ from 0 to 1 according to Equation 2.12.

$$\rho = \frac{2}{1 + e^{-10\, l_p}} - 1 \tag{2.12}$$

The learning rate is reduced after each epoch to ensure a smooth convergence using a learning rate schedule, likewise adopted from the original DANN paper (Ganin et al., 2016). The proposed schedule is shown in Equation :

$$\alpha = \frac{\alpha_0}{(1 + 10\, l_p)^{0.75}} \tag{2.13}$$

where l_p once again describes the linear training progress. Unlike the gradient reversal factor, which only applies to DANN-based models, the learning rate schedule is used for all models. The initial learning rate α_0 is found using a hyperparameter search.

The hyperparameters and training specifications mentioned above were primarily selected based on prior work and applied to all models without further refinement. Other hyperparameters were explicitly tuned for each domain adaptation method, such as the learning rate, momentum, and the trade-off between multiple loss objectives. Table 2.2 summarizes the performed grid searches for each method and indicates the optimal hyperparameters found for each method. It is important to emphasize that for all methods, the grid search was only performed on the $S \rightarrow L$ adaptation task. The resulting optimal hyperparameters were then used for the other two adaptation tasks. The $S \rightarrow L$ task was selected to perform the grid search because it has the largest domain gap and is considered as the most challenging task. For AdaBN, the learning rate and momentum found on the $S \rightarrow L$ adaptation task were likely too big for the $M \rightarrow L$ adaptation task and did not lead to convergence. Only in this case, the second best set of hyperparameters found on the $S \rightarrow L$ was used for the $M \rightarrow L$ task, which is also indicated in Table 2.2.

2.4.6 Evaluation Metrics

In this work, all experiments were evaluated using the two common evaluation metrics used for RUL prediction (Saxena et al., 2008): the root mean square error (RMSE) and the NASA scoring function.

The RMSE is defined as follows:

$$RMSE = \sqrt{\frac{1}{n_T}\sum_{i=1}^{n_T}(\hat{y}^i - y^i)^2} \tag{2.14}$$

NASA's scoring metric is not symmetric and penalizes over-estimation more than under-estimation and is defined as follows:

$$s = \sum_{i=1}^{n_T} \exp\left(\alpha \left|\hat{y}^i - y^i\right|\right) \quad , \quad \alpha = \begin{cases} \frac{1}{10}, & \text{if } \hat{y} - y \geq 0, \\ \frac{1}{13}, & \text{otherwise.} \end{cases} \tag{2.15}$$

2.4.7 Representation Transferability

The *Proxy A-distance (PAD)* proposed in (Ben-David et al., 2010) is a suitable metric to measure the divergence for domain adaptation tasks. Thus, we can use it to evaluate the transferability of representations Using the PAD, the divergence between two domains is computed by evaluating how well a classifier can separate the source from the target domain. If separation is easy, the two domains are likely dissimilar, and their discrepancy is large. On the other hand, if samples can hardly be discriminated between the two domains, they will likely have a small discrepancy. Using the classification error ϵ, the PAD $\hat{d}_{\mathcal{A}}$ can be found according to Equation 2.16.

$$\hat{d}_{\mathcal{A}} = 2\,(1 - 2\epsilon) \tag{2.16}$$

In general, all DA methods that contain a domain-invariant feature learning part aim somewhat to minimize the divergence between the source and the target domain. For example, DANN uses the PAD as a discrepancy measure and aims to minimize it during training in the alignment component (Ganin et al., 2016).

2.5 Results

In this section, we compare the prediction accuracy of the considered models in terms of the RMSE and the NASA score. The results are shown in Table 2.3. Additionally, the upper bound RMSE for the medium-haul and long-haul flight domains are 3.16 and 2.17, respectively.

In addition, to evaluating the performance, we also evaluate the PAD (qualitatively measuring the divergence between the domains), calculated from the learned feature

Table 2.3: The results for the three considered Domain Adaptation tasks.

Name	**S → L**		**S → M**		**M → L**	
	RMSE Mean ± Std	s-score Mean ± Std	RMSE Mean ± Std	s-score Mean ± Std	RMSE Mean ± Std	s-score Mean ± Std
Baseline	7.34 ± 2.12	1.74 ± 0.24	4.59 ± 0.51	1.38 ± 0.07	2.63 ± 0.19	1.24 ± 0.03
AdaBN	7.13 ± 0.72	1.82 ± 0.18	5.03 ± 0.66	1.49 ± 0.09	3.23 ± 0.26	1.26 ± 0.02
MK-MMD	4.76 ± 0.57	1.47 ± 0.04	4.40 ± 0.42	1.36 ± 0.04	2.64 ± 0.20	1.22 ± 0.04
DANN	4.38 ± 0.56	1.43 ± 0.11	4.07 ± 0.23	1.35 ± 0.04	2.75 ± 0.35	**1.20 ± 0.01**
OPS-DANN Hard	**3.90 ± 0.56**	**1.39 ± 0.10**	**3.71 ± 0.20**	**1.31 ± 0.04**	2.66 ± 0.23	**1.20 ± 0.00**
OPS-DANN Soft	4.00 ± 0.33	1.40 ± 0.11	3.86 ± 0.18	1.33 ± 0.03	2.72 ± 0.20	1.21 ± 0.02
Multi-Class OPS-DANN	3.97 ± 0.65	1.40 ± 0.11	3.93 ± 0.46	1.33 ± 0.03	**2.58 ± 0.20**	**1.20 ± 0.01**

embeddings. As described in the section 2.4.7, the PAD is used to evaluate the ability of the applied DA methods to extract domain-invariant features. In Figure 2.9, the A-distance varies from one task to another, showing a larger domain gap for the task $S \rightarrow L$, a smaller one for the task $S \rightarrow M$ and almost no domain gap for the task $M \rightarrow L$

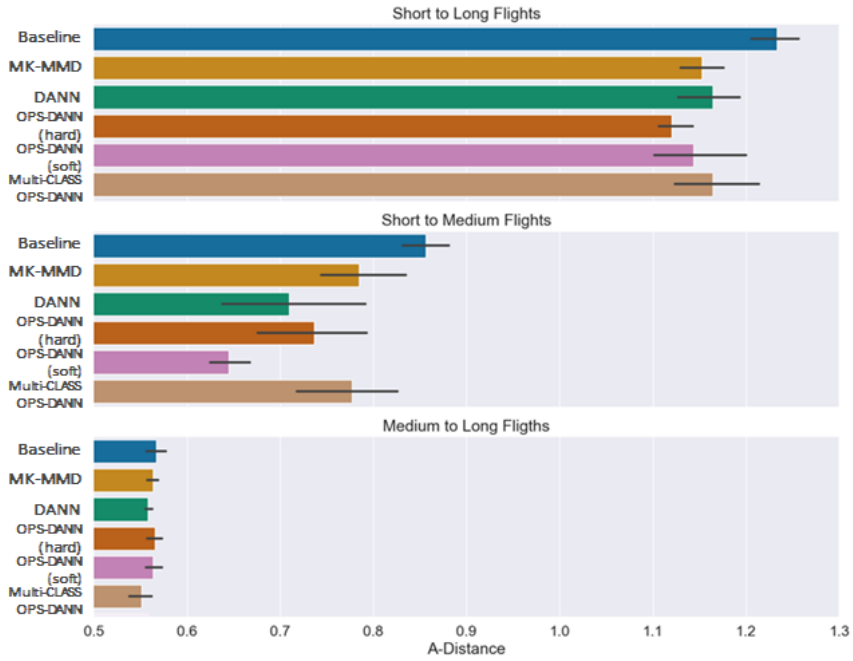


Figure 2.9: A-distance for each method and task respectively

$S \rightarrow L$ task

Performance evaluation: The adaptation task from the short to the long flight domain is the most challenging because it has the largest domain gap. The baseline model trained solely on source data for this task (without any adaptation) exhibits an RMSE value that is 240% larger than the one of the models trained on the target domain. As seen in Table 2.3, the MK-MMD and DANN methods improved the baseline methods in terms of RMSE

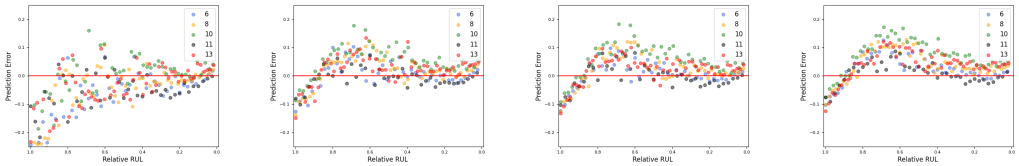
by 35 and 40 %, respectively. The improvements by DA methods are also visible in the NASA score. AdaBN is a simple method without any additional parameters. It improves the model performance with respect to RMSE only slightly. However, it leads to a higher NASA score compared to the baseline due to the overestimation of the RUL predictions. For the $S \rightarrow L$ task, all three proposed OPS-DANN variants improve substantially upon the baseline by at least 45% in terms of RMSE and by 20% in terms of S-score.

It is similarly important to compare the performance of a novel DA method to other state-of-the-art methods. Even though DANN has the lowest RMSE out of the three considered baseline methods, the proposed OPS-DANN methods outperform DANN by at least 9% in RMSE and 2% in s-score. Thus, all models benefit from unlabeled target domain data, as expected in this setup. However, comparisons with the upper bound RMSE of 2.17 show that by using labeled short-haul flight data for adaptation, there is still a considerable performance gap to models trained with the labeled long-haul flight.

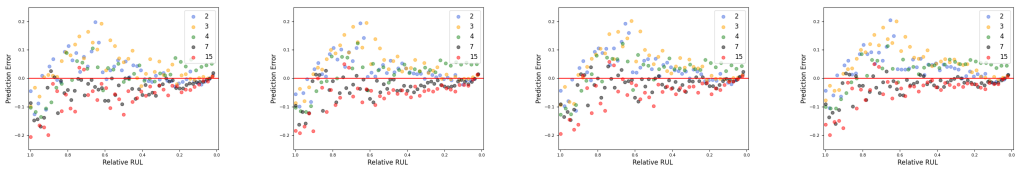
Evaluation of the alignment: Figure 2.9, shows that all three proposed OPS-DANN methods can extract features that contain less discriminative information about the two domains than the baseline on the $S \rightarrow M$ DA task. The two methods using individual domain discriminators (OPS-DANN soft and hard) reach the lowest PAD values for this task, indicating that the usage of a dedicated domain discriminator per operating phase can support the extraction of features that are more domain-invariant compared to domain adaptation techniques aligning the entire marginal distributions of the two domains.

$S \rightarrow M$ task

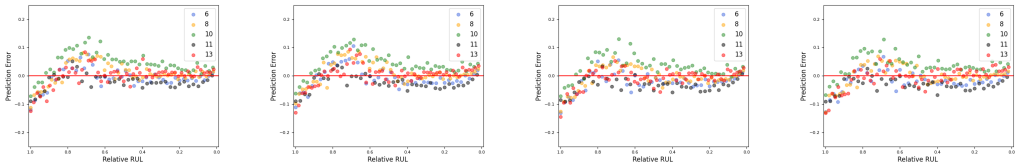
Performance evaluation: Compared to the $S \rightarrow L$ task, the $S \rightarrow M$ task has a smaller domain gap. The baseline model trained without any adaptation solely based on source data for this task results in an RMSE that is 45% larger than the one from the model trained on the target domain. Due to the smaller domain gap in the $S \rightarrow M$ task compared to the $S \rightarrow L$ task, the source domain is more similar to the target domain and, therefore, it allows learning models with a similar performance level as models trained on the target data. As seen in Table 2.3, the MK-MMD and DANN methods are able to improve the baseline methods slightly in terms of RMSE by 4 and 11%, respectively. However, in terms of the s-score, the results decreased slightly from 1.38 to 1.36 and 1.35, respectively. AdaBN approach performed worse than the baseline in this case. Also, in this task, DANN is the best-performing comparison model. Nevertheless, all three proposed OPS-DANN methods outperform the original DANN. While for the Multi-Class OPS-DANN, the improvement is 3% compared to the DANN performance, the OPS-DANN (soft) and OPS-DANN (hard) reduce the RMSE by 5% and 9%, respectively, and the s-score by 1% and 3%



(a) $S \rightarrow L$ DANN (b) $S \rightarrow L$ OPS-DANN Hard (c) $S \rightarrow L$ OPS-DANN Soft (d) $S \rightarrow L$ Multi-Class OPS



(e) $S \rightarrow M$ DANN (f) $S \rightarrow M$ OPS-DANN Hard (g) $S \rightarrow M$ OPS-DANN Soft (h) $S \rightarrow M$ Multi-Class OPS



(i) $M \rightarrow L$ DANN (j) $M \rightarrow L$ OPS-DANN Hard (k) $M \rightarrow L$ OPS-DANN Soft (l) $M \rightarrow L$ Multi-Class OPS

Figure 2.10: Prediction error of the different models as a function of the relative lifetime.

Evaluation of the alignment: On the $S \rightarrow M$ task, all methods reach considerably lower PAD values, even the lower bound, without any adaptation. This makes it more challenging to differentiate between these two domains. However, it is worth noting that all the proposed methods in the Operation Profile-specific Domain Adaptation Network (OPS-DANN) have considerably lower PAD values compared to the baseline. However, only the OPS-DANN (soft) method can extract more domain-invariant features than DANN, while the other two proposed methods have slightly higher PAD values. Even though OPS-DANN (hard) has a higher PAD than both DANN and OPS-DANN (soft), it reaches a lower RMSE demonstrating that the sole ability to extract domain-invariant features does not guarantee a superior performance on the regression task.

$M \rightarrow L$ task

Performance evaluation: The third considered task uses medium-haul flights as the source domain and aims to adapt a model to perform well on long-haul flights. Compared to the two previous tasks, the discrepancy between these two domains is much smaller. This observation is also confirmed by the performance of the baseline method that has

a value RMSE that is only 21% larger than the one of a model trained on the target domain. All domain adaptation methods performed slightly worse than the baseline model. Using the MK-MMD and DANN methods appears to result in under-estimated predictions, thus achieving a lower s-score than the baseline. Out of the three proposed OPS-DANN models, only one of them (Multi-Class OPS-DANN) has a slightly lower RMSE value compared to the baseline (2.58 compared to 2.63). The other two approaches are at a similar performance level as the baseline model, however, achieving a lower s-score (1.20 compared to 1.24). The observed slight differences between the models are likely insignificant and are potentially caused by the randomness of the training procedure. Overall, none of the comparison methods DA methods could improve this task's target-free baseline significantly. None of the methods are able to reach the upper bound RMSE value of 2.17, which is achieved when the model is trained on the labeled target dataset.

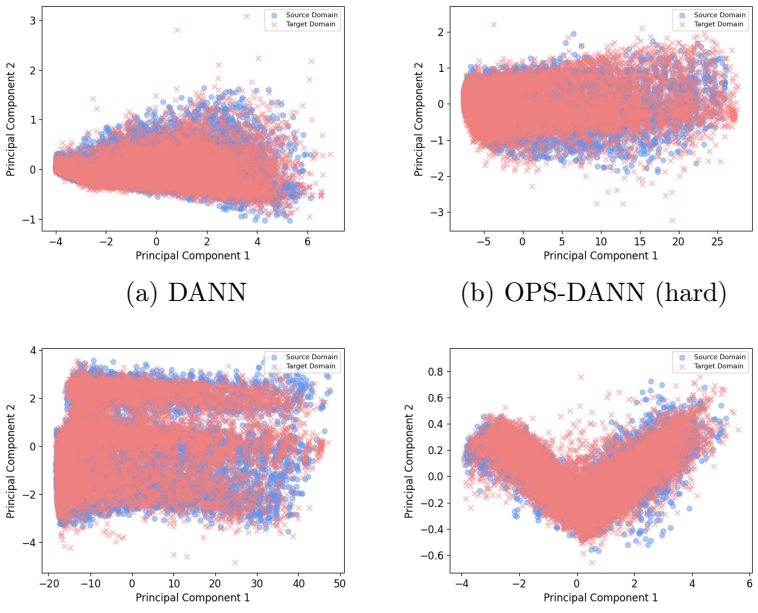
Visualizations: We provide additional visualizations, in Figure 2.10, by showing the test RUL evolution over time for the three adaptation approaches ($S \rightarrow L$, $S \rightarrow M$, $M \rightarrow L$) using DANN and the proposed three methods. The figures confirm our initial expectations. As the units approach the end of their lifetime, the predictions tend to converge with the ground truth RUL value.

Evaluation of the alignment: Comparing the PAD values of all applied methods in Figure 2.9, it becomes apparent that none of the methods are able to reduce substantially the domain distinctiveness compared to the models trained without adaptation.

2.6 Discussion

The findings of this study demonstrate that, on two out of the three tasks, the proposed three variants of the operation phase alignment methods perform similarly well and are able to achieve better results compared to both the DANN model as well as the other domain adaptation methods. This demonstrates the effectiveness of the proposed methods in adapting to different domains, particularly in addressing the unique challenges posed by multimodal feature distributions and the variability of operating conditions and system configurations. Furthermore, it should be noted that DANN is already an effective method for domain adaptation. However, the proposed operation phase alignment methods are able to improve upon the performance of DANN and other traditional domain adaptation methods in the context of PHM.

In addition to evaluating the performance of the proposed methods for RUL prediction, we also investigate the impact of each method on the embeddings generated by the alignment process. To enable 2D visualizations, we applied a principal component analysis (PCA) and present the first two principal components in Figures 2.11 and 2.12. This visualization provides insights into how the different phases of the operation profile are represented in the embeddings and how the proposed methods affect this representation.



(a) DANN (b) OPS-DANN (hard)

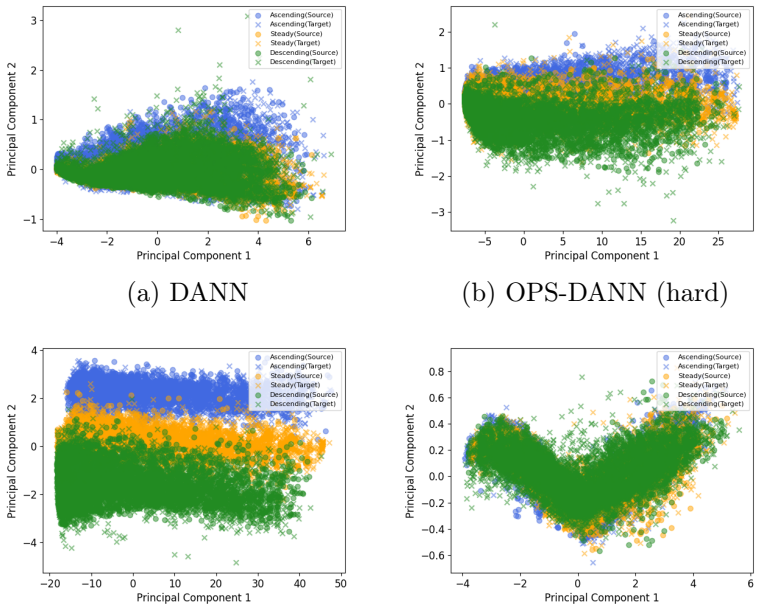
(c) OPS-DANN (soft) (d) Multi-class OPS-DANN

Figure 2.11: Visualizing the Impact of the Operation Profile Alignment on **Domain** alignment: A Comparison of DANN, OPS-DANN (soft), OPS-DANN (soft), and Multi-Class OPS-DANN Models using PCA on the Embedding space

As shown in Figure 2.11 DANN and our proposed approaches are able to effectively align the source and target embeddings, as demonstrated by the overlapping of the source and target data points in the figures. However, it is worth noting that there is a distinct difference between the methods with respect to how distinguishable the operation phases are in the embedding space.

In Figures 2.11 and 2.12, the OPS-DANN soft and hard approaches have phases that can be distinguished but that are more pronounced for the soft approach. On the contrary, the Multi-Class OPS-DANN approach has only one cluster that is similar to the one obtained with the DANN approach which is not able to distinguish between the operating phases. However, the difference between DANN and Multi-Class OPS-DANN is that they have different shapes in the embedding space. Moreover, DANN aims to learn domain invariant features, and Multi-Class OPS-DANN aims to learn domain and operating-condition invariant features. This difference in the ability to distinguish between the operation phases highlights the unique characteristics of each proposed approach and their ability to align the operating characteristics and phases of the different domains.

It should be noted that the Multi-Class OPS-DANN uses the operation phase labels in a distinctly different way than the other two proposed methods. Instead of aligning the marginal distributions of all three operation phases separately, it aims to extract



(a) DANN

(b) OPS-DANN (hard)

(c) OPS-DANN (soft)

(d) Multi-class OPS-DANN

Figure 2.12: Visualizing the Impact of the Operation Profile Alignment on the **Phase** alignment: A Comparison of DANN, OPS-DANN (soft), OPS-DANN (soft), and Multi-Class OPS-DANN Models using PCA on the Embedding space

features invariant to both the domain and the operation phase of the input data simultaneously. Figure 2.12 shows that Multi-Class OCS-DANN successfully learns such a feature representation that is invariant to the operation phases.

Contrary to the Multi-Class OPS-DANN, the OCS-DANN (soft) models contain an additional classifier that aims to learn how to distinguish between different operating phases while simultaneously learning to extract features that are domain invariant. Figure 2.12 illustrates this behavior: each operating phase forms a separate cluster while the source and target domains overlap to a large extent. This behavior is similar to the OCS-DANN (hard).

This observation highlights the effectiveness of the proposed approach in aligning the operating characteristics and conditions of the different domains, leading to improved performance for RUL prediction.

2.7 Conclusion

In this paper, we propose a novel approach that utilizes domain adaptation techniques to align the operating phases to improve the accuracy of RUL predictions.

The main novelty of our proposed approach is the integration of the information on the different phases of the operation profile into the alignment process. The proposed approaches align the marginal distributions of each phase of the operation profile in the labeled source domain with its counterpart in the unlabeled target domain. Two novel domain adaptation approaches are proposed based on an adversarial domain adaptation by considering the different phases of the operation profile separately.

The proposed methods have shown to be effective in improving the performance of deep learning models for RUL prediction by effectively transferring the models between sub-fleets that are operated under different conditions. The results of this study demonstrate the potential of these methods to improve the accuracy and reliability of prognostics and health management in real-world applications. Furthermore, the proposed methods have a better performance compared to state-of-the-art domain adaptation methods such as DANN, MK-MMD, and AdaBN.

There are three interesting potential future research directions resulting from this research. First, the proposed methods can be extended to enable them to tackle challenges arising from imbalanced operating conditions, which is a very common case in practical applications. Second, it would be interesting to investigate whether learning a soft assignment before performing adaptation may be beneficial by training an operating phase classifier separately from the DA model. Thirdly, while this research focused on regression, the proposed methodology can be easily extended to classification tasks such as for example for fault diagnostics problems. We leave the aforementioned open research directions for future work.

Acknowledgments: This work was supported by the Swiss National Science Foundation under Grant PP00P2_176878.

3 DARE-GRAM: Unsupervised Domain Adaptation Regression by Aligning Inverse Gram Matrices

This chapter corresponds to the published article[I]:
Nejjar, Ismail, Qin Wang, and Olga Fink. "DARE-GRAM: Unsupervised domain adaptation regression by aligning inverse gram matrices." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023.

Abstract: Unsupervised Domain Adaptation Regression (DAR) aims to bridge the domain gap between a labeled source dataset and an unlabelled target dataset for regression problems. Recent works mostly focus on learning a deep feature encoder by minimizing the discrepancy between source and target features. In this work, we present a different perspective for the DAR problem by analyzing the closed-form ordinary least square (OLS) solution to the linear regressor in the deep domain adaptation context. Rather than aligning the original feature embedding space, we propose to align the inverse Gram matrix of the features, which is motivated by its presence in the OLS solution and the Gram matrix's ability to capture the feature correlations. Specifically, we propose a simple yet effective DAR method which leverages the pseudo-inverse low-rank property to align the scale and angle in a selected subspace generated by the pseudo-inverse Gram matrix of the two domains. We evaluate our method on three domain adaptation regression benchmarks. Experimental results demonstrate that our method achieves state-of-the-art performance.

[I]Please note, this is the author's version of the manuscript published in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. Changes resulting from the publishing process, namely editing, corrections, final formatting for printed or online publication, and other modifications resulting from quality control procedures may have been subsequently added.

3.1 Introduction

Regression problems, in which models learn to predict continuous variables, are one fundamental paradigm in machine learning. Regression problems are omnipresent in many different applications, including computer vision tasks, such as head-pose estimation (T.-Y. Yang et al., 2019), facial landmark detection (H. Li et al., 2022), human pose estimation (Zheng et al., 2021), depth estimation (Godard et al., 2019) and eye-tracking problems(Shenoy et al., 2021), and also widely in industrial applications, such as product quality prediction and condition monitoring (Q. Wang et al., 2019a). Nevertheless, real-world applications are often subject to the environmental conditions under which the data are collected and other influencing factors, hence domain gaps between datasets are inevitable.

Unsupervised Domain adaptation (UDA) aims to overcome the distributional shift between a labeled source domain and an unlabelled target domain. Many UDA methods have been proposed to alleviate the domain shift problem. One common UDA direction is feature alignment by adversarial learning (Long et al., 2018) or explicit losses such as maximum mean discrepancy (Long et al., 2015) to learn domain-invariant representations. Input alignment (Y. Yang & Soatto, 2020) and self-training using pseudo-label refinement(Lian et al., 2019) are also popular UDA directions. While many DA methods have been developed and evaluated for classification and segmentation problems, some are not directly transferable to DA regression (X. Chen et al., 2021). Pioneer works in Domain Adaptation Regression (DAR) (Cortes & Mohri, 2011b; Mansour et al., 2009a) introduced theoretical analysis for the problem. A few algorithms were proposed to tackle DAR. For example, importance weighting (de Mathelin et al., 2021; Yamada et al., 2014) and feature alignment (B. Cao et al., 2010; S. J. Pan et al., 2010) have shown improved results over learning only from the source.

Most recent unsupervised DAR methods (X. Chen et al., 2021; Singh & Chakraborty, 2020) use the deep learning framework and focus on learning a shared deep feature extractor by directly minimizing the discrepancy between source and target features. By doing so, it is implicitly assumed that if the feature discrepancy is small, a shared linear regressor can be easily learned from the source supervision. This formulation used by existing works focuses solely on the feature extractor.

In this work, we propose to look at the DAR problem from a different perspective. In particular, we pay close attention to the linear regressor, which is attached directly after the feature extractor. Motivated by the closed-form ordinary least squares (OLS) regression solution, we analyze the potential optimal regressor for each domain. We reveal in Section 3.3.2 that even when the discrepancy between source and target features is small, the learning of a shared linear regressor could still be difficult because of the *inverse Gram matrix* term in the OLS solution.

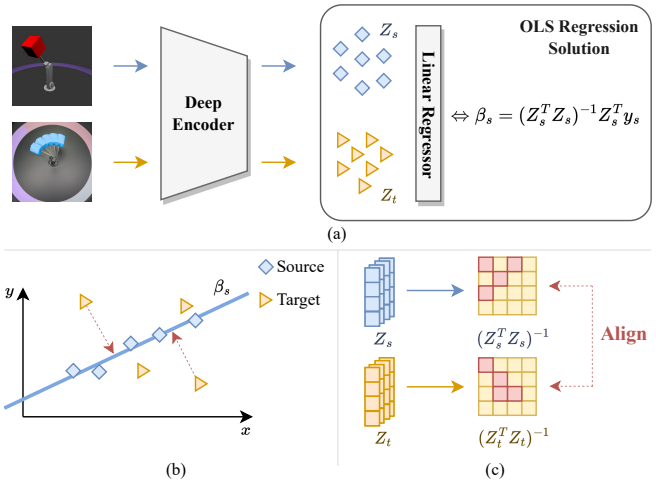


Figure 3.1: Illustration of the UDA for regression setup and our main motivation. (a) Deep domain adaptation networks commonly use a shared deep feature encoder and a shared linear regression layer. We propose to pay close attention to the linear regressor, where the ordinary least square (OLS) solution is well-known. (b) Given a trained source linear regressor β_s, the target features Z_t may not be calibrated to β_s. (c) Unlike previous adaptation methods which align in the original feature embedding space, we propose to align the inverse Gram matrix of the features $(Z^T Z)^{-1}$, which is motivated by its presence in the OLS solution and the Gram matrix's ability to capture the feature correlations.

In light of this, we propose an ordinary least squares inspired deep domain adaptation method for regression called Domain Adaptation Regression by aligning the inverse GRAM matrices (DARE-GRAM). As shown in Figure 3.1, unlike previous methods, which directly align the features, we align the inverse Gram matrix of the features. This is motivated by its presence in the closed-form solution of the ordinary least squares. More specifically, we leverage the low-rank property of the pseudo-inverse to align a selected subspace in scale and angle engendered by the Gram Matrix, which represents the intensity and pairwise interactions between different features for the source and target domains. The scale and angle alignment based on the Gram matrix can lead to a better-calibrated regressor with regard to both source and target data. The contributions of this work are as follows:

- We offer a new perspective to understand the UDA for regression problems by leveraging the well-known closed-form solutions to the linear regression problem.
- Rather than aligning the original feature embedding space, we propose to align the inverse Gram matrix of the features.
- Empirical results on three benchmarks validate the superiority of the DARE-GRAM over baseline methods.

3.2 Related Work

Unsupervised Domain Adaptation. The goal of unsupervised domain adaptation (UDA) (Patel et al., 2015) is to address the domain-shift problem between a labeled source and an unlabeled target domain. UDA has been widely studied for classification and segmentation problems (Tsai et al., 2018; Vu et al., 2019a) to mitigate the gap between features across different domains. Early works addressed this problem via instance weighting (J. Huang et al., 2006; Sugiyama et al., 2007), feature transformation (S. J. Pan et al., 2010), and feature space alignment (Fernando et al., 2013). More recently, unsupervised domain adaptation has shown impressive results (Hoyer et al., 2022; Na et al., 2022; Wilson & Cook, 2020a). Discrepancy minimization (Kang et al., 2019; Long et al., 2015) and domain adversarial learning (Ganin & Lempitsky, 2015; Hoffman et al., 2018) have been widely used within UDA methods to mitigate the gap between features across different domains. Moreover, feature regularization-based approaches (X. Chen et al., 2019) and domain-specific normalization-based methods (M. Chen et al., 2019; Y. Li et al., 2016b) have also demonstrated good performance. While most approaches perform feature alignment in the encoding feature space, some works proposed to carry out alignment in the input space (Y. Yang & Soatto, 2020). More recently, self-training has also demonstrated encouraging results by training the network with gradually improved target pseudo-label (Liang et al., 2022; Q. Wang et al., 2022; P. Zhang et al., 2021; Y. Zhang & Davison, 2021). Existing UDA methods mostly focus on classification and segmentation problems. While some UDA techniques can directly be applied to regression problems, recent works have shown that many do not perform well in the regression setup (Y. Bao et al., 2022a; X. Chen et al., 2021).

Domain Adaptation for Regression. Domain Adaptation for Regression (DAR) has received relatively little attention in comparison to classification problems. Early theoretical properties for DAR were introduced in (Cortes & Mohri, 2011b; Mansour et al., 2009a). Different algorithms were proposed to tackle DAR (Redko et al., 2020). Unfortunately, most algorithms require access to a labeled target domain and are unsuitable for UDA regression. For instance, Boosting strategies have been explored (Pardoe & Stone, 2010; X. Wang et al., 2019) to extend previous classification domain adaptation methods based on AdaBoost (Margineantu & Dietterich, 1997) to regression tasks. Other instance weighting methods in the shallow regime, (de Mathelin et al., 2021; Yamada et al., 2012, 2014) have been explored for a different range of applications. Some specific vision applications have been explored in the context of UDA (J. Jiang et al., 2021; D. Kim et al., 2022; C. Li & Lee, 2021; Ohkawa et al., 2022) , such as monocular depth estimation (Akada et al., 2022; Bhattacharjee et al., 2022; Lo et al., 2022; Tonioni et al., 2019) or gaze estimation (Y. Bao et al., 2022b; Guo et al., 2020). However, these methods aim at improving upon a specific task and not for regression tasks in general. Recent works for UDA regression were proposed (X. Chen et al., 2021; Singh & Chakraborty, 2020; J. Wu et al., 2022a). A key finding in RSD (X. Chen et al., 2021) is that in regression problems, deep neural networks

are less robust to feature scaling than classification, and aligning the distributions of deep representations will alter feature scale and impede domain adaptation regression. To tackle this challenge, the authors of (X. Chen et al., 2021) proposed to match the orthogonal bases of both domains to close domain shifts without altering their feature scale by introducing a new geometrical distance. While RSD-based methods have shown improved results for DAR, matching only the eigenvectors can have some disadvantages, such as more loose numerical error bound (Anderson et al., 1999) and may not satisfy the more strict conditions for distribution estimation (Knowles & Yin, 2013). In contrast to RSD, we propose using the inverse Gram Matrix, which carries the necessary information to align the source and target features while being less sensitive to the batch size.

Gram Matrix and Subspace Alignment. Distribution alignment approaches have been used for domain adaptation (Fernando et al., 2013; B. Sun & Saenko, 2016; Wei et al., 2021). Subspace-based domain adaptation has demonstrated good performance in visual domain adaptation (B. Gong et al., 2012; Gopalan et al., 2011), modeling distribution change by finding the best intermediate subspaces. The methods first independently compute a domain-specific d-dimensional subspace for the source and target data. Then project the source and target data into intermediate ones along the shortest geodesic path connecting the two d-dimensional subspaces on the Grassmann manifold. Instead of computing a large number of intermediate subspaces, the authors of (Fernando et al., 2013) directly aligns the two subspaces. Furthermore, the authors in (B. Sun & Saenko, 2015) proposed to incorporate distribution alignment into subspace adaptation to align the source and target features. Given their close relation, distribution alignment approaches have also been used for Neural Style Transfer (NST) (Bousmalis et al., 2017; Kalischek et al., 2021). Early works on NST (Gatys et al., 2016) introduced the Gram Matrix as the statistics of feature maps to extract style-specific attributes. Although the connection between aligning distributions and NST may not be straightforward, it was demonstrated in (Y. Li et al., 2017) that the style loss in (Gatys et al., 2016) may be expressed as an unbiased empirical estimate of the Maximum Mean Discrepancy (MMD)(Gretton et al., 2012) with a quadratic kernel. Unlike previous works in neural style transfer which directly aligns the Gram matrix, we propose to align the inverse Gram matrix as it is presented in the OLS solution. We will show in our method and our ablation study that this is critical for regression problems.

3.3 Methods

3.3.1 Problem Definition

In UDA, we are given labeled samples $\chi_s = \{(x_s^i, y_s^i)\}_{i=1}^{N_s}$ from the source domain and unlabeled samples $\chi_t = \{(x_t^i)\}_{i=1}^{N_t}$ from the target domain, where N_s and N_t denote the number of samples in χ_s and χ_t. In contrast to the discrete labels $\mathcal{Y}$ in classification problems, this work focuses on the regression problem where $\mathcal{Y} \subset \mathbb{R}^{N_r}$ is multidimensional

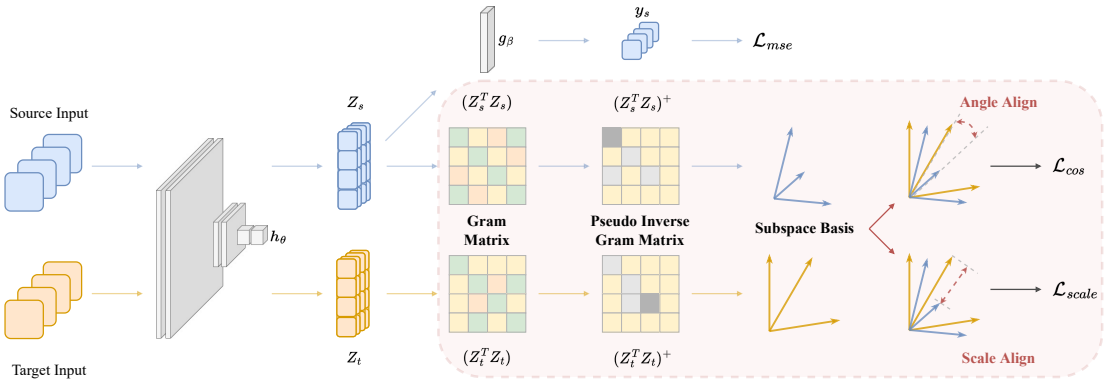


Figure 3.2: An overview of the proposed DARE-GRAM approach for domain adaptive regression problems. Rather than aligning the features Z, we align the inverse Gram matrix, which is motivated by the ordinary least square solution. To achieve this, we compute the pseudo-inverse Gram Matrices for source and target features and align their angle and scale.

and continuous, and N_r correspond to the number of regression tasks. The discrepancy between $P(\chi_s)$ and $P(\chi_t)$ is one of the main challenges for UDA. We aim to learn a model $F : x \mapsto y$, which can generalize well on the target domain. Formally, we want to minimize the expected error on the target data:

$$\arg\min_F \mathbb{E}_{(x^t, y^t)} \|F(x^t), y^t\|_2^2, \tag{3.1}$$

where y^t is not known during the training.

A source-only baseline can be learned by using the supervision from the source data by minimizing the Mean square error loss (MSE) between the prediction and the ground truth label on the source samples:

$$\mathcal{L}_{src} = \frac{1}{N_s} \sum_{i=1}^{N_s} \|\tilde{y}_s^i - y_s^i\|_2^2, \tag{3.2}$$

where $\tilde{y}_s^n = F(x_s^i)$ is the predicted value for the training source image x_s^i. To overcome the distribution gap between the source and target, additional constraints should be given.

3.3.2 Motivation

In deep domain adaptation models, given an input image x, a feature encoder h_θ is used to learn the deep representation $z = h_\theta(x)$ of p dimensions. A linear layer g_β is then applied on z to make the final prediction:

$$\tilde{y} = F(x) = g_\beta(h_\theta(x)) = g_\beta(z). \tag{3.3}$$

During training, the feature matrix is $Z = [z^1, ..., z^b]$ where $Z \in \mathbb{R}^{b \times p}$ for a batch of b images. For many adaptation methods(X. Chen et al., 2021; Singh & Chakraborty, 2020), the focus has been on minimizing the distribution difference between source features Z_s and target features Z_t. Given the aligned features, it is often assumed that they will then lead to a good performance on the target domain. However, this formulation focuses solely on the feature extractor h_θ and does not take the discrimination ability of the final linear layer g_β into account. Target features aligned with the source domain may not be adapted to the linear layer. This can be especially dangerous for regression problems because it has been demonstrated empirically (X. Chen et al., 2021) that in the DA for regression context, the models can be sensitive to feature scale differences.

In this work, we propose to take the linear prediction layer g_β into account for the distribution alignment in domain adaptation regression problems. The proposed research is motivated by the question *How to find a feature space, on which a shared linear regressor can easily learn*?

Fortunately, for the linear regression problem, a closed-form solution exists and is well-studied. Given the feature Z and regression ground truth label Y, the problem of estimating the parameter β for a linear layer $Y = Z\beta$ has

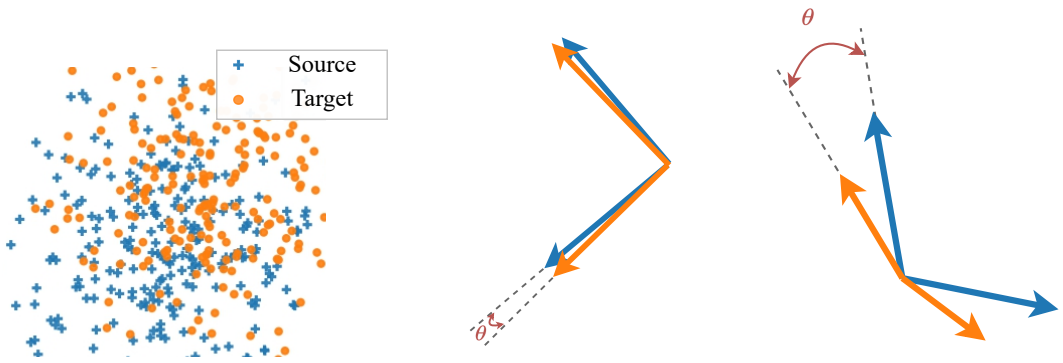
the ordinary least-squared (OLS) closed-form solution (Goldberger et al., 1964):

$$\hat{\beta} = (Z^T Z)^{-1} Z^T Y \tag{3.4}$$

where $(Z^T Z)^{-1} \in \mathbb{R}^{p \times p}$ is the inverse of the Gram Matrix. Entries are then the inner products of the basis functions of the finite-dimensional subspace. $Z^T Y \in \mathbb{R}^{p \times N_r}$ projects features to the label space.

The final linear prediction layer g_β is shared by the source and target domains. Therefore, the estimated value from the two domains should be similar $\hat{\beta}_s \sim \hat{\beta}_t$ where

$$\begin{aligned} \hat{\beta}_s &= (Z_s^T Z_s)^{-1}\ Z_s^T Y_s, \\ \hat{\beta}_t &= (Z_t^T Z_t)^{-1}\ Z_t^T Y_t. \end{aligned} \tag{3.5}$$



(a) Two distributions (b) Z (c) $(Z^T Z)^{-1}$

Figure 3.3: Illustration of the impact the inverse Gram operation in the OLS solution. (a) Assume that the features of the source and target follow two Gaussian distributions with slightly different mean and variance. (b) Under the representation subspace distance, the feature subspaces of Z_s and Z_t are well aligned with a very small angle difference. (c) However, because of the inverse Gram operation, the difference in terms of the inverse Gram matrix $(Z^T Z)^{-1}$ can still be large in terms of both angle and scale.

Most of the previous DAR works regularize the neural network by minimizing the distance between source and target data in the feature representation subspace of Z, i.e. aligning Z_s and Z_t. However, because of the inverse operation in Equation 3.5, even if the distance between Z_s and Z_t is small, the distance in terms of $(Z^T Z)^{-1}$ can be large, as seen in Figure 3.3. This can further lead to distinct $\hat{\beta}_s$ and $\hat{\beta}_t$ and makes it potentially infeasible to learn a common regressor that performs well for both domains. Given this observation and motivated by the closed-form OLS solution, we propose to focus on the subspace of inverse Gram matrix $(Z^T Z)^{-1}$ for the alignment. More specifically, we propose to align the angle between the source and target pseudo-inverse Gram Matrix, formed by a subset of the eigenspace. In addition, we propose to ensure the same scale of Z for the source and target reflected by the Gram matrix $(Z^T Z)$ by minimizing the distance between selected eigenvalues of both domains.

3.3.3 Angle Alignment for Gram Matrix Inverse

The first term in Equation 3.5 concerns the Gram matrix. The Gram matrix is sometimes regarded as a style representation as it calculates the correlations between the different features. It can also be seen as an unbiased empirical estimate of the MMD with a quadratic kernel (Gatys et al., 2016). The inverse operation is also essential because it first relates the variance of the unbiased estimator β to the eigenvalues of $(Z^T Z)$. Particular attention must be paid to the small eigenvalues, which have a maximum inflationary effect on the variance of the least squares estimator by significantly destabilizing the estimator when it approaches zero. Secondly, the ill-conditioned Gram matrix motivates using a low-rank inverse approximation (Chung et al., 2015), which allows obtaining regularised basis to be aligned for the source and target domain.

However, such an alignment is non-trivial because the Gram matrix can be non-invertible in deep learning models. During training, the batch size b is generally smaller than the embedding dimension p. Given a feature matrix $Z \in \mathbb{R}^{b \times p}$, with $b < p$, the Gram Matrix $(Z^T Z) \in \mathbb{R}^{p \times p}$, has rank r smaller or equal to b. Hence the Gram Matrix is not fully ranked and thus not invertible. The Moore-Penrose pseudo-inverse in this case can generalize the concept of matrix inverse when the matrix may not be invertible.

We propose to consider only a selected subspace of the Gram matrix to solve this problem. As not all basis vectors contribute equally, the basis vectors with the highest eigenvalues are the most influential. Therefore, we only consider the most dominant basis vectors in the alignment process.

This step has two main objectives : (i) maximize the mutual information between the two distributions by considering only a selected subset, (ii) avoid numerical instability when not considering degenerate eigenspace. Concretely, given the singular value decomposition (SVD) (Van Loan & Golub, 1996) of the feature matrix Z defined by $Z = UDV^T$. The Gram matrix $(Z^T Z)$, can be decomposed using the SVD of Z as :

$$\begin{aligned} (Z^T Z) = (UDV^T)^T (UDV^T) = V \Lambda V^T, \\ \lambda_k := \Lambda_{k,k} = D^2_{k,k} \quad \text{for} \quad k = 1, ..., p. \end{aligned} \tag{3.6}$$

where the orthogonal matrix $V \in \mathbb{R}^{p \times p}$ is identical to the matrix in the SVD of Z and $\Lambda \in \mathbb{R}^{p \times p}$ is the diagonal matrix containing the squared eigenvalues of Z.

Given the ordered eigenvalues of the Matrix $(Z^T Z)$ $\lambda_1 \geq ... \geq \lambda_k \geq ... \geq \lambda_p \geq 0$, the Moore-Penrose pseudo-inverse (Penrose, 1955) can be derived by discarding the singular values that are below λ_k and treating them as zero. The pseudo-inverse of $(Z^T Z)$ can be expressed as:

$$(Z^T Z)^+ = V^T \Lambda^+ V = V^T \left(\begin{array}{ccc|c} \frac{1}{\lambda_1} & & & \\ & \ddots & & 0 \\ & & \frac{1}{\lambda_k} & \\ \hline & 0 & & 0 \end{array} \right) V \tag{3.7}$$

The operation is equivalent to removing the dimensions with the largest singular value in the inverse matrix. This is in line with (X. Chen et al., 2019) as it has been shown that penalizing high eigenvalues is beneficial in domain adaptation.

The selection of k (the number of principal components used) can be achieved through

a threshold on the cumulative sum of the eigenvalues of $(Z^T Z)$. Since the smaller eigenvalues do not contribute significantly to the cumulative sum, the corresponding principal components may be continued to be dropped as long as the desired threshold limit is not exceeded. Given λ_s and λ_t respectively the eigenvalues of the matrix $(Z_s^T Z_s)$ and $(Z_t^T Z_t)$, the goal is to find k, s.t.

$$\frac{\sum_{i=0}^{k} \lambda_{s,i}}{\sum_{i=0}^{p} \lambda_{s,i}} > T \quad \text{and} \quad \frac{\sum_{i=0}^{k} \lambda_{t,i}}{\sum_{i=0}^{p} \lambda_{t,i}} > T, \tag{3.8}$$

where T is a threshold controlling the proportion of explained variance by the first k principal components. In the following, the pseudo-inverse with respect to k of the Gram matrix for source and target is denoted as $G_s^+ = (Z_s^T Z_s)^+$ and $G_t^+ = (Z_t^T Z_t)^+$, respectively.

Following (X. Chen et al., 2021), the cosine similarity is used to calculate the angle difference between source and target. Unlike previous methods, the angle calculation directly uses the column space of G_s^+ and G_t^+, forming a subspace of $\mathbb{R}^p$ spanned by the column vectors of G_s^+ and G_t^+. A direct measurement of the principal angles is defined as follows:

$$\cos(\theta_i^{S \leftrightarrow T}) = \frac{G_{s,i}^+ \cdot G_{t,i}^+}{\|G_{s,i}^+\| \cdot \|G_{t,i}^+\|} \tag{3.9}$$

where $i \in [1, p]$, and G_i^+ represent the ith column of the inverse Gram matrix G^+. The cosine similarity between the span of the subspace for both the source and target feature are stored in $M = [\cos(\theta_1^{S \leftrightarrow T}), \ldots, \cos(\theta_p^{S \leftrightarrow T})]$. The loss to align the selected basis from the pseudo-inverse of the Gram matrix can be written as:

$$\mathcal{L}_{cos}(Z^S, Z^T) = \|\mathbf{I} - M\|_1^1 \tag{3.10}$$

with $\mathbf{I}$ a vector of ones, of shape p. Minimizing the above term maximizes the cosine similarity between the source and target representation subspace by reducing the angle between the basis of both domains.

Discussion The proposed method is also more robust and stable compared to the direct feature alignment of Z (*e.g.* RSD (X. Chen et al., 2021)). An important difference between RSD and DARE-GRAM lies on the choice of subspace for the alignment. RSD relies on the U basis derived from the SVD decomposition of Z. However, the vectors U are first, not unique for a matrix with repeated singular values and, secondly, may be numerically unstable since the gradient depends on $\frac{1}{\lambda_i - \lambda_j}$. Morever, A drawback of RSD is that a large batch size $b \geq p$ can result in full space, causing the principal

angles(RSD (X. Chen et al., 2021)-Eq.2) between two subspace to become zero. In this case, no alignment can be performed by RSD. Our method does not have this drawback.

3.3.4 Scale Alignment

Preserving the source feature scale is critical in domain adaptive regression problems (X. Chen et al., 2021). In addition to the angle alignment presented in the previous section, we propose to explicitly align the scale of the target subspace to the source.

More specifically, the scale of the matrix Z can be estimated by its trace norm $\|Z\|_1 = \mathrm{Tr}(\sqrt{Z^T Z}) = \sum_{i=1}^{N} \sqrt{\lambda_i}$, where the last term is the sum of the singular values of Z. The scale of Z is therefore the sum of the diagonal elements of the Gram-Matrix. The scale distance between source and target feature is regularized by minimizing the difference between the k-principal eigenvalues:

$$\mathcal{L}_{scale}(Z^S, Z^T) = \|\lambda_{s,i=1,\dots,k} - \lambda_{t,i=1,\dots,k}\|_2. \tag{3.11}$$

Unlike the previous methods (X. Chen et al., 2021), which explicitly avoid aligning the feature scale, the pseudo-inverse Gram columns that form the basis of our subspace are not necessarily orthonormal. Therefore, matching the source and target basis scale is also essential to complete the alignment process. As a note, the eigenvectors from the SVD decomposition are orthonormal, and the length of the vectors is fixed and set to one, as shown in Figure 3.3(b).

3.3.5 Overview

Combining our angle alignment for the inverse gram and scale alignment, the total loss used for the end-to-end training can be written as:

$$\mathcal{L}_{total}(Z^S, Z^T) = \mathcal{L}_{src} + \alpha_{cos}\mathcal{L}_{cos}(Z^S, Z^T) + \gamma_{scale}\mathcal{L}_{scale}(Z^S, Z^T), \tag{3.12}$$

where $\alpha_{cos}, \gamma_{scale}$ are hyper-parameters controlling the effect of the angle and scale alignment. An overall of our method is presented in Figure 3.2.

3.4 Experiments

3.4.1 Experimental setup

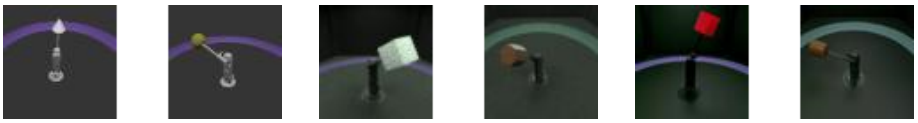
We evaluate our proposed method on three domain adaptations for regression benchmark datasets: dSprites, MPI3D, and Biwi Kinect.

dSprites (Matthey et al., 2017) is a synthetic 2D dataset generated from five ground truth independent latent factors. Following common practice (X. Chen et al., 2021), we treat the three variants of the datasets as three different domains. They are generated by adding Color (**C**) or background noise such as Scream (**S**) and Noise (**N**), shown in Figure 3.4. These three domains comprise 737,280 images each. Dsprites can be used as a benchmark for regression domain adaptation, especially if we consider scale, position X, and Y. Similarly to the setup in (X. Chen et al., 2021), the orientation factor is excluded from consideration. We evaluate all methods on the three sub-regression tasks on six adaptation directions: **C** $\rightarrow$ **N**, **C** $\rightarrow$ **S**, **N** $\rightarrow$ **C**, **N** $\rightarrow$ **S**, **S** $\rightarrow$ **C**, and **S** $\rightarrow$ **N**.

(a) Noise (b) Color (c) Scream

Figure 3.4: Sample example of different domains in dSprites.

MPI3D (Gondal et al., 2019) is a benchmark dataset that consists of 1,036,800 examples of 3D objects from three different domain : Toy (**T**), RealistiC (**RC**) and ReaL (**RL**), as shown in Figure 3.5. This real-world robotics dataset allows the investigation of the domain gap between real data and simulated ones. This dataset was recorded in a controlled environment, defined by seven factors of variation such as object color, shape, size and position, camera height, background color, and two degrees of freedom of motion of a robotic arm The task is to predict these intrinsic factors from the input image. For this paper, we evaluate our method on six transfer tasks: **RL** $\rightarrow$ **RC**, **RL** $\rightarrow$ **T**, **RC** $\rightarrow$ **T**,**RC** $\rightarrow$ **RL**, **T** $\rightarrow$ **RL** and **T** $\rightarrow$ **RC**. We only considered the two regression tasks, rotation about a vertical and horizontal axis.

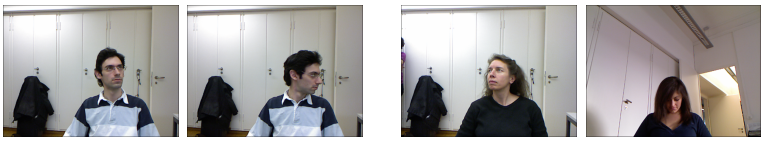


(a) Toy (b) Realistic (c) Real

Figure 3.5: Sample example of different domains in MPI3D.

Biwi kinect (Fanelli et al., 2013) is a real-word dataset containing over 15K images of 20 people, 6 Females (**F**) with 5874 images and 14 Males (**M**) with 9804 images, recorded with a Microsoft Kinect sensor while turning their heads around freely. The example images are shown in Figure 3.6. The three factors of variations used to evaluate our method are yaw, pitch, and roll angles. We evaluate our method on two transfer tasks: **M** $\rightarrow$ **F** and **F** $\rightarrow$ **M**.

Implementation Details. A pre-trained ResNet-18 (K. He et al., 2015) on ImageNet is used as the backbone for all methods. For all the experiments, the different tasks



(a) Male (b) Female

Figure 3.6: Sample example of different domains in Biwi kinect

share the same encoder but a separated single linear regressor with a Sigmoid activation function. The source and target labels were scaled in the range $[0, 1]$ to eliminate the effects of diverse scales in regression values. We use the SGD (Ruder, 2016) optimizer with a momentum of 0.9. The weight decay is set to $1e^{-3}$ for the loss optimization. The newly added layers are trained with a learning rate ten times that of the pre-trained layers, which is initialized to $\eta_0 = 1e^{-2}$. We further adopt the same learning rate scheduler $\eta = \eta_0 \cdot (1 + 0.0001 \cdot p)^{-0.75}$ as (Ganin & Lempitsky, 2015; Long et al., 2018), where p is the number of iterations changing from 0 to the maximum number of iterations. The images are resized to 224×224 and concatenated into batches of size $b = 36$. The number of iterations was set as in (X. Chen et al., 2021) to 20,000, 10,000, and 1,500 iterations for dSprites, MPI3D, and Biwi Kinect, respectively. These setup choices are identical to RSD (X. Chen et al., 2021). An NVIDIA RTX 3090 GPU was used for all the experiments.

Compared Methods We compare our method with a range of adaptation methods: (i) Domain Adaptation via Transfer Component Analysis (**TCA**) (S. J. Pan et al., 2010) (ii) Maximum Classifier Discrepancy (**MCD**) (Saito, Watanabe, et al., 2018) (iii) Joint Distribution Optimal Transportation for Domain Adaptation (**JDOT**) Courty, Flamary, Habrard, and Rakotomamonjy, 2017 (iv) Adaptive Feature Norm (**AFN**) R. Xu et al., 2019 (v) Deep Adaptation Network (**DAN**) (Long et al., 2015) (vi) Deep Adaptation Neural Network (**DANN**) (Ganin et al., 2016) and (vii) Representation Subspace Distance for Domain Adaptation Regression (**RSD**) (X. Chen et al., 2021).

3.4.2 Results

Evaluation metrics. Following previous works (X. Chen et al., 2021; W. Li et al., 2021), Mean Absolute Error (MAE) is used as our evaluation metric across all the regression tasks. Each experiment is repeated three times, and the average results are reported.

Evaluation on dSprites: As shown in Table 3.1, our model achieves the best performance among all competing methods. Specifically, our method outperforms the previous state-of-the-art regression-based method RSD(X. Chen et al., 2021) by 30.8% in terms of average MSE over all directions. On the three difficult adaptation directions $\mathbf{C} \rightarrow \mathbf{N}$, $\mathbf{C} \rightarrow \mathbf{S}$, $\mathbf{N} \rightarrow \mathbf{S}$. **DARE-GRAM** also improves the performance over the RSD. The improvement is especially significant on the direction $\mathbf{C} \rightarrow \mathbf{S}$ and $\mathbf{N} \rightarrow \mathbf{S}$. The improvement is

Table 3.1: Comparisons with previous works on the dSprites regression tasks. All results are shown in sum of MAE with the ResNet-18.

Method	C → N	C → S	N → C	N → S	S → C	S → N	Avg
Resnet-18	0.94	0.90	0.16	0.65	0.08	0.26	0.498
TCA	0.94	0.87	0.19	0.66	0.10	0.23	0.498
MCD	0.81	0.81	0.17	0.65	0.07	0.19	0.450
JDOT	0.86	0.79	0.19	0.64	0.10	0.23	0.468
AFN	1.00	0.96	0.16	0.62	0.08	0.32	0.523
DAN	0.70	0.77	0.12	0.50	0.06	0.11	0.377
DANN	0.47	0.46	0.16	0.65	**0.05**	0.10	0.315
RSD	0.31	0.31	0.12	0.53	0.07	0.08	0.237
DARE-GRAM (ours)	**0.30**	**0.20**	**0.11**	**0.25**	**0.05**	**0.07**	**0.164**

Table 3.2: Comparisons with related works on the MPI3D regression tasks. All results are shown in sum of MAE with the ResNet-18.

Methods	RL → RC	RL → T	RC → RL	RC → T	T → RL	T → RC	Avg
Resnet-18	0.17	0.44	0.19	0.45	0.51	0.50	0.377
TCA	0.17	0.42	0.19	0.42	0.50	0.50	0.373
MCD	0.13	0.40	0.15	0.45	0.52	0.50	0.358
JDOT	0.16	0.41	0.16	0.41	0.47	0.47	0.353
AFN	0.18	0.45	0.20	0.46	0.53	0.53	0.390
DAN	0.12	0.35	0.12	0.27	0.40	0.41	0.278
DANN	0.09	0.24	0.11	0.41	0.48	0.37	0.283
RSD	**0.09**	0.19	**0.08**	0.15	0.36	0.36	0.205
DARE-GRAM (ours)	**0.09**	**0.15**	0.10	**0.14**	**0.24**	**0.24**	**0.160**

by 33.3% and 52.8%, respectively. **Evaluation on MPI3D:** We further evaluate the effectiveness of our method on this more complex simulation-real data set. As shown in Table 3.2, in average, our method outperforms all previous methods. The improvement over the previous state-of-the-art RSD is more than 21.9%. The improvement is especially significant in the four hard adaptation directions **T** → **RL**, **T** → **RC**, **RL** → **T**, and **RC** → **T**. The performance is comparable with RSD on the **RC**/**RL** pair. This might be because the domain gap is relatively small between the pair and the performance (≈ 0.1) could be close to saturation.

Evaluation on Biwi Kinect: Given the much smaller size of the dataset (15,000 images compared to the number of samples in the scale of millions in the other two datasets), the Biwi Kinect regression task is particularly challenging. Additionally, the high imbalance between the two domains and the lack of separation of training and testing sets makes it more difficult and closer to real-world scenarios for DAR. The results reported in Table 3.3 demonstrate that our model can also consistently improve over previous methods on both directions on this more challenging task.

The performance improvement on the three datasets of very different natures demonstrates the effectiveness of our proposed method.

Table 3.3: Comparisons with previous works on Biwi Kinect.

Method	M → F	F → M	Avg
Resnet-18	0.29	0.38	0.335
TCA	0.31	0.39	0.350
MCD	0.31	0.37	0.340
JDOT	0.29	0.39	0.340
AFN	0.32	0.41	0.365
DAN	0.28	0.37	0.325
DANN	0.30	0.37	0.335
RSD	0.26	0.30	0.280
DARE-GRAM (ours)	**0.23**	**0.29**	**0.260**

3.4.3 Discussion and Analysis

To provide more insights on the proposed Unsupervised Domain Adaptation Regression by Aligning Inverse Gram Matrices, we provide a detailed analysis of the different components of the methodology.

Angle Alignment and Scale Alignment In the first ablation, we study the impact of our angle alignment on the inverse Gram matrix and our scale alignment on the eigenvalues. The **C** → **S** and **N** → **S** tasks in dSprites are used for the ablation here. As shown in Table 3.4, both components of the proposed methodology are able to improve over the baseline. Minimizing the angle between the pseudo-inverse of the gram Matrix can reduce the MSE over the source-only baseline by 70% on the **C** → **S** task. We also compare the alternate angle alignment on the Gram matrix and truncated Gram matrix without considering the inverse. The MAE in both cases is significantly worst than our inverse version. This demonstrates the significant impact of the inverse operation (as in Equation 3.5) on the regression layer and verifies our motivation for considering the closed-form solution of OLS regression problems. In addition, as discussed in Section 3.3.4, the scaling constraints provide essential additional supervision on the alignment and further improve the model performance. Both terms are effective in improving the performance.

Table 3.4: Ablation study of different components in our proposed method on C → S and N → S task from dSprites. All results are shown in sum of MAE.

Method	C → S	N → S
Resnet-18 (source only)	0.90	0.65
RSD	0.31	0.53
Angle Alignment for Gram	0.88	0.55
Angle Alignment for truncated Gram	0.89	0.52
Angle Alignment for Gram Inverse (ours)	0.27	0.36
Scale Alignment (ours)	0.23	0.60
DARE-GRAM (ours, angle + scale)	**0.20**	**0.25**

Effect of a Larger Batch Size Given the direct relationship between the number of samples in a batch to the variance of the estimated β in Equation 3.5, we studied the effect of the different training batch sizes in Figure 3.7 (using the same hyperparameters). As the batch size increases, our approach results in lower MAE. Furthermore, the RSD approach is more sensitive to the batch size and leads to numerical errors for batches bigger than 64. In this paper, we used only the same batch size of 36 to have a fair comparison with RSD. However, better results can be achieved with our method by further increasing the batch size to 256.

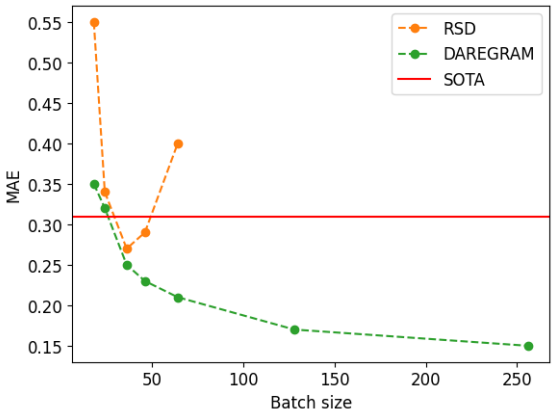
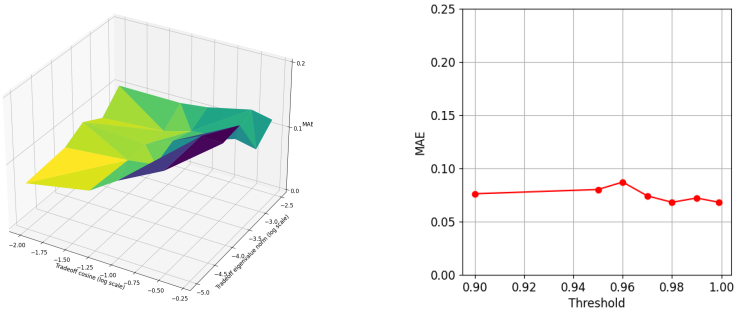


Figure 3.7: Batch size sensitivity on transfer task $\mathbf{C} \rightarrow \mathbf{S}$. For RSD, larger batch sizes lead to numerical errors, thus results not shown. Our method DARE-GRAM is able to achieve better performance with large batch sizes because the feature correlations can be better captured by the Gram matrix with larger number of samples.



(a) Impact of α_{cos} and γ_{scale} in log scale on the sum of MAE.

(b) Impact of Threshold T on the sum of MAE

Figure 3.8: Hyperparameter sensitivity of our method on dSprites transfer task $\mathbf{S} \rightarrow \mathbf{N}$.

Effect of alignment factors: We conduct additional experiments to evaluate the impact on the performance when using different values of hyperparameter α_{cos}, γ_{scale} and the threshold T. As shown in Figure 3.8, results confirm that our method is not sensitive to hyperparameters.

Alignment performance of Z and $(Z^T Z)^{-1}$ To further validate the proposed method, we examined the cosine similarity of the k-principal components of Z_s, Z_t, as well as $(Z_s^T Z_s)^{-1}, (Z_t^T Z_t)^{-1}$, after applying our proposed method and RSD. The results are shown in Table A2 in the supplementary material. Our results demonstrate that aligning Z by RSD can lead to poorly aligned inverse Gram matrix $(Z^T Z)^{-1}$. In contrast, by aligning $(Z^T Z)^{-1}$, our proposed method leads to a well-aligned Z, providing further empirical support for the effectiveness of our proposed method.

3.5 Conclusion

In this paper, we have presented a new domain adaptative regression method called DARE-GRAM. We tackled the domain adaptation for regression problems from a different perspective analyzing the ordinary least square solution to the linear regressor in the deep domain adaptation context. Rather than aligning the original feature embedding space, we aligned a selected subspace of the pseudo-inverse Gram matrix, leveraging the pseudo-inverse low-rank property. Finally, two new regularization terms were proposed to align the scale and angle in a selected subspace generated by the Gram matrix of the two domains. Experimental results show that DARE-GRAM achieves significant improvement in three benchmark regression datasets while ensuring the stability and robustness of the training procedure.

Acknowledgments: This work was supported by the Swiss National Science Foundation under Grant PP00P2_176878.

4 Uncertainty-Guided Alignment for Unsupervised Domain Adaptation in Regression

This chapter corresponds to the published article[I]:
Nejjar, Ismail, Gaetan Frusque, Florent Forest and Olga Fink, "Uncertainty-Guided Alignment for Unsupervised Domain Adaptation in Regression.", 2024.

Abstract: Unsupervised Domain Adaptation for Regression aims to adapt models from a labeled source domain to an unlabeled target domain for regression tasks. Traditional feature alignment methods, successful in classification, often prove ineffective for regression due to the correlated nature of regression features. To address this challenge, we propose Uncertainty-Guided Alignment (UGA), a novel method that integrates predictive uncertainty into the feature alignment process. UGA employs Evidential Deep Learning to predict both target values and their associated uncertainties. This uncertainty information guides the alignment process and fuses information within the embedding space, effectively mitigating issues such as feature collapse in out-of-distribution scenarios. We evaluate UGA on two computer vision benchmarks and a real-world battery state-of-charge prediction across different manufacturers and operating temperatures. Across 52 transfer tasks, UGA on average outperforms existing state-of-the-art methods. Our approach not only improves adaptation performance but also provides well-calibrated uncertainty.

4.1 Introduction

Effective transfer of regression models across diverse domains is crucial for applications such as computer vision and prognostics. However, this task remains challenging because variations in data distributions and feature relationships between domains can significantly degrade model performance when applied to new environments. In prognostics, domain gaps often arise from differences between industrial asset units, operating conditions, and usage patterns, while in computer vision, these gaps are typically caused by variations in lighting, camera settings, and scene compositions.

[I]Please note, this is the author's version of the manuscript that is currently under review.

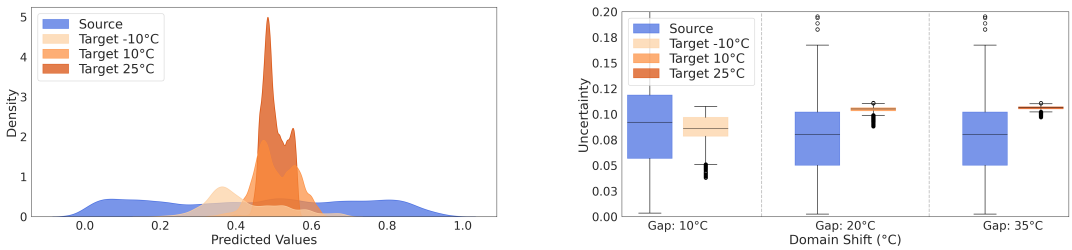
Unsupervised Domain Adaptation (UDA) has emerged as a promising approach to address this challenge, allowing the transfer of knowledge from a labeled source domain to an unlabeled target domain (Oza et al., 2023; Singhal et al., 2023). While UDA has achieved significant success in classification tasks, its application to regression problems remains less explored and presents unique challenges (Mansour et al., 2009a). In particular, traditional UDA methods designed for classification, such as correlation alignment (B. Sun & Saenko, 2016), maximum mean discrepancy (Borgwardt et al., 2006; Long et al., 2015), and adversarial training (Ganin et al., 2016), have shown limited effectiveness when applied to regression tasks (X. Chen et al., 2021). Nevertheless, regression tasks are prevalent in various fields, including computer vision (e.g., depth estimation (Kong et al., 2024; Piccinelli et al., 2023), age prediction (Nejjar, Ahmed, & Fink, 2024; X. Wen et al., 2020)) and prognostics (e.g., remaining useful life prediction of complex industrial and infrastructure assets (Fink et al., 2020; Nejjar, Geissmann, et al., 2024)).

The challenges of UDA in regression compared to classification arise from the fundamental differences between classification and regression tasks. The first challenge is related to the structure of the feature space. In classification, learned features are typically distributed across various dimensions of the embedding space, forming distinct clusters for each class (Hinton et al., 1986; LeCun et al., 2015). This rich and dispersed feature representation facilitates easier alignment between source and target domains, which is crucial for traditional alignment-based UDA methods. In contrast, regression tasks tend to generate more correlated features, with meaningful information concentrated in lower-dimensional subspaces. This phenomenon is often due to deep learning models developing *lazy* representations (Kunin et al., 2024), where the model relies on a limited subset of features to make predictions. These inherent differences require specialized approaches for UDAR, with recent state-of-the-art methods focusing on subspace alignment rather than aligning the entire feature space (X. Chen et al., 2021; Nejjar, Wang, & Fink, 2023a). By targeting these lower-dimensional informative subspaces, these methods better address the unique challenges posed by regression tasks.

A second major challenge in UDAR is linked to the quantification of prediction uncertainty. In classification tasks, models inherently can provide confidence estimates through class-wise probabilities (Mukhoti et al., 2023). Recent advancements in classification-based domain adaptation methods heavily rely on there confidence or entropy measures to generate accurate pseudo-labels for self-training (Forest & Fink, 2023; Hao et al., 2024; Zou, Yu, Kumar, & Wang, 2018). In contrast, regression models typically output point estimates without associated uncertainty measures, presenting a unique challenge for UDAR. Although there is no direct equivalent to classification confidence in regression, incorporating uncertainty estimates could serve a similar function. These uncertainty measures can help identify samples where the model is more confident, thereby guiding the alignment process during domain adaptation. In critical applications, understanding the confidence levels of predictions not only provides valuable insights for decision-making but could also be effective for domain adaptation strategies for regression tasks by focusing

on reliable predictions. Leveraging uncertainty in regression thus holds the potential to improve the effectiveness of UDAR methods, ensuring more robust and trustworthy model performance across diverse domains.

Most well-known methods of uncertainty quantification in deep learning require multiple forward passes at test time, such as Monte Carlo Dropout (MCD) (Gal & Ghahramani, 2016). Among these, Deep Ensembles have generally performed best in uncertainty prediction (Ovadia et al., 2019), but their significant memory and compute burden at training and test time hinders their adoption in real-life and mobile applications. Consequently, there has been increased interest in uncertainty quantification using deterministic single forward-pass neural networks, which have a smaller footprint and lower latency. Two popular works in single forward-pass uncertainty, deterministic uncertainty quantification (DUQ) (Van Amersfoort et al., 2020) and Spectral-normalized Neural Gaussian Process (SNGP) (J. Liu et al., 2020), propose distance-aware output layers and introduce additional inductive biases in the feature extractor to encourage smoothness and sensitivity in the latent space. These methods perform well and are almost competitive with Deep Ensembles on Out-of-Distribution benchmarks. However, they require substantial changes to training and introduce additional hyper-parameters due to the specialized output layers used at training. A simpler approach is Evidential Deep Learning (L. Huang et al., 2023; Sensoy et al., 2018; Zhou et al., 2023), which aims to learn to predict the parameters of a higher-order distribution, allowing for direct uncertainty estimation without the need for multiple forward passes, and requires only a simple change in the loss function.



(a) Density plot comparing model predictions for battery State of charge between the source domain at -20°C and target domains of increasing temperature, showing the effect of domain shift on prediction patterns.

(b) Box plot comparing uncertainty distributions between source domain (-20°C) and target domains of increasing temperature, showing the impact of domain shift on model uncertainty.

Figure 4.1: Visualization of domain shift effects on prediction distribution and uncertainty in regression tasks for battery state of charge prediction. (a) Density plot of model predictions. (b) Box plot of uncertainty distributions.

To illustrate the challenges discussed above, we present two visualizations in Figure 4.1, where the domains are defined by battery operating temperature and the task is to predict the state of charge of a battery. Figure 4.1 (a) illustrates the density of model predictions

under varying degrees of domain shift. As the temperature difference (domain gap) increases, the distribution of predicted values becomes narrower, indicating a tendency toward less diverse predictions for out-of-distribution data. This phenomenon, known as feature collapse (Van Amersfoort et al., 2020), occurs when feature extractors map the entire target domain to a confined region within the source domain feature space. Figure 4.1 (b) demonstrates the differences in uncertainty distributions between the source domain (-20°C) and target domains with progressively higher temperatures, using a model trained with Deep Evidential Regression (Amini et al., 2020). This visualization highlights the role of uncertainty as an indicator of domain shift in regression tasks. Notably, as the domain gap widens with increasing the temperatures, the uncertainty distributions become narrower, further suggesting feature collapse within the target domain.

The visualizations in Figure 4.1 underscore these two key challenges in UDAR: (1) the need for effective feature alignment strategies that account for the unique characteristics of regression tasks, and (2) the importance of quantifying and leveraging uncertainty in the adaptation process. To address both challenges, we propose a novel approach named Uncertainty-Guided Alignment (UGA) for unsupervised domain adaptation in regression tasks. Our method leverages Evidential Deep Learning to effectively guide the adaptation process. In UGA, we use uncertainty information in two distinct ways:

1. Guiding the alignment process: We focus on features that are more certain in the source domain and potentially transferable to the target domain.

2. Embedding space regularization: We encourage the model to learn more robust and generalizable features.

We evaluate our method on standard computer vision regression tasks, including the dSprites and MPI3D datasets, demonstrating its effectiveness across different types of domain shifts. Additionally, we apply UGA to a real-world prognostics problem: battery state-of-charge estimation under varying temperature conditions and across different manufacturers. In this real-world scenario, we observe that the UGA model's performance remains consistently superior across different domain gaps compared to baseline methods. The model demonstrates well-calibrated uncertainty estimates throughout the battery state-of-charge, with higher uncertainty at the beginning of a new battery cycle that gradually decreases as the battery ages and predictions become more accurate. Across 52 transfer tasks, our proposed method outperforms existing state-of-the-art approaches on average, demonstrating its effectiveness in improving adaptation performance and providing valuable confidence measures for regression predictions in unsupervised domain adaptation tasks.

4.2 Related Work

4.2.1 Unsupervised Domain Adaptation

In the realm of UDA, two main directions have emerged. The first direction aims to reduce the discrepancy between domains by using optimal transport (Chang et al., 2022; Courty, Flamary, Tuia, & Rakotomamonjy, 2017; Fatras et al., 2021; C.-X. Ren et al., 2024) or matching moments of the distributions through the Maximum Mean Discrepancy (MMD) (C. Chen et al., 2020; Long et al., 2015; Y.-W. Luo & Ren, 2021; Shi et al., 2023). The second direction in UDA revolves around learning domain-invariant representations through adversarial learning techniques (L. Chen, Chen, et al., 2022; Ganin et al., 2016; Rangwani et al., 2022). In the context of classification, leveraging self-training with confident pseudo-labels has proven to be effective (Hao et al., 2024; Hoyer et al., 2023b; Huo et al., 2024; Karim et al., 2023; T. Sun et al., 2022; Zou, Yu, Kumar, & Wang, 2018). Furthermore, in the context of UDA, various mixing strategies have been developed to foster domain-invariant representations within the embedding space (T. Jing et al., 2023; Y. Wu et al., 2020; M. Xu et al., 2019), ultimately improving the model's generalization capabilities. A recent line of works leverage foundation models such as CLIP to perform the adaptation (Ge et al., 2023b).

The field of Unsupervised Domain Adaptation for Regression has received relatively less attention than classification. Early theoretical works have derived generalization bounds (Cortes & Mohri, 2011a; Mansour et al., 2009b; Redko et al., 2022). More recently, (J. Wu et al., 2022a) proposed a framework for domain adaptation in regression, specifically designed for scenarios where target labels are accessible. This framework encompasses popular approaches, including domain-invariant representation learning, reweighting, and adaptive Gaussian processes. Recent endeavors have focused on addressing the unique challenges within the unsupervised setting for domain adaptation for regression (X. Chen et al., 2021; Dhaini et al., 2023; Nejjar, Geissmann, et al., 2023; Nejjar, Wang, & Fink, 2023a). For instance, RSD (X. Chen et al., 2021) addresses the challenge of feature scale changes during domain adaptation by aligning representation subspaces using a geometrical distance measure. Following a similar motivation, DARE-GRAM (Nejjar, Wang, & Fink, 2023a) aligns the angle and scale of the inverse Gram matrices for the source and target features. The authors address ill-conditioned inverse Gram matrices due to the ratio between batch size and the embedding dimension by proposing the use of the pseudo-inverse of the Gram matrix, achieved by selecting a specific subspace. Although traditional feature alignment methods can be applied in UDAR, their effectiveness is limited in this context. Regression models tend to learn focused representations, highly correlated with the target output, causing feature alignment to fail, as information about the domain shift in the input is not prominently present in the learned embeddings. This motivates the need to enrich the feature representation to guide feature alignment.

4.2.2 Uncertainty-guided domain adaptation

Reducing prediction uncertainty on the target domain is a technique that has been explored for domain adaptation. The main intuition is that uncertainty for non-adapted target features is expected to be high, while it is expected to be low for source samples. In the setting of classification, entropy minimization and the related technique of self-training with pseudo-labels, effectively encourage cross-domain feature alignment (L. Han et al., 2019; Zou, Yu, Kumar, & Wang, 2018). (J. Wen et al., 2019) proposed to minimize the uncertainty discrepancy between domains, where uncertainty is measured by the prediction entropy or variance using Monte-Carlo Dropout as an approximate Bayesian variational inference (Gal & Ghahramani, 2016). The approach proposed in (Liang et al., 2019) aligns the conditional distributions by minimizing the MMD between the mean embeddings of each class. The target distribution class mean embeddings are estimated by a sum of the elements weighted by their posterior probability. In the source-free DA setting, (Roy et al., 2022) proposed a method based on the information maximization loss but with uncertainty weighting of the samples, using a last-layer Laplace approximation to estimate the uncertainty of the pre-trained model. While our work is closely related to (J. Wen et al., 2019) which was applied to classification tasks, its extension to regression tasks has not been explored. Moreover, the existing studies focused on a narrow scenario, limited to Bayesian neural networks, particularly Monte Carlo Dropout, and adversarial alignment. This paper demonstrates that various uncertainty frameworks, including Gaussian Processes and evidential neural networks, can effectively guide alignment in regression tasks.

4.3 Preliminaries

Deep Evidential Regression (DER). To address the challenge of uncertainty quantification in UDAR, we adopt Deep Evidential Regression (DER), a method that simultaneously performs prediction and uncertainty estimation. Unlike more commonly applied uncertainty quantification regression methods for deep learning, DER models both aleatoric (data-inherent) and epistemic (model-related) uncertainties.

From a maximum likelihood perspective, the likelihood of the parameters $\boldsymbol{\theta}$ given the N observations $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ and regression targets $\mathbf{y} = \{y_1, \ldots, y_N\}$ is defined as:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \prod_{i=1}^{N} p(y_i|\mathbf{x_i}, \boldsymbol{\theta}) \tag{4.1}$$

In practice, it is commonly assumed that the target variable y is drawn from a Gaussian distribution with mean $\mu = f(\mathbf{x})$ and variance σ^2. To effectively learn both aleatoric and epistemic uncertainties, the approach proposed by (Amini et al., 2020) assumes that the mean and variance of the Gaussian distribution are themselves drawn from higher-order distributions: the mean from a Gaussian distribution and the variance from an Inverse-

Gamma distribution. This hierarchical modeling choice allows for the incorporation of prior knowledge about the parameters. Specifically, the Normal Inverse-Gamma distribution with parameters $(\gamma, \nu, \alpha, \beta)$ is used as a conjugate prior:

$$y \sim \mathcal{N}(\mu, \sigma^2),\ \mu \sim \mathcal{N}(\gamma, \sigma^2/\nu),\ \sigma^2 \sim \Gamma^{-1}(\alpha, \beta), \tag{4.2}$$

where $\gamma \in \mathrm{R}$ represents the predicted mean, $\nu > 0$ is the precision of the mean, which is inversely related to epistemic uncertainty, $\alpha > 1$ and $\beta > 0$ control the shape and scale of the inverse-gamma distribution, respectively, and are associated with aleatoric uncertainty. This formulation allows the model to quantify uncertainty in both its prediction (μ) and the inherent noise in the data (σ^2). The training process aims to minimize the negative log-likelihood (NLL) loss given by:

$$\mathcal{L}_{\mathrm{NLL}} = \tfrac{1}{2}\log\left(\tfrac{\pi}{\nu}\right) - \alpha\log(2\beta(1+\nu)) + \log(\psi) + \left(\alpha + \tfrac{1}{2}\right)\log\left((y-\gamma)^2\nu + 2\beta(1+\nu)\right) \tag{4.3}$$

where $\psi = \Gamma(\alpha)/\Gamma(\alpha+1/2)$. The total evidential loss is defined as the sum of the negative log-likelihood loss and a regularization term:

$$\mathcal{L}_{\mathrm{EVI}} = \mathcal{L}_{\mathrm{NLL}} + \lambda_{\mathrm{EVI}}\mathcal{L}_R, \tag{4.4}$$

where $\mathcal{L}_R = |y-\gamma|(2\nu+\alpha)$ penalizes evidence of prediction errors, and the hyperparameter λ_{EVI} balances the trade-off between fitting the model to the data and increasing the uncertainty measures.

In practice, this loss is minimized using standard gradient descent methods. This allows the neural network to learn not only point estimates for predictions but also full probability distributions, capturing the uncertainty associated with each prediction.

4.4 Methods

4.4.1 Problem setting

Let $\mathcal{X}$ and $\mathcal{Y}$ denote the input space and output spaces, respectively. We define the source domain S as consisting of N_S labeled samples $\{(\mathbf{x}_i^S, y_i^S)\}_{i=1}^{N_S}$ drawn from the joint distribution P_S. Conversely, the target domain as T comprises N_T unlabeled samples $\{\mathbf{x}_i^T\}_{i=1}^{N_T}$ drawn from the from marginal distribution P_T. Our primary objective is to predict the continuous-valued outputs y^T for unlabeled samples in the target domain by leveraging the labeled data from the source domain. To achieve this, we develop a model $f(\cdot)$ composed of two neural networks: a feature extractor $g(\cdot)$ that transforms the input data into meaningful feature representations, and a regression head $r(\cdot)$ that maps these

extracted features to the continuous output space. Our objective is to train $f : \mathbf{x} \to y$, parameterized by θ, to minimize the expected error on the target data using the ℓ_2 loss function. We assume that the labels in both domains cover the same range.

4.4.2 Theoretical Motivation

In Unsupervised Domain Adaptation for Regression (UDAR), a common approach is to align features between the source and target domains. However, perfect feature alignment can lead to suboptimal performance, as it may eliminate essential domain-specific information. To address this challenge, we present a theoretical framework that quantifies this limitation and motivates our uncertainty-guided approach. Central to this framework is the concept of the adaptation gap (Dong, Nejjar, et al., 2024; Q. Jiang et al., 2023), which measures the difference in predictive power between the source and target domains:

Definition 4.4.1 (Adaptation Gap) *Let $\mathbf{z}^S, y^S$ and $\mathbf{z}^T, y^T$ denote the features extracted by the feature extractor $g(\cdot)$ and the labels for the source and the target domains, respectively. The adaptation gap Δ_p is defined as:*

$$\Delta_p := I(\mathbf{z}^S; y^S) - I(\mathbf{z}^T; y^T) \tag{4.5}$$

where $I(\mathbf{z}; y)$ represents the mutual information between $\mathbf{z}$ and y. We assume that $I(\mathbf{z}^S; y^S) \geq I(\mathbf{z}^T; y^T)$, ensuring $\Delta_p \geq 0$, given that the feature extractor has been trained on the source domain, involving a loss of information. In other words, the features extracted by $g(\cdot)$ provide less information about y in the target domain than in the original source domain.

Theorem 4.4.1 *Let $g(\cdot)$ be a deterministic feature extractor, and let $\mathbf{z}^S = g(x^S)$, $\mathbf{z}^T = g(x^T)$ denote the extracted features for the source and target domains, respectively. Assuming that $y^S|\mathbf{z}^S$ and $y^T|\mathbf{z}^T$ follow Gaussian distributions, if the features are perfectly aligned, i.e., $\mathbf{z}^S = \mathbf{z}^T = \mathbf{z}$, then:*

$$\inf_f \mathrm{E}_{p_T}[\ell(f(\mathbf{z}^T), y^T)] - \inf_{f'} \mathrm{E}_{p_S}[\ell(f'(\mathbf{z}^S), y^S)] \geq \sigma^2_{y^S|x^S}(e^{2\Delta_p} - 1)$$

where ℓ denotes the mean squared error loss, p_S and p_T are the source and target domain distributions respectively, and $\sigma^2_{y^S|x^S}$ is the conditional variance of y^S given x^S.

Our theorem highlights a fundamental limitation of perfect feature alignment in UDAR: as the adaptation gap Δ_p increases, the lower bound on expected loss grows exponentially. This result indicates a need for an approach that balances feature alignment with

the preservation of domain-specific information. This finding motivates our proposed Uncertainty-Guided Alignment approach.

As shown in Figure 4.0a, models trained to predict uncertainty using evidential learning can distinguish between source and target domains in a single forward pass, while retaining domain-specific information. UGA leverages this capability to guide the alignment process by: (1) selectively aligning features based on uncertainty estimates, ensuring that domain-specific information is preserved where necessary; (2) incorporating uncertainty as a regularization term, encouraging the model to learn more robust and transferable features. In addition, UGA is providing a mechanism to express confidence in the model's predictions.

4.4.3 Overview of the proposed method

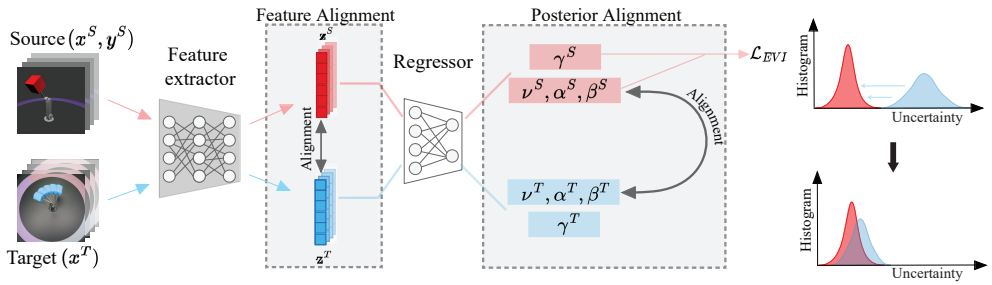


Figure 4.2: Overview of our proposed Uncertainty-Guided Alignment (UGA) framework for unsupervised domain adaptation in regression. UGA leverages the Deep Evidential Regression (DER) uncertainty framework. We propose aligning augmented feature representations with uncertainty to guide traditional feature alignment. We also introduce Posterior alignment, an approximation method aligning the evidential parameters.

Our proposed method, Uncertainty Guided Adaptation (UGA), addresses the challenge of unsupervised domain adaptation in regression by leveraging uncertainty to guide the alignment of source and target domain distributions. UGA introduces two core strategies: Uncertainty-Guided Feature Alignment and Uncertainty-Guided Posterior Alignment.

In the feature alignment approach, we align the distributions of feature embeddings between source and target domains, considering both the predicted values and their associated uncertainties. This ensures a comprehensive alignment that accounts for both regression outputs and the model's confidence in those predictions. Alternatively, the posterior alignment approach directly aligns the uncertainty distributions between domains at the posterior level. This serves as an approximation of feature alignment, focusing on matching uncertainty estimates across domains.

While our framework is flexible and can be combined with various feature alignment techniques and uncertainty quantification methods, we demonstrate it using a combination

of Maximum Mean Discrepancy (MMD) and Deep Evidential Regression (DER) for simplicity and consistency. Figure 4.2 provides an overview of the proposed framework.

This choice is motivated by the fact that Bayesian neural networks (BNNs), traditionally a standard for uncertainty quantification treating model weights as distributions, are computationally expensive due to the requirement of multiple input samples to estimate output variance. To overcome this limitation, we adopt evidential deep learning, which frames learning as an evidence-acquisition process. This approach uses a standard neural network architecture, significantly improving computational efficiency compared to BNNs by eliminating the need for sampling. By leveraging evidential learning, we can estimate uncertainties in a single forward pass, enabling more efficient and scalable uncertainty-guided adaptation.

Uncertainty-Guided Feature Alignment

Uncertainty-guided Feature Alignment aligns the embeddings of source and target domains, following a similar approach to (Long et al., 2015), but with the addition of uncertainty guidance. Let the feature embeddings for the source and target domains be denoted as $\mathbf{z}_i^S = g(\mathbf{x}_i^S)$ and $\mathbf{z}_i^T = g(\mathbf{x}_i^T)$, respectively. To quantify the discrepancy between these distributions, we compute the MMD as:

$$\mathrm{MMD}^2_{\text{feature}} = \Big|\Big| \frac{1}{N_S} \sum_{i=1}^{N_S} \phi(\mathbf{z}_i^S) - \frac{1}{N_T} \sum_{i=1}^{N_T} \phi(\mathbf{z}_i^T) \Big|\Big|^2_{\mathcal{H}_k}. \tag{4.6}$$

Here, the feature embeddings z_i incorporate both the predicted target values and their associated uncertainties. By leveraging this uncertainty information, we can more effectively guide the minimization of feature discrepancies between the source and target domains. The final objective function for this alignment method is formulated as:

$$\mathcal{L}_{\text{feature}} = \mathcal{L}_{\text{EVI}} + \lambda \cdot \mathrm{MMD}^2_{\text{feature}} \tag{4.7}$$

where λ is a hyperparameter that controls the relative weight of the feature alignment term, and $\mathcal{L}_{\text{EVI}}$ represents the supervised loss of the source domain.

Uncertainty-Guided Posterior Alignment

The Uncertainty-Guided Posterior Alignment method takes a different approach by directly aligning the uncertainty distributions between the source and target domains. Unlike feature alignment, which focuses on aligning feature representations, this method focuses on aligning the uncertainty estimates.

To achieve this, we compute the MMD loss between the higher-order parameters of the uncertainty distributions produced by the DER model. The evidential network outputs for the source and target domains are denoted as $\gamma_i^S, \nu_i^S, \alpha_i^S, \beta_i^S = f(\mathbf{x}_i^S)$ and $\gamma_i^T, \nu_i^T, \alpha_i^T, \beta_i^T = f(\mathbf{x}_i^T)$, respectively. In this context, γ represents the predicted mean, while ν, α, and β parameterize the uncertainty distribution. Alignment is achieved by minimizing the squared MMD loss:

$$\text{MMD}^2_{\text{posterior}} = \left\| \frac{1}{N_S} \sum_{i=1}^{N_S} \phi([\nu_i^S, \alpha_i^S, \beta_i^S]) - \frac{1}{N_T} \sum_{i=1}^{N_T} \phi([\nu_i^T, \alpha_i^T, \beta_i^T]) \right\|^2_{\mathcal{H}_k} \tag{4.8}$$

This alignment ensures that the uncertainty estimates captured by the evidential network, represented by the parameters ν, α, and β, remain consistent and transferable across different domains. The final objective function to minimize is defined as follows:

$$\mathcal{L}_{\text{posterior}} = \mathcal{L}_{\text{EVI}} + \lambda \cdot \text{MMD}^2_{\text{posterior}}. \tag{4.9}$$

4.5 Experiments

We evaluate the performance of our proposed method using two computer vision benchmarks and a PHM task focused on battery State of Charge prediction. This section details our experimental setup, datasets, and evaluation metrics utilized.

4.5.1 Datasets and Tasks

Computer Vision Benchmarks

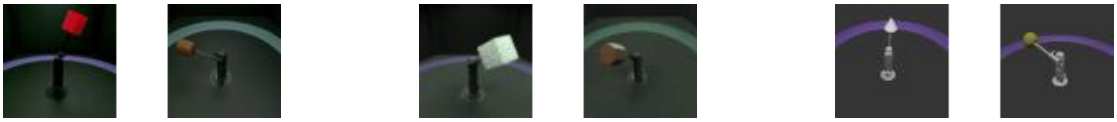
dSprites (Matthey et al., 2017) is a synthetic 2D image dataset with five independent latent factors. We use three variants as separate domains: Color (**C**), Scream (**S**), and Noise (**N**), as illustrated in Figure 4.3. Each domain consists of 737,280 images. Our tasks focus on regression for scale and position.

(a) Noise (b) Color (c) Scream

Figure 4.3: Sample example of different domains in dSprites.

MPI3D (Gondal et al., 2019) contains 1,036,800 examples of 3D objects from three domains: Toy (**T**), RealistiC (**RC**), and ReaL (**RL**). This dataset enables the study of domain gaps between real and simulated data in a robotics context. We focus on regression tasks involving rotation around vertical and horizontal axes. Figure 4.4 showcases sample

examples from each domain, highlighting the variations in appearance and characteristics.



(a) Real (b) Realistic (c) Toy

Figure 4.4: Sample example of different domains in MPI3D.

PHM task: State-of-Charge prediction

For the state-of-charge (SOC) prediction task, we use two publicly available datasets:

- LG 18650HG2 dataset (Kollmeyer et al., 2020), examples are seen in Figure 4.5.
- Panasonic 18650PF dataset (Kollmeyer, 2018),examples are seen in Figure 4.6.

The LG dataset captures the discharge behavior of a 3Ah LG HG2 cell tested at temperatures ranging from -20°C to 40°C, while the Panasonic dataset uses a 2.9Ah 18650PF cell tested from -20°C to 25°C. Both datasets provide time-series measurements of voltage, current, battery case temperature, amp-hours, watt-hours, and power for each test under various standardized drive cycles. These drive cycles simulate different driving conditions and are summarized in Table 4.1. These datasets were collected specifically for State-of-Charge estimation in lithium-ion batteries.

Table 4.1: Description of Drive Cycles. NN: Neural Network cycle (Panasonic dataset only). Mixed: Random combinations of other cycles (LG dataset only).

Drive Cycle	Description
US06	Aggressive, high speed and/or high acceleration
HWFET	Highway driving conditions under 60 mph
UDDS	City driving conditions with frequent stops
LA92	More aggressive urban driving than UDDS
NN	Combination of US06 and LA92
Mixed	Random combinations of above cycles

Our study focuses on adapting SOC prediction models across different lithium-ion battery types and operating conditions. We identify two domain gaps:

1. Manufacturer difference: LG and Panasonic batteries have unique chemical compositions and performance characteristics.
2. Temperature variations: Battery behavior changes significantly based on temperature conditions.

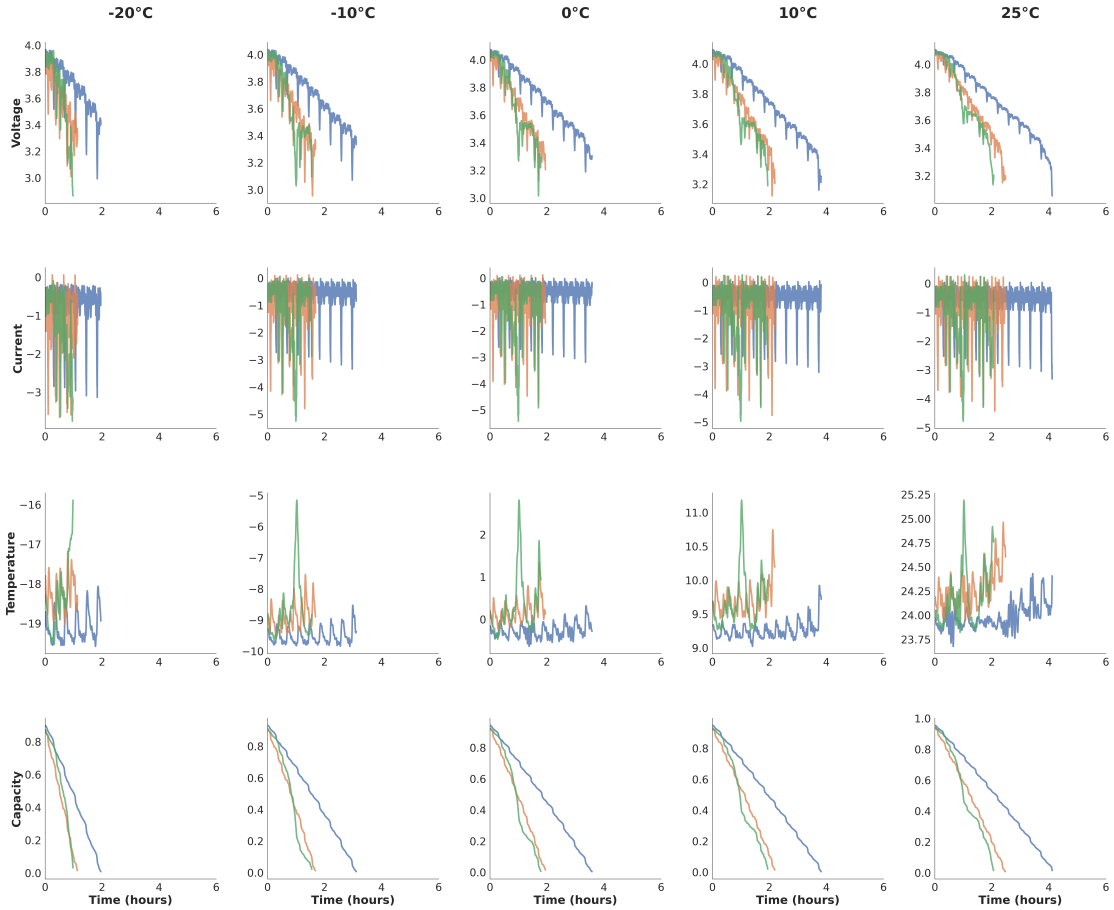


Figure 4.5: Example of State of Charge (SoC) curves and associated features (voltage, current, temperature, and time) for LG batteries that have been charged and discharged until reaching 0% capacity. The plots illustrate the impact of temperature on battery capacity, discharge rate, and overall performance throughout the battery's lifecycle.

Our objective is to adapt a SOC prediction model trained on one battery type at a specific temperature to another battery type at a different temperature (eg. adapting a model trained on LG batteries at 25°C to Panasonic at 0°C). This task presents a challenging unsupervised domain adaptation problem as it involves simultaneously bridging both domain gaps (manufacturer differences and temperature variations).

During data preprocessing, we downsample the dataset to 1Hz to reduce computational complexity while maintaining adequate temporal resolution. We select voltage, current, and temperature as input features, with SOC as the target variable.

For our train-test split, we reserve specific drive cycles exclusively for testing and use the remaining cycles for training. Specifically, for the Panasonic dataset, US06, LA92, and NN cycles are used for testing, while for the LG dataset, US06, LA92, and HWFET cycles serve as the test set. This partitioning ensures that our models are evaluated on a diverse range of driving conditions that were not seen during training.

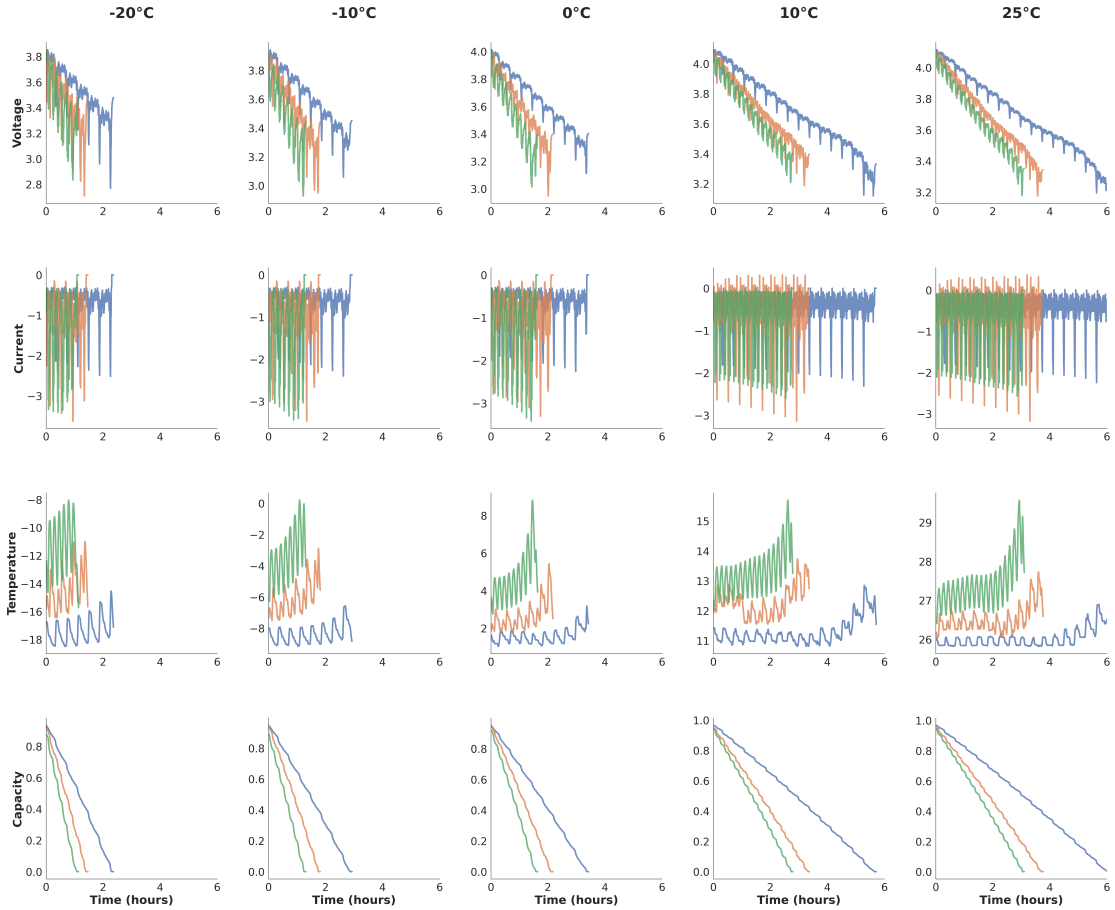


Figure 4.6: Comparison of SoC curves and associated features (voltage, current, temperature, and time) for Panasonic batteries that have been charged and discharged until reaching 0% capacity. The plots highlight the differences in battery performance between the two manufacturers, including variations in capacity, discharge rate, and temperature sensitivity throughout the battery's lifecycle.

4.5.2 Experimental Setup

Computer Vision Experiments

We use a pre-trained ResNet-18 (K. He et al., 2015) as the backbone, modified by disabling batch normalization and adding dropout (p=0.1) after each block. For DER, we use a single linear layer. We employ the Adam optimizer with a learning rate of $2e^{-5}$ and weight decay of $1e^{-3}$. The number of iterations is set to 20,000 for dSprites and 10,000 for MPI3D (2,000 for the S→N task), following (X. Chen et al., 2021) with a batch size of $b = 36$. We normalize the source and target labels to a common range of $[0, 1]$. The results are reported using the Mean Absolute Error (MAE) as the primary evaluation metric, in line with previous studies (X. Chen et al., 2021; W. Li et al., 2021).

PHM Experiments

For battery SOC prediction, we use a two-layer LSTM followed by a single linear layer. The model processes three inputs (Voltage, Current, and Temperature) provided as time series with a sequence length of 100. We set the hidden dimension to 64. Training is conducted using the SGD optimizer with a learning rate of $1e^{-1}$ for the final layers and $1e^{-2}$ for the LSTM, with weight decay of $1e^{-4}$ and a batch size of 128. All models are trained for 30 epochs. We report the Mean Square Error (MSE) as the primary metric and include coefficient of determination (R^2) for interpretability, given that the state of charge trajectory over time approximates a near-linear curve.

4.5.3 Hyperparameters

For both computer vision and PHM tasks, we set the hyperparameter λ in Equations 4.7 and 4.9 to increase from 0 to 1 during training, following $\lambda = \frac{2}{1+\exp(-10\mathrm{p})} - 1$, where p represents the training progress. We set λ_{EVI} to 1 for computer vision tasks and 0.1 for the PHM task.

4.6 Results and Discussion

4.6.1 Computer Vision Results

Table 4.2: Comparisons on dSprites regression tasks (sum of MAE across three targets). ×: no uncertainty (MSE loss), DER: Deep Evidential Regression. We compare JDOT, CORAL, and MMD with/without uncertainty-guided alignment.

Method	Unc.	C→N	C→S	N→C	N→S	S→C	S→N	Avg
Source only	×	0.94	0.90	0.16	0.65	0.08	0.26	0.498
DANN	×	0.47	0.46	0.16	0.65	0.05	0.10	0.315
RSD	×	0.31	0.31	0.12	0.53	0.07	0.08	0.237
DARE-GRAM	×	0.30	0.20	0.11	0.25	0.05	0.07	0.164
JDOT	×	0.86	0.79	0.19	0.64	0.10	0.23	0.468
UGA-Post. JDOT	DER	0.09	0.13	0.11	0.18	0.05	0.12	0.114
UGA-Feat. JDOT	DER	0.08	0.11	0.08	0.17	0.05	0.07	0.093
CORAL	×	0.95	0.87	0.13	0.57	0.05	0.12	0.448
UGA-Post. CORAL	DER	0.56	0.40	0.24	0.94	0.04	0.28	0.410
UGA-Feat. CORAL	DER	0.06	0.24	0.08	0.32	**0.02**	0.03	0.125
MMD	×	0.70	0.77	0.12	0.50	0.06	0.11	0.377
UGA-Post. MMD	DER	**0.04**	**0.10**	0.04	**0.16**	0.03	0.03	0.067
UGA-Feat. MMD	DER	**0.04**	**0.10**	**0.02**	**0.16**	**0.02**	**0.02**	**0.060**

dSprites: Table 4.2 presents the results obtained on the dSprites dataset, comparing our proposed methods with existing domain adaptation approaches. Specifically, we consider

Table 4.3: Comparisons on MPI3D regression tasks.

Methods	Unc.	RL→RC	RL→T	RC→RL	RC→T	T→RL	T→RC	Avg
Source only	×	0.17	0.44	0.19	0.45	0.51	0.50	0.38
JDOT	×	0.16	0.41	0.16	0.41	0.47	0.47	0.35
DANN	×	0.09	0.24	0.11	0.41	0.48	0.37	0.28
RSD	×	0.09	0.19	0.08	0.15	0.36	0.36	0.21
DARE-GRAM	×	0.08	0.15	0.10	0.14	**0.24**	**0.24**	**0.16**
MMD	×	0.12	0.35	0.12	0.27	0.40	0.41	0.28
UGA-Post. MMD	DER	0.08	0.45	0.08	0.18	0.41	0.30	0.25
UGA-Feat. MMD	DER	**0.07**	**0.13**	**0.07**	**0.11**	0.34	**0.24**	**0.16**

DANN, RSD, and DARE-GRAM as state-of-the-art methods for UDAR. Additionally, we use three feature alignment methods JDOT, CORAL, and MMD. For each case, we report results obtained without uncertainty guidance and with the proposed frameworks: Feature alignment (UGA-Feature) and Posterior alignment (UGA-Posterior).

Our Uncertainty-Guided Alignment methods consistently outperform existing approaches across all transfer tasks. The UGA-Feature approach, in particular, shows substantial improvements when combined with various feature alignment methods. JDOT's average MAE improved from 0.468 to 0.093, CORAL's from 0.448 to 0.125, and MMD's from 0.377 to 0.060 when combined with UGA-Feature. These results significantly surpass the performance of state-of-the-art methods like RSD (0.237) and DARE-GRAM (0.164). The UGA-Posterior approach also shows notable improvements, especially when combined with JDOT and MMD, achieving at least a 34% relative improvement over RSD and 4% over DARE-GRAM. Interestingly, while UGA-Posterior CORAL shows only a modest improvement from 0.448 to 0.410, UGA-Feature CORAL achieves a substantial gain to 0.125. This discrepancy might be attributed to CORAL's simplicity, which may not effectively align higher-order evidence in the posterior.

MPI3D: Table 4.3 presents the results obtained on the MPI3D dataset. Our proposed methods exhibit superior performance in five out of six tasks and achieve the best average performance. Across the six tasks, posterior alignment achieves an average MAE of 0.250, while feature alignment achieves an average MAE of 0.160. However, in the T→RL task, DARE-GRAM outperforms our proposed methods, with an MAE of 0.24, compared to UGA-Feature's MAE of 0.34. In the appendix, we present the histogram of predicted values and observe that for the T→RL task, the model struggles to predict values near the boundary (0 or 1).

Table 4.4: Results for source LG and target Panasonic

Transfer	Source only		CORAL		DAN		JDOT		DARE-GRAM		MMD		MMD+EVI	
Task	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2
-20 → -10	0.06	0.15	0.06	0.15	0.10	-0.39	0.02	0.77	0.02	0.79	0.01	0.85	0.02	0.76
-20 → 0	0.07	0.05	0.07	0.05	0.15	-0.89	0.01	0.89	0.01	0.93	0.01	0.87	0.00	0.95
-20 → 10	0.08	-0.00	0.08	-0.00	0.13	-0.66	0.00	0.95	0.01	0.86	0.01	0.93	0.00	0.94
-20 → 25	0.08	-0.03	0.08	-0.03	0.10	-0.30	0.00	0.95	0.06	0.18	0.08	-0.01	0.01	0.89
-10 → -20	0.10	-0.33	0.10	-0.32	0.06	0.18	0.04	0.50	0.04	0.41	0.03	0.59	0.03	0.55
-10 → 0	0.05	0.31	0.05	0.31	0.07	0.06	0.01	0.88	0.01	0.91	0.01	0.87	0.01	0.94
-10 → 10	0.08	0.00	0.08	-0.00	0.11	-0.39	0.01	0.93	0.00	0.94	0.01	0.90	0.00	0.95
-10 → 25	0.09	-0.11	0.09	-0.16	0.08	0.01	0.01	0.86	0.01	0.92	0.08	0.03	0.00	0.98
0 → -20	0.06	0.16	0.06	0.16	0.08	-0.13	0.04	0.51	0.05	0.34	0.04	0.41	0.05	0.39
0 → -10	0.07	0.11	0.07	0.12	0.04	0.49	0.03	0.64	0.03	0.61	0.02	0.71	0.02	0.70
0 → 10	0.04	0.46	0.04	0.48	0.08	-0.08	0.00	0.95	0.01	0.92	0.00	0.94	0.00	0.96
0 → 25	0.07	0.07	0.06	0.27	0.08	-0.05	0.00	0.96	0.02	0.79	0.00	0.94	0.00	0.97
10 → -20	0.16	-1.13	0.12	-0.66	0.13	-0.79	0.04	0.47	0.07	0.10	0.10	-0.32	0.04	0.42
10 → -10	0.05	0.30	0.05	0.36	0.10	-0.29	0.02	0.73	0.03	0.57	0.03	0.60	0.02	0.72
10 → 0	0.02	0.72	0.02	0.73	0.04	0.43	0.01	0.82	0.02	0.78	0.01	0.83	0.01	0.87
10 → 25	0.01	0.81	0.01	0.82	0.08	-0.10	0.00	0.96	0.01	0.90	0.00	0.94	0.00	0.97
25 → -20	0.31	-3.15	0.31	-3.15	0.12	-0.67	0.10	-0.27	0.11	-0.42	0.09	-0.14	0.07	0.10
25 → -10	0.29	-2.83	0.29	-2.83	0.12	-0.65	0.04	0.40	0.07	0.03	0.08	-0.06	0.03	0.56
25 → 0	0.27	-2.49	0.27	-2.48	0.08	-0.00	0.01	0.87	0.08	-0.07	0.08	-0.08	0.01	0.87
25 → 10	0.21	-1.72	0.20	-1.63	0.10	-0.27	0.01	0.86	0.01	0.88	0.01	0.88	0.01	0.92
Average	0.11	-0.43	0.11	-0.39	0.09	-0.23	**0.02**	0.73	0.03	0.57	0.04	0.53	**0.02**	**0.77**

4.6.2 PHM Results

Tables 4.4 and 4.5 present the results of the battery State of Charge prediction task, which involves adapting models between different battery manufacturers (LG to Panasonic and vice versa) and a range of operating temperatures (-20°C to 25°C).

LG to Panasonic Transfer: In Table 4.4, the source-only baseline model yields an average MSE of 0.11 and a negative R^2 of -0.43, highlighting the difficulty of adapting across different battery manufacturers and operating temperatures. Our proposed MMD+EVI approach significantly improves the SOC prediction performance, achieving an average MSE of 0.02 and an R^2 of 0.77. In contrast, the standard MMD method, which does not leverage uncertainty guidance, results in a higher MSE of 0.04 and a lower R^2 of 0.53. The incorporation of uncertainty guidance in MMD+EVI results in a 45% relative improvement in R^2 compared to the standard MMD method. Other methods such as CORAL and DAN exhibit limited improvements over the baseline, with CORAL performing similarly to source-only (R^2 of -0.39) and DAN offering slight gains (R^2 of -0.23). JDOT shows strong alignment capabilities, closely following MMD+EVI with an R^2 of 0.73. DARE-GRAM also performs well, achieving an R^2 of 0.57, although it does not achieve the level of performance achieved by MMD+EVI.

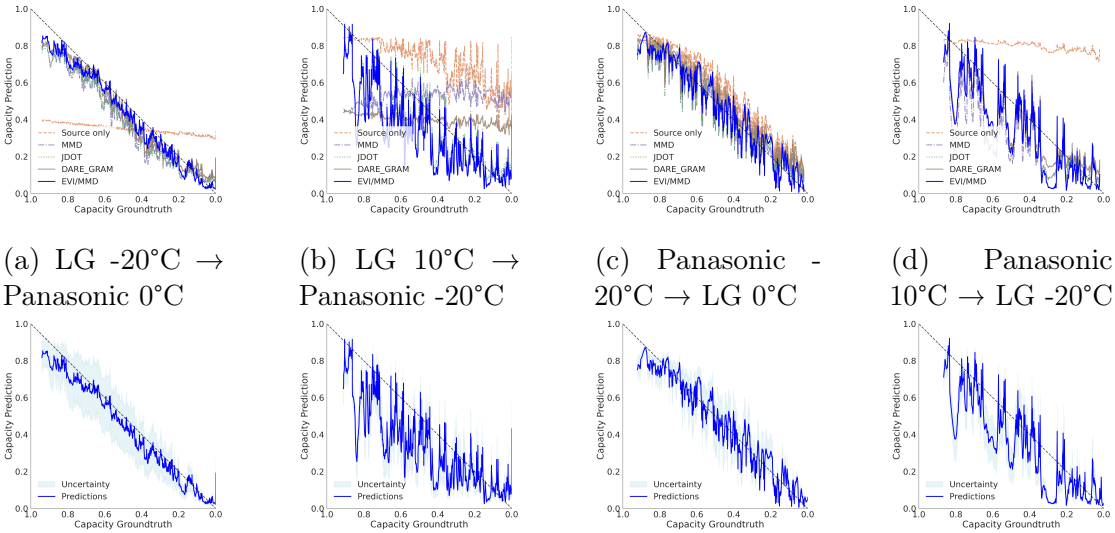
Table 4.5: Results for source Panasonic and target LG

Transfer	Source only		CORAL		DAN		JDOT		DARE-GRAM		MMD		MMD+EVI	
Task	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2
-20 → -10	0.03	0.47	0.03	0.47	0.02	0.69	0.03	0.54	0.03	0.53	0.04	0.43	0.02	0.62
-20 → 0	0.02	0.72	0.02	0.72	0.01	0.87	0.02	0.72	0.02	0.77	0.02	0.74	0.01	0.82
-20 → 10	0.01	0.79	0.01	0.79	0.01	0.79	0.02	0.71	0.02	0.75	0.02	0.77	0.01	0.83
-20 → 25	0.05	0.33	0.05	0.33	0.06	0.08	0.02	0.71	0.02	0.75	0.01	0.85	0.02	0.77
-10 → -20	0.01	0.90	0.01	0.90	0.01	0.83	0.01	0.89	0.01	0.88	0.01	0.90	0.01	0.86
-10 → 0	0.03	0.59	0.03	0.59	0.01	0.90	0.02	0.74	0.02	0.66	0.02	0.67	0.02	0.62
-10 → 10	0.01	0.78	0.01	0.78	0.01	0.83	0.02	0.76	0.01	0.84	0.01	0.82	0.01	0.78
-10 → 25	0.04	0.43	0.04	0.44	0.05	0.32	0.02	0.76	0.01	0.82	0.01	0.86	0.01	0.87
0 → -20	0.02	0.66	0.02	0.66	0.12	-1.11	0.02	0.74	0.01	0.78	0.01	0.78	0.01	0.76
0 → -10	0.01	0.83	0.01	0.83	0.02	0.76	0.01	0.91	0.00	0.92	0.01	0.92	0.01	0.88
0 → 10	0.02	0.68	0.02	0.68	0.00	0.93	0.01	0.82	0.01	0.80	0.01	0.81	0.01	0.78
0 → 25	0.02	0.69	0.02	0.69	0.16	-1.35	0.01	0.82	0.01	0.86	0.01	0.92	0.01	0.91
10 → -20	0.20	-2.44	0.20	-2.43	0.09	-0.62	0.02	0.62	0.02	0.63	0.04	0.28	0.03	0.50
10 → -10	0.12	-0.85	0.12	-0.83	0.09	-0.33	0.01	0.85	0.01	0.88	0.01	0.87	0.01	0.85
10 → 0	0.05	0.29	0.05	0.30	0.05	0.29	0.00	0.94	0.00	0.96	0.00	0.94	0.00	0.94
10 → 25	0.01	0.89	0.01	0.89	0.00	0.94	0.00	0.95	0.00	0.94	0.00	0.95	0.00	0.94
25 → -20	0.25	-3.26	0.25	-3.25	0.16	-1.73	0.05	0.21	0.02	0.58	0.07	-0.13	0.03	0.49
25 → -10	0.22	-2.32	0.22	-2.32	0.16	-1.50	0.02	0.76	0.03	0.52	0.02	0.65	0.01	0.82
25 → 0	0.19	-1.83	0.19	-1.82	0.13	-0.97	0.01	0.88	0.02	0.75	0.01	0.88	0.01	0.90
25 → 10	0.12	-0.84	0.12	-0.79	0.08	-0.20	0.01	0.92	0.00	0.97	0.00	0.93	0.00	0.93
Average	0.07	-0.13	0.07	-0.12	0.06	0.02	0.02	0.76	0.01	0.78	0.02	0.74	**0.01**	**0.79**

Panasonic to LG Transfer: Table 4.5 presents the reverse adaptation scenario, where the baseline achieves a moderate performance with an MSE of 0.07 and an R^2 of -0.13, indicating a less severe domain shift compared to the LG to Panasonic direction. Our proposed MMD+EVI method excels in this setting, attaining the lowest MSE (0.01) and the highest R^2 of 0.79, which constitutes a 6.8% relative improvement over the standard MMD method that achieves an R^2 of 0.74. JDOT remains competitive with an R^2 of 0.76, while DARE-GRAM closely follows with an R^2 of 0.78. Additionally, CORAL and DAN slightly outperform the baseline, achieving R^2 values of -0.12 and 0.02, respectively.

Highlights: Our proposed MMD+EVI method excels in scenarios involving extreme temperature alternative (e.g., shifting from 25°C to -20°C), as it handles these larger domain shifts more effectively than other approaches. For example, in the LG to Panasonic transfer at 25°C → -20°C, MMD+EVI achieves an R^2 of 0.10, outperforming standard MMD (-0.14), CORAL (-3.15), and DAN (-0.67). A similar trend is observed in the Panasonic to LG 25°C → -20°C transfer, where MMD+EVI achieves an R^2 of 0.49, with the standard MMD, CORAL, and DAN achieving -0.13, -3.25, and -1.73, respectively. These results underline the importance of handling extreme domain shifts and the effectiveness of uncertainty-guided adaptation in mitigating these challenges.

Qualitative results: Figure 4.7 illustrates the qualitative performance of our proposed method and baselines on various transfer tasks in battery SoC prediction.



(a) LG -20°C → Panasonic 0°C

(b) LG 10°C → Panasonic -20°C

(c) Panasonic -20°C → LG 0°C

(d) Panasonic 10°C → LG -20°C

(e) LG -20°C → Panasonic 0°C (MMD+EVI)

(f) LG 10°C → Panasonic -20°C (MMD+EVI)

(g) Panasonic -20°C → LG 0°C (MMD+EVI)

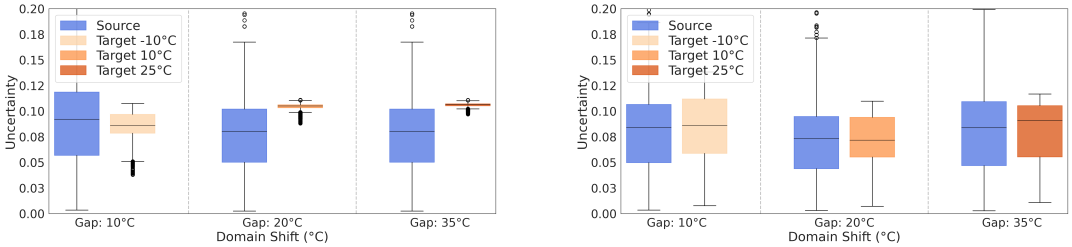
(h) Panasonic 10°C → LG -20°C (MMD+EVI)

Figure 4.7: Prediction trajectories for battery SoC across different transfer tasks. (a-d) Comparison of various methods. (e-h) MMD+EVI predictions with uncertainty estimates (shaded area represents 2σ).

Figures 4.7(a-d) illustrate the prediction trajectories of different methods across various transfer tasks. Our proposed MMD+EVI approach consistently achieves closer alignment with the ground truth compared to other approaches, particularly in challenging scenarios involving large temperature differences (b and d). Additionally, Figures 4.7(e-h) showcase the MMD+EVI predictions with their associated uncertainty estimates. A clear trend is observed in the uncertainty behavior: it is initially high at the start of the battery's lifecycle and progressively decreases as the battery ages. This pattern aligns with the expectation that prediction confidence increases with the accumulation of more data over the battery's lifespan, enhancing the model's reliability over time.

The effectiveness of our uncertainty-guided approach is further illustrated in Figure 4.8, which compares uncertainty distributions between source and target domains.

Figure 4.8(a) shows a significant discrepancy in uncertainty distributions between source and target domains when no alignment is applied. In contrast, Figure 4.8(b) demonstrates that our UGA-Feature alignment method effectively minimizes this discrepancy, resulting in highly comparable uncertainty distributions across both domains. This alignment indicates that our proposed approach successfully calibrates the model's confidence under varying operating conditions, thereby enhancing its cross-domain performance.



(a) DER without alignment

(b) UGA-Feature alignment

Figure 4.8: Uncertainty distribution comparison between source (-20°C) and target domains of increasing temperature for LG batteries, (a) without alignment and (b) with UGA-Feature alignment.

4.7 Ablation study

To further investigate the impact of different uncertainty quantification methods within our approach, we conducted an ablation study on the dSprites dataset.

Table 4.6: Ablation study on dSprites regression tasks comparing different uncertainty quantification methods. Results are presented as the sum of MAE across three regression targets. ×indicates no uncertainty used (MSE loss for training), GP denotes Gaussian Processes, and DER represents Deep Evidential Regression.

Method	Uncertainty	C → N	C → S	N → C	N → S	S → C	S → N	Avg
MMD (Long et al., 2015)	×	0.70	0.77	0.12	0.50	0.06	0.11	0.38
UGA-Post. MMD	GP	0.05	0.21	0.07	0.55	0.03	0.04	0.16
UGA-Post. MMD	DER	0.04	0.10	0.04	0.16	0.03	0.03	0.07
UGA-Feat. MMD	GP	0.05	0.19	0.05	0.40	0.03	0.03	0.13
UGA-Feat. MMD	DER	**0.04**	**0.10**	**0.02**	**0.16**	**0.02**	**0.02**	**0.06**

The results in Table 4.6 clearly demonstrate the advantages of incorporating uncertainty in the domain adaptation process. Both Gaussian Process (GP) and Deep Evidential Regression (DER) uncertainty frameworks significantly improve upon the standard MMD approach across all transfer tasks. However, DER consistently outperforms GP, particularly in challenging transfer scenarios such as N → S and S → N. This superior performance of DER can be attributed to its ability to model both aleatoric and epistemic uncertainty, providing a more comprehensive and robust guide for alignment for domain adaptation tasks. Furthermore, the results consistently show that feature alignment (UGA-Feat.) outperforms posterior alignment (UGA-Post.), regardless of the uncertainty quantification method used.

4.8 Conclusion

This paper introduces Uncertainty-Guided Alignement (UGA), a novel approach for Unsupervised Domain Adaptation in Regression tasks. By integrating predictive uncertainty into the feature alignment process, UGA effectively addresses the challenges arising from the interdependencies among regression features. We provide a theoretical foundation for expanding the embedding space to incorporate input information through uncertainty estimation, demonstrating how this approach overcomes the limitations of traditional feature alignment methods in regression contexts. Through comprehensive experiments on multiple computer vision benchmarks and a real-world battery life prediction task, we show that UGA consistently outperforms state-of-the-art methods across 52 diverse transfer tasks. Incorporating uncertainty into the alignment process significantly improves adaptation performance, particularly in challenging scenarios with substantial domain gaps, such as battery life prediction under extreme temperature variations. Moreover, UGA's well-calibrated uncertainty estimates offer valuable insights into the severity of domain shifts, facilitating more informed decision-making in critical applications such as prognostics and health management. We also demonstrate that while uncertainty estimation alone can effectively quantify the magnitude of domain shift, our proposed alignment method further improves adaptation performance.

Although UGA assumes similar target label ranges between domains, potentially limiting its applicability in certain scenarios, future research could explore leveraging uncertainty for test-time adaptation in regression tasks.

5 IM-Context: In-Context Learning for Imbalanced Regression Tasks

This chapter corresponds to the published article[I]:
Nejjar, Ismail, Faez Ahmed, and Olga Fink. "IM-Context: In-Context Learning for Imbalanced Regression Tasks." *Transactions on Machine Learning Research*, 2024.

Abstract: Regression models often fail to generalize effectively in regions characterized by highly imbalanced label distributions. Previous methods for deep imbalanced regression rely on gradient-based weight updates, which tend to overfit in underrepresented regions. This paper proposes a paradigm shift towards in-context learning as an effective alternative to conventional in-weight learning methods, particularly for addressing imbalanced regression. In-context learning refers to the ability of a model to condition itself, given a prompt sequence composed of in-context samples (input-label pairs) alongside a new query input to generate predictions, without requiring any parameter updates. In this paper, we study the impact of the prompt sequence on the model performance from both theoretical and empirical perspectives. We emphasize the importance of localized context in reducing bias within regions of high imbalance. Empirical evaluations across a variety of real-world datasets demonstrate that in-context learning substantially outperforms existing in-weight learning methods in scenarios with high levels of imbalance.

5.1 Introduction

Imbalanced data distributions, common in the real world, pose significant challenges to the generalization of conventional deep learning models due to variance across minority labels and bias toward majority labels (Y. Zhang, Kang, et al., 2023). While numerous studies have addressed learning from imbalanced data, most of them focus on classification

[I]Please note, this is the author's version of the manuscript published in Transactions on Machine Learning Research. Changes resulting from the publishing process, namely editing, corrections, final formatting for printed or online publication, and other modifications resulting from quality control procedures may have been subsequently added.

tasks (Z. Wang et al., 2024; Y. Yang & Xu, 2020; Z. Yang et al., 2024; Y. Zhang et al., 2022). Recent work emphasizes that the continuity of the label space, and the relationship between features and labels (Shin et al., 2022), make imbalanced regression a fundamentally different problem from imbalanced classification (J. Ren et al., 2022; Y. Yang et al., 2021). Imbalanced regression problems are critical in many fields. For instance, in computer vision, age estimation datasets are often imbalanced, with fewer samples for younger age groups due to legal and ethical limitations, while older age groups are underrepresented as the population naturally declines with age. Similarly, in engineering design, the distribution of designs is often skewed, with the most desirable characteristics at the tail end of the distribution. This means that the region of greatest interest lies in the minority samples of the distribution. Enhancing model performance in these low-data regions is essential for achieving more accurate and reliable outcomes across these applications.

To address the specificities of deep imbalanced regression, two main approaches have been proposed: sample re-weighting and embedding space regularization. Sample re-weighting techniques, as those proposed by Y. Yang et al., 2021 and Steininger et al., 2021, apply kernel density estimation to smooth the label distribution. This method leverages local dependencies by enforcing similarity between nearby labels. Alternatively, embedding space regularization techniques, such as ranking (Y. Gong et al., 2022) and contrastive learning (Keramati et al., 2024), preserve both local and global dependencies by leveraging label similarity rankings within the feature space. These methods, commonly referred to as **in-weight learning**, rely on gradient updates to adjust model weights, assuming that models can effectively generalize from limited data points in tail regions. However, with inherently less information in minority regions of the label distribution, these models are prone to overfitting samples from these regions (K. Cao et al., 2019).

A different learning paradigm, the recently proposed **in-context learning**, has the potential to overcome the limitations of **in-weight learning** in minority regions. In-context learning refers to the ability of a model to rapidly generalize to the new concepts on which it has not been previously trained (Bietti et al., 2024; Reddy, 2023), using only a few examples — often referred to as few-shot (Brown et al., 2020; Webb et al., 2023). This approach is particularly notable because it does not require any parameter updates on the model's weights.

Recent work has explored **in-context learning for regression tasks**. For instance, the recently proposed 'Prior-data Fitted Networks' (Müller et al., 2022) were trained to mimic Bayesian inference, similar to Gaussian processes and Bayesian neural networks, and applied to Bayesian optimization (Müller et al., 2023). Additionally, it has been empirically shown that transformers can learn from scratch using randomly sampled data from function classes, achieving performance comparable to that of the optimal least squares estimator in linear function scenarios (Garg et al., 2022). Moreover, transformers can even learn from more complex function classes. Interestingly, Chan et al., 2022

highlights that in-context learning particularly emerges in training distributions with properties such as long-tailedness. *Unlike **in-weight learning**, which needs to be trained from scratch on specific datasets, an **in-context learning** model is pre-trained on diverse, often synthetic, data and can adapt to different tasks using context examples. This means a single model can perform multiple tasks without needing a separate model for each one.*

In this paper, we introduce In-Context Learning for IMbalanced Regression Tasks, referred to as IM-Context, to explore this new learning paradigm from an imbalanced regression perspective. We present the counter-intuitive finding that increasing the context size can negatively impact imbalanced regression performance in minority regions. We establish a theoretical error bound for the expected error when in-context learning is applied to imbalanced regression, demonstrating that a large number of context examples can theoretically approximate the true predictive distribution. However, in the case of imbalanced regression, using the entire training set as context biases the model toward the majority region, leading to inferior performance on rare labels. As an alternative, we propose a localized approach where only the 'closest' in-context samples to a new query are used. This method mitigates bias and reduces the sequence length of context samples, thereby reducing the memory requirement for each inference.

In this localized setting, our findings reveal that in regions with dense label distributions (where the majority of samples are located), the expected error remains tightly bounded and relatively flat, indicating insensitivity to the number of context samples. Conversely, in regions with sparse label distributions (minority of samples), we observe a shift in behavior where more context samples hurt the performances in the minority region: the bound widens as more context samples are added. This highlights the need for locality, as indiscriminately adding more context examples can skew the context label distribution and worsen predictions. However, retrieving semantically relevant examples as context in practice is challenging, particularly for imbalanced datasets, because the probability of retrieving samples from the majority region is higher. To address this challenge, we propose creating a second training set, where the number of samples in each region is inversely proportional to its representation in the original set: majority regions have fewer points, while minority regions are overrepresented, we refer to it as inverse density dataset. Given a new query sample, we retrieve neighboring samples from both training sets. The proposed strategy, referred to as Augmented, serves to mitigate potential biases toward the majority region; N.B. alternative sampling strategies could be applied.

To empirically validate these findings, we use two pre-trained models (Garg et al., 2022; Müller et al., 2023), and evaluate our methodology across eight imbalanced regression tasks. This comprises three benchmark datasets covering two facial age estimation tasks, one text similarity task, and six tabular datasets with different degrees of imbalance. Experiments in several real-world datasets show that our in-context learning approach consistently outperforms state-of-the-art in-weight learning methods, particularly in regions with high imbalance.

5.2 Related works

Imbalanced regression: The continuity in label space makes imbalanced regression different from imbalanced classification. One potential option is to discretize the continuous label space and apply methods typically used in imbalanced classification. However, this requires specifying a minimum acceptable error threshold for the discretization process (J. Ren et al., 2022). Moreover, the continuity of the label space can provide additional information about the relationships between data instances, contrasting with the categorical distinctions in classification tasks (Y. Yang et al., 2021).

Various strategies have been proposed to refine how relationships in both label and feature spaces are handled in regression. Label Distribution Smoothing (LDS) (Y. Yang et al., 2021) and DenseLoss (Steininger et al., 2021) employed kernel density estimation to derive the 'true' label density from empirical data, thereby smoothing and reweighting the data labels. Feature Distribution Smoothing (FDS) (Y. Yang et al., 2021) applied similar principles to the feature space and performs distribution smoothing by transferring the first two feature statistics between nearby target labels. A different direction to overcome deep imbalanced regression is to learn additional tasks to regularise the model feature representation. Different types of tasks have been proposed to capture the relationships between features and labels at local and global levels (Y. Gong et al., 2022; Keramati et al., 2024; Z. Wang & Wang, 2024). For instance, Y. Gong et al., 2022 proposed ranking similarity regularization, while Keramati et al., 2024 adapted contrastive learning to regression tasks by improving the continuity of the feature space. In a recent study, Z. Wang and Wang, 2024 leveraged deep evidential regression (Amini et al., 2020)(a framework that learns to predict uncertainty in a single forward pass) and proposed leveraging data with similar target labels to compute the variational distribution of the latent representation, imposing probabilistic reweighting on imbalanced data.

However, these deep learning methods predominantly relied on **in-weight learning**, wherein achieving high-quality representations in a long-tailed setting is difficult because the features in the minority can be easily overfitted. This issue was highlighted by J. Ren et al., 2022, as the Mean Square Error (MSE) loss used for training in imbalanced regression introduces biases to the majority region. Alternative learning strategies may potentially help overcome the limitation of in-weight learning in minority regions.

In-context Learning (ICL) in regression: ICL, based on Transformer architecture, enables a pre-trained model to learn a new task with minimal examples (Brown et al., 2020; Vaswani et al., 2017). The remarkable success of transformers and their ability to do in-context learning shown in natural language processing has inspired a line of research exploring the algorithmic power of transformers. Notably, Garg et al., 2022 investigated transformers in in-context regression tasks, ranging from learning linear regression to more complex function classes. Subsequent works have suggested that the attention mechanism in the transformer may mimic various optimization algorithms such

as gradient (Ahn et al., 2024; Akyurek et al., 2023; Y. Bai et al., 2023; Mahankali et al., 2024; Von Oswald et al., 2023).

From a Bayesian perspective, S. M. Xie et al., 2022 suggested that in-context learning is a form of implicit Bayesian inference. Building on this idea, Müller et al., 2022 trained transformer models using 'Prior Fitted Networks', where data sampled from a prior distribution enables these models to approximate the posterior predictive distribution during inference for tabular data (van Breugel & van der Schaar, 2024). A theoretical foundation for these models was further developed by Nagler, 2023 with a focus on classification tasks. However, the practical implementation of ICL in regression, particularly under conditions of extreme data imbalance or when dealing with real-world data, remains underexplored.

5.3 Methodology

5.3.1 Problem Setting

We address a regression problem where an input feature $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$ is mapped to a label $y \in \mathcal{Y} = \mathbb{R}$. More specifically, in the imbalanced regression scenario, the training set $D_s = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_s}$ and the test set $D_t = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_t}$ are sampled from distinct joint distributions, $p_s(\mathbf{x}, y)$ and $p_t(\mathbf{x}, y)$, referred to as the source and target distributions, respectively. In our setup, the source distribution typically exhibits a skewed label distribution $p_s(y)$, while the target distribution is uniformly spread across the range of target values $p_t(y)$. The goal is to improve the prediction on the tail regions.

In-Context Learning Consider a transformer model f_θ which is capable of in-context learning. This capability is demonstrated when the model can accurately approximate the output y_{query} for a new query input $\mathbf{x}_{query}$ based on a sequence of in-context examples. This sequence, called a prompt, is given by $(\mathbf{x}_1, y_1, \ldots, \mathbf{x}_n, y_n, \mathbf{x}_{query})$. The training set D_s can serve as a basis for the prompt sequence. Given a new query sample $\mathbf{x}_{query}$ from the test set D_t, the model predicts $\hat{y}_{query} = f_\theta(\mathbf{x}_{query} \mid D_s)$. In the following sections, we evaluate the circumstances under which $\hat{y}_{query}$ can be accurately similar to the true label y_{query}.

5.3.2 Convergence

In the following sections, we delve into the theoretical analysis of the model convergence in the general case. We show that in imbalance scenarios, using the entire training set can introduce bias, primarily due to the overrepresentation of the majority samples. We introduce a localized version aimed at alleviating this bias and provide a theoretical bound for the expected prediction error, showing that the behavior differs depending on whether points are from majority or minority regions.

General convergence

Given a pre-trained transformer capable of in-context learning f_θ, let us consider the case where we sample points D_n from a realization of the random training distribution D_s. We can decompose the model's Expected Prediction Error for a new sample $(\mathbf{x}, y)$ as :

$$\begin{aligned}\mathrm{EPE}_{f_\theta}(x) &= \mathbb{E}_{D_n}\left[(f_\theta(\mathbf{x} \mid D_n) - \mathbb{E}[y \mid \mathbf{x}])^2\right] \\ &= \mathbb{E}_{D_n}[\Big((f_\theta(\mathbf{x} \mid D_n) - \mathbb{E}_{D_n}[f_\theta(\mathbf{x} \mid D_n)]) + (\mathbb{E}_{D_n}[f_\theta(\mathbf{x} \mid D_n)] - \mathbb{E}[y \mid \mathbf{x}])\Big)^2] \\ &= \mathrm{Var}_{D_n}[f_\theta(\mathbf{x} \mid D_n)] + \left(\mathrm{Bias}^2_{D_n}[f_\theta(\mathbf{x} \mid D_n)]\right) + \sigma^2\end{aligned} \tag{5.1}$$

Proposition 5.3.1 ***c-Lipschitz Continuity:*** *Consider an infinitely large training set D_s, from which subsets D_n and D'_n are independently sampled. Assume that the label noise is constant. Given the observation from Garg et al., 2022 that error decreases as more samples are given as context, we can consider that the model f_θ, is c-Lipschitz, with $c = (c_1, \ldots, c_n) \in \mathbb{R}^n_+$ where $c_i = \delta i^{-\alpha}$ for each $i \in \{1, \ldots, n\}$, with δ as a positive constant and $\alpha > 0.5$:*

$$\left|f_\theta(\mathbf{x} \mid D_n) - f_\theta(\mathbf{x} \mid D'_n)\right| \leq \sum_{i=1}^{n} c_i \mathbf{1}_{\{\mathbf{x}_i \neq \mathbf{x}'_i\}} \tag{5.2}$$

Theorem 5.3.2 ***Application of McDiarmid's Inequality McDiarmid et al., 1989:*** *Given that f_θ is c-Lipschitz under the defined conditions, for any $t > 0$, the tail probability is bounded by:*

$$\Pr(|f_\theta(\mathbf{x} \mid D_n) - \mathbb{E}[f_\theta(\mathbf{x} \mid D_n)]| \geq t) \leq 2\exp\left(-\frac{2t^2}{\|c\|_2^2}\right), \tag{5.3}$$

Since $\|c\|_2^2 = \sum_{i=1}^{\infty}(\delta i^{-\alpha})^2$, which converges given $\alpha > 0.5$, by the Borel-Cantelli lemma, $f_\theta(x \mid D_n)$ convergences to the expected prediction almost surely:

$$f_\theta(\mathbf{x} \mid D_n) \xrightarrow{\text{a.s.}} \mathbb{E}[f_\theta(\mathbf{x} \mid D_n)] \quad \text{as} \quad n \to \infty, \tag{5.4}$$

Variance : Under the assumption of **Proposition 5.1**, the variance of the model's prediction should decrease as more samples are provided.

Proposition 5.3.3 ***Convergence to training label distribution:*** *If we assume that the attention blocks in model f_θ are 'mimicking' gradient descent steps (Ahn et al., 2024; Von Oswald et al., 2023) on the context example D_n, the model's behavior should resemble that of a neural network trained with MSE loss. Consequently, we can expect:*

$$\mathbb{E}_{D_n}\left[f_\theta(\mathbf{x} \mid D_n)\right] \to \mathbb{E}_{p_s}[y \mid \mathbf{x}] \quad \text{as} \quad n \to \infty \tag{5.5}$$

Bias : While **Proposition 5.3.3** suggests that the model converges to the conditional probability of p_s, the behavior of f_θ crucially depends on p_s. If the samples from D_n are independently and identically distributed, f_θ can be considered an unbiased estimator of $\mathbb{E}_{p_s}[y \mid \mathbf{x}]$. Conversely, if D_n contains biased or non-representative samples, as often occurs in imbalanced regression scenarios, the model's predictions will inherently reflect these biases, and the model will tend to underestimate rare labels.

Local convergence

Imbalanced regression : A simple approach to reduce model bias in the tail regions of the label distribution is to select a finite set of neighboring samples of $\mathbf{x}$ from D_n. Selecting only the neighboring sample for context is equivalent to adjusting the distribution of labels $p_s(y \mid \mathbf{x})$ so that it corresponds locally to $p_t(y \mid \mathbf{x})$. This reflects the proportional relationship $\frac{p_s(y|\mathbf{x})}{p_t(y|\mathbf{x})} \approx \frac{p_s(y)}{p_t(y)}$.

While previous propositions ensure convergences for infinitely large datasets, it is also important to consider the expected prediction error in the case of small and finite numbers of context examples. Under this condition, we can assume that, in the worst case, the model will simply average the labels of the context samples. To the best of our knowledge, there is no existing work that specifically addresses this issue in the context of imbalanced regression with k-NN. This assumption is in line with previous works, for example Garg et al., 2022 showed that transformers always outperform sample averaging, while Bhattamishra et al., 2023 demonstrated that a frozen GPT-2 can mimic the nearest neighbor algorithm. Formally, with k samples from D_n closest to $\mathbf{x}$, denoted as D_k, the expected error bound is given by:

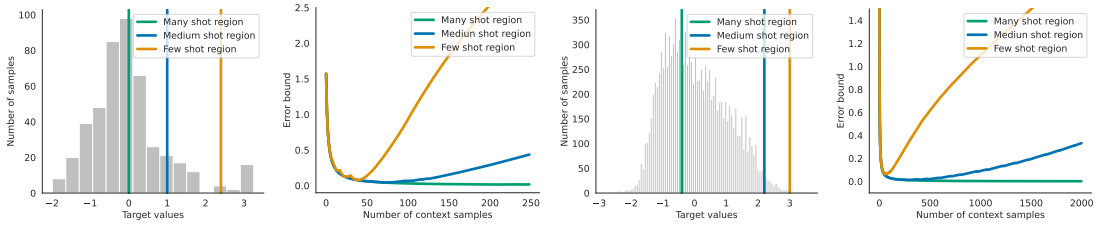
$$\mathbb{E}_{D_k}\left[(\mathbb{E}[y \mid \mathbf{x}] - f_\theta(\mathbf{x} \mid D_k))^2\right] \leq \mathbb{E}_{D_k}\left[(\mathbb{E}[y \mid \mathbf{x}] - \tilde{f}(\mathbf{x} \mid D_k))^2\right] \tag{5.6}$$

where $\tilde{f}(\mathbf{x} \mid D_k) = \frac{1}{k}\sum_{i=1}^{k} y_i$ represents the average of in-context labels, with $k > 1$. The associated error can be decomposed following Hastie et al., 2009 into:

$$\mathbb{E}_{D_k}\left[(\mathbb{E}[y \mid \mathbf{x}] - f_\theta(\mathbf{x} \mid D_k))^2\right] \leq \underbrace{\left(y - \frac{1}{k}\sum_{i=1}^{k} y_i\right)^2}_{\text{Bias}} + \underbrace{\frac{\sigma^2}{k}}_{\text{Variance}} + \sigma^2 \tag{5.7}$$

In the case of imbalanced regression, the variance term depends on the number of selected neighbors k and should decrease as more context examples are selected, similar to before. The bias for the minority regions will increase with k as the averaging estimator will

predict the mean of the context examples, which is also in agreement with the previous propositions. Figure 5.1 illustrates this behavior by plotting the expected error for data points across different regions, for the Boston and AgeDB datasets. For points in the majority region, the expected error decreases as more context examples are given. **However, for points in the tail region, the error initially decreases but then exhibits a U-shaped curve as k increases.** Thus, for minority samples, the error bound is tighter for a small number of selected neighbors, further emphasizing the need for localized context.



(a) Boston dataset distribution and Error bound (b) Age dataset distribution and Error bound

Figure 5.1: Training distribution of two datasets: Boston (a) and AgeDB (b). The empirical expectations of errors are computed assuming ideal retrieval of neighbors for a new query sample (i.e., retrieving samples that are semantically relevant in terms of the target variable, rather than just close in the input space). The behavior of examples from different shot regions varies distinctly with the number of context examples. In the many-shot regions, the error bound stabilizes as more examples are provided, whereas in the few-shot regions, additional context examples lead to an increase in the error bound.

Empirical validation

To validate the error bound in practice, we compared the results obtained on two datasets: Boston and AgeDB, across three regions: many, medium, and few-shot regions. These regions are defined based on the number of training samples per label in each discretized bin, with 'Few' having fewer than 20 samples, 'Median' between 20 and 100 samples, and 'Many' exceeding 100 samples.

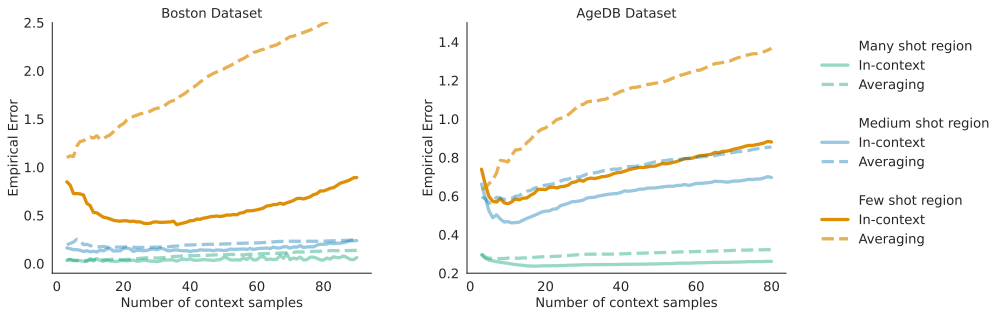


Figure 5.2: Empirical Error: Averaging vs In-context learning (using GPT-2 model from (Garg et al., 2022)).

For each test sample $\mathbf{x}_{query}$, we retrieve its k nearest neighbors $(\mathbf{x}_1, ..., \mathbf{x}_k)$ from the training set. Throughout the document, we use cosine similarity as a measure for retrieving neighboring points to the query point. Figure 5.2 empirically validates on the two datasets that in-context learning can perform better than simply averaging the context labels. Interestingly, for the few shot regions in both datasets, we observe that the empirical error of averaging increases rapidly with the number of context examples. This underscores the challenge of retrieving neighboring samples that are most useful for prediction in this region, even when using a small number of context examples. As highlighted by Steck et al., 2024, samples that are close in feature space according to a given distance metric (e.g., cosine similarity) do not necessarily have similar target values. In imbalanced datasets, this discrepancy between proximity in feature space and similarity in target values makes it difficult to select representative samples for the region, as the difference in density makes it more likely to retrieve samples from the majority region.

5.3.3 Retrieving Neighbours

Retrieving representative samples as context in practice is challenging, particularly for imbalanced datasets, because the probability of retrieving samples from the majority region is higher. To address this challenge, we propose creating a second training set, where the number of samples in each region is inversely proportional to its representation in the original set — majority regions have fewer points, while minority regions are overrepresented, we refer to this dataset as inverse density. For each new query sample, we retrieve $k = k'_s + \tilde{k}_s$ neighboring examples from both (1) the training set (k'_s neighbors) and (2) and inverse density dataset ($\tilde{k}_s$ neighbors), to ensure balanced representation and reduce the risk of bias toward the majority region. We refer to this version as augmented. In Figure 5.3, we compare the performance of in-context learning by retrieving neighboring context examples from the original training dataset and the augmented version. We can clearly see that this strategy decreases the error on both medium and few-shot regions, and accentuates the U-shape curve, as expected theoretically. Other resampling strategies, such as Smoter and undersampling, are discussed in the ablation.

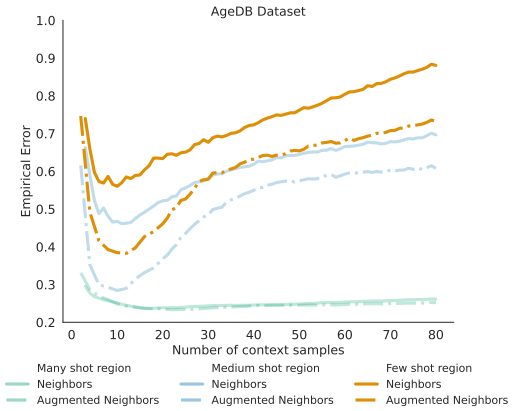


Figure 5.3: Impact of neighbors retrieval on performances (using GPT-2).

5.4 Experiments

5.4.1 Experimental Setup

Datasets: We use three benchmark datasets curated by Y. Yang et al., 2021 specifically for imbalanced regression tasks: AgeDB-DIR, derived from the AgeDB dataset for age estimation Moschoglou et al., 2017; IMDB-WIKI-DIR, another age estimation dataset sourced from IMDB-WIKI Rothe et al., 2018; and STS-B-DIR, which measures text similarity between two sentences based on the Semantic Textual Similarity Benchmark A. Wang et al., 2018. Additionally, we use six tabular datasets, namely Boston Harrison and Rubinfeld, 1978, Concrete Yeh, 2007, Abalone Nash et al., 1995, Communities Redmond, 2009, Kin8nm, and an engineering design dataset: Airfoil W. Chen et al., 2019, with inherent imbalances. For these tabular datasets, we created a balanced test set to evaluate model performance, to assess model performance in scenarios with a small number of training samples, different feature sizes, and varying levels of imbalance.

Evaluation Metrics: Consistent with established protocols by Y. Yang et al., 2021, we assess model performance using Mean Absolute Error (MAE) and Geometric Mean (GM) for the AgeDB-DIR and IMDB-WIKI-DIR datasets. For the other datasets, we use Mean Squared Error (MSE) as the primary metric, consistent with Y. Yang et al., 2021. We present our results across four predefined shot regions—All, Many, Median, and Few—which categorize the subsets of the datasets based on the number of training samples available per label within each discretized bin. Specifically, the 'Few' category includes bins with fewer than 20 samples, 'Median' encompasses bins with 20 to 100 samples, and 'Many' refers to bins with over 100 samples per label.

Preprocessing of inputs for In-Context Learning: In this paper, we propose an in-context learning approach as opposed to in-weight learning, which means we do not train any models directly. For the benchmark datasets, we preprocess images and text into embeddings using the Hugging Face implementation of CLIP (Radford et al., 2021) and BERT ("all-mpnet-base-v2" model) (Reimers & Gurevych, 2019). Specifically, for each image from the training and testing datasets, we extract embeddings from the CLIP image encoder before pooling, resulting in 768-dimensional features. A similar process is applied to the STS-B-DIR dataset, where we use BERT's text embeddings for each sentence, also resulting in 768-dimensional features. For the tabular datasets, we use the feature representations directly as inputs to the transformer. Moreover, we consider two in-context learning models: the first from Garg et al., 2022, referred to as GPT2, which uses the GPT-2 architecture and is trained on non-linear data points; the second model from Müller et al., 2023, referred to as Prior Fitted Networks (PFN). They accept 20-dimensional and 18-dimensional inputs, respectively. When the number of dimensions exceeds the model's input size, we split the input feature representation into non-overlapping chunks and ensemble the predictions from each chunk. To ensure the robustness and reproducibility of our results, we conducted three separate experiments

using different random seeds, between 0 and 3. We report the average results over all three seeds to account for the randomness introduced by inverse density sampling for the second training set. In all experiments, for the GPT2 model from Garg et al., 2022, we retrieve $k'_s = \tilde{k}_s = 10$ nearest neighbors, for the PFN model we retrieve $k'_s = \tilde{k}_s = 15$, except for the IMDB dataset where k'_s was equal to 5. This choice is motivated by the preliminary experiments conducted in section 5.3.3 as observed in Figure 5.3 where the minimum error in all regions is achieved for 10 neighbors. An NVIDIA RTX 2080 GPU was used for all the experiments.

5.4.2 Experimental Results

AgeDB: In Table 5.1, we report the results obtained on the AgeDB-DIR benchmarks. We compared the best combination of previously reported results to our approach. Given the small size of the training dataset (12.2K images), we observe that in-context learning outperforms all in-weight learning baselines in all shot regions, surprisingly even in the many-shot region. In the few-shot setting, our localized version using the Garg et al., 2022 model achieves the lowest mean absolute error of 7.83, which is an improvement of 1.4 error points over the combination ConR+LDS+FDS+Ranksim proposed in Keramati et al., 2024. Furthermore, on the AgeDB dataset, in-weight learning achieves the highest performance on all eight shot-metrics combinations with the best MAE results of 6.05 on the overall test set.

Table 5.1: Main results for AgeDB-DIR. Best results are in **bold** and second-best results are underlined.

	Learning IWL	ICL								
Metrics			**MAE ↓**				**GM ↓**			
Shot			all	many	medium	few	all	many	medium	few
MSE loss	✓	×	7.77	6.62	9.55	13.67	5.05	4.23	7.01	10.75
LDS + FDS (Y. Yang et al., 2021)	✓	×	7.55	7.01	8.24	10.79	4.72	4.36	5.45	6.79
RankSim (Y. Gong et al., 2022)	✓	×	7.02	6.49	7.84	9.68	4.53	4.13	5.37	6.89
VIR (Z. Wang & Wang, 2024)	✓	×	6.99	6.39	7.47	9.51	4.41	4.07	5.05	6.23
ConR + LDS + FDS + RankSim (Keramati et al., 2024)	✓	×	6.81	6.32	7.45	9.21	4.39	3.81	5.01	6.02
PFN - localized (Müller et al., 2023) (Ours)	×	✓	6.58	**5.61**	8.49	10.49	4.29	**3.58**	6.30	8.19
GPT2 - localized (Garg et al., 2022) (Ours)	×	✓	**6.05**	5.67	**6.71**	**7.83**	**3.79**	3.59	**4.17**	**4.90**

IMDB: In Table 5.2, we report the results obtained on the IMDB-WIKI-DIR benchmarks. This is the largest dataset used in this study with 191.5K training samples. Both models (Garg et al., 2022; Müller et al., 2023) using our localized approach achieve the lowest error in the few-shot and medium-shot regions. The best-performing in-context model is (Müller et al., 2023), which achieves an MAE of 16.33 and 10.79, improving over state-of-the-art methods on age estimation by 34.63 and 1.02 error points, respectively. In the many-shot region, in-weight learning outperforms in-context learning due to the

Table 5.2: Main results for IMDB-WIKI-DIR. Best results are in **bold** and second-best results are underlined.

	Learning IWL	ICL								
Metrics			**MAE ↓**				**GM ↓**			
Shot			all	many	medium	few	all	many	medium	few
MSE loss	✓	×	8.06	7.23	15.12	26.33	4.57	4.17	10.59	20.46
LDS + FDS (Y. Yang et al., 2021)	✓	×	7.78	7.20	12.61	22.19	4.37	4.12	7.39	12.61
RankSim + FDS (Y. Gong et al., 2022)	✓	×	7.35	6.81	11.50	22.75	4.05	3.85	6.05	14.68
ConR + FDS (Keramati et al., 2024)	✓	×	7.29	6.90	12.01	21.72	4.02	3.83	6.71	12.59
VIR (Z. Wang & Wang, 2024)	✓	×	**7.19**	**6.56**	11.81	20.96	**3.85**	**3.63**	6.51	12.23
PFN - localized (Müller et al., 2023) (Ours)	×	✓	8.96	8.71	**10.79**	**16.33**	5.26	5.17	**6.00**	**9.42**
GPT2 - localized (Garg et al., 2022) (Ours)	×	✓	7.76	7.35	11.15	17.71	4.29	4.13	5.96	11.00

abundance of training samples (almost 150k samples), which facilitates more effective learning and better generalization.

Table 5.3: Main results for STS-B-DIR. Best results are in **bold**.

Metrics	**MSE ↓**			
Shot	all	many	medium	few
MSE loss	0.974	0.851	1.520	0.984
LDS + FDS	0.903	0.806	1.323	0.936
VIR	0.892	0.795	0.899	0.780
RankSim	0.865	0.876	0.867	0.670
PFN - localized	0.544	0.536	0.547	0.618
GPT2 - localized	**0.528**	**0.524**	**0.527**	**0.566**

STS: Table 5.3 presents the results on the STS-B-DIR dataset. In this text modality, both models using our localized method consistently and substantially improve the results over state-of-the-art methods in all regions, achieving the best MSE of 0.528 on the overall test set and 0.566 in the few-shot region.

Tabular: Table 5.4 shows the average results on six tabular datasets. While the previous results on the previous datasets rely on features extracted from pre-trained models, a direct comparison of in-weight learning vs in-context learning can be made using tabular datasets. The results clearly show that in-context learning outperforms different machine learning methods in the medium and few shot regions. Across the six datasets the proposed method using (Garg et al., 2022) model is almost consistently ranked first across all those datasets on the few shot region, and is highly competitive in comparison to other methods in the medium shot region.

5.5 Ablations

Representation learning. In our experiment with the age datasets, we used CLIP embeddings across all age datasets to assess the effectiveness of in-context learning compared to in-weight learning. The methods in Table 5.1 were trained to learn representations directly from the images, whereas we used pre-extracted CLIP embeddings. We trained a three-layer MLP with 256 neurons in the hidden layers using these embeddings. Addition-

Table 5.4: Average results on the Tabular datasets. Best results are in **bold**. (med.: medium)

	Learning IWL	ICL								
Metrics			**RMSE ↓**				**Rank ↓**			
Shot			all	many	medium	few	all	many	med.	few
KNN	×	✓	3.96 ± 0.23	1.72 ± 0.30	3.07 ± 0.48	6.41 ± 0.47	6.2	4.5	6.0	6.5
Linear Regression	×	✓	4.00 ± 0.25	2.10 ± 0.24	3.28 ± 0.47	6.18 ± 0.54	5.3	5.3	6.2	5.5
Decision Tree	✓	×	3.03 ± 0.32	1.90 ± 0.42	2.72 ± 0.63	4.29 ± 0.48	7.0	7.3	7.2	5.5
Gradient Boosting	✓	×	2.50 ± 0.07	1.26 ± 0.21	2.04 ± 0.17	4.05 ± 0.13	3.8	3.2	3.8	5.0
Random Forest	×	✓	2.57 ± 0.16	**1.22 ± 0.35**	2.07 ± 0.30	4.16 ± 0.35	3.5	**2.5**	3.5	4.3
Neural Networks	✓	×	2.49 ± 0.21	1.44 ± 0.25	2.13 ± 0.21	3.79 ± 0.51	2.7	3.2	3.2	2.6
PFN - localized (Ours)	×	✓	2.67 ± 0.19	1.37 ± 0.23	2.03 ± 0.27	4.38 ± 0.26	5.2	4.5	3.3	5.3
GPT2 - localized (Ours)	×	✓	**2.34 ± 0.19**	1.72 ± 0.35	**1.95 ± 0.22**	**3.31 ± 0.39**	**2.3**	5.5	**2.8**	**1.2**

ally, we conducted an experiment where we fine-tuned the GPT2 model on the AgeDB dataset. Results presented in Table 5.5 indicate that also in this scenario in-context learning consistently outperforms in-weight learning. Interestingly, the fine-tuned GPT2 model achieves almost similar performance to the in-context learning approach. These results suggest that in-context learning is a particularly effective solution for relatively small datasets with the presence of scarce data, such as AgeDB.

Table 5.5: Ablation results for AgeDB-DIR benchmark.

	Learning IWL	ICL								
Metrics			**MAE ↓**				**GM ↓**			
Shot			all	many	medium	few	all	many	medium	few
MLP	✓	×	6.67	5.97	8.22	9.01	4.29	3.86	5.48	5.84
GPT2 - Finetuned (Garg et al., 2022)	✓	×	6.14	5.73	**6.69**	8.09	4.01	3.70	4.59	5.39
GPT2 - localized (Garg et al., 2022) (Ours)	×	✓	**6.05**	**5.67**	6.71	**7.83**	**3.79**	**3.59**	**4.17**	**4.90**

All Context vs. Localized Approach. A key rationale for using the PFN model from Müller et al., 2023 is its lack of positional embedding. This allows for a direct comparison of the model's performance using the entire training set versus a localized version as context. Figure 5.4 presents results across the first five tabular datasets. The engineering dataset, which has more than 30k training samples, could not fit in memory and was therefore excluded from this ablation. The results show a significant performance improvement with the localized approach in in-context learning compared to the non-localized version across all regions, with a particularly significant improvement in the few-shot region.

Sampling strategy: In Section 5.3.3, we presented a strategy to mitigate the under-representation of minority points when retrieving neighbors. In this section, we compare different sampling strategies on the tabular dataset. The 'Vanilla' approach retrieves neighbors from the original training set. 'Downsampling' reduces the majority region,

creating a balanced training set and retrieving neighbors only from this balanced set. SMOTER (Branco et al., 2017) generates new synthetic data points for the minority region, and neighbors are retrieved exclusively from this augmented dataset. 'Inverse' uses a dataset where the number of samples in each region is inversely proportional to its representation in the original set. Lastly, our strategy involves creating an Augmented dataset to ensure balanced representation while preserving diversity in the majority region. As seen in Table 5.6, across the datasets, our method demonstrates the best trade-off for mitigating bias in all regions.

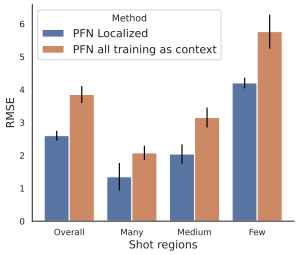


Figure 5.4: Localization vs all training set set as context

Metrics	**RMSE ↓**			
Shot	all	many	medium	few
Vanilla	2.43	**1.41**	1.99	3.70
Downsampling	2.65	2.42	2.38	3.17
Inverse	2.79	2.93	2.24	**3.13**
SMOTER	2.43	1.68	2.09	3.37
Augmented (Ours)	**2.34**	1.72	**1.95**	3.31

Table 5.6: Ablation results on the Tabular datasets using alternative sampling strategy.

5.6 Conclusion

In this work, we proposed the use of in-context learning to address the challenge of imbalanced regression, where a model can learn from context examples during inference without any additional training on the model weights. We show that a single in-context learning model can adapt to multiple tasks and improve performance in regions with high imbalance, unlike in-weight regression models, which often fail to generalize in minority regions and require specific training for each task. Through theoretical and empirical analyses, we highlighted the importance of localized context—using a small number of selected neighbors—in mitigating the bias inherent in imbalanced label distributions. Our evaluation across several benchmark datasets with diverse modalities revealed that in-context learning achieves superior results in regions of high imbalance.

Our comparative analysis demonstrated that two in-context learning models (Garg et al., 2022; Müller et al., 2023) based on Transformer architectures can achieve strong performance in imbalanced regression tasks for one-dimensional labels. While models are currently trained to predict a single value for a given query, future work will focus on extending our approach to multi-dimensional labels, such as depth estimation. Future work could also explore other model variants, and further investigate the factors contributing to performance differences between different in-context learning models for minority and majority regions.

6 Recall and Refine: A Simple but Effective Source-free Open-set Domain Adaptation Framework.

This chapter corresponds to the published article[I]:
Nejjar, Ismail, Hao Dong, and Olga Fink. "Recall and Refine: A Simple but Effective Source-free Open-set Domain Adaptation Framework.",2024.

Abstract: Open-set Domain Adaptation (OSDA) aims to adapt a model from a labeled source domain to an unlabeled target domain, where novel classes — also referred to as target-private unknown classes — are present. Source-free Open-set Domain Adaptation (SF-OSDA) methods address OSDA without accessing labeled source data, making them particularly relevant under privacy constraints. However, SF-OSDA presents significant challenges due to distribution shifts and the introduction of novel classes. Existing SF-OSDA methods typically rely on thresholding the prediction entropy of a sample to identify it as either a known or unknown class but fail to explicitly learn discriminative features for the target-private unknown classes. We propose Recall and Refine (RRDA), a novel SF-OSDA framework designed to address these limitations by explicitly learning features for target-private unknown classes. RRDA employs a two-step process. First, we enhance the model's capacity to recognize unknown classes by training a target classifier with an additional decision boundary, guided by synthetic samples generated from target domain features. This enables the classifier to effectively separate known and unknown classes. In the second step, we adapt the entire model to the target domain, addressing both domain shifts and improving generalization to unknown classes. Any off-the-shelf source-free domain adaptation method (e.g., SHOT, AaD) can be seamlessly integrated into our framework at this stage. Extensive experiments on three benchmark datasets demonstrate that RRDA significantly outperforms existing SF-OSDA and OSDA methods.

[I]Please note, this is the author's version of the manuscript that is currently under review.

6.1 Introduction

Unsupervised Domain Adaptation (UDA) (Ben-David et al., 2010; Ganin & Lempitsky, 2015; Long et al., 2015) adapts a model from a labeled source domain to an unlabeled target domain (Oza et al., 2023), effectively addressing the issue of domain shift where the source and target distributions differ. UDA strategies typically align feature distributions between domains using metric learning techniques (Kang et al., 2019; Long et al., 2015) or adversarial training (Ganin & Lempitsky, 2015; Y. Luo et al., 2019; Tzeng et al., 2017), and more recently, self-training approaches (Hoyer et al., 2023a; T. Sun et al., 2022; Zhu et al., 2023). Despite their success, most current domain adaptation approaches operate under the assumption of a shared label set between the source and target domains (i.e., $C_s = C_t$), referred to as Closed-set Domain Adaptation (Saenko et al., 2010). However, this assumption is often impractical in real-world scenarios.

In contrast, Open-set Domain Adaptation (OSDA) extends the target label space beyond that of the source domain (i.e., $C_s \subset C_t$) (H. Liu et al., 2019; Saito, Yamamoto, et al., 2018), thereby adding complexity to the DA task. OSDA aims to align target samples from known classes with those from the source domain while effectively identifying target samples belonging to categories not observed in the source domain, referred to as unknown classes (Bucci et al., 2020; Jang et al., 2022; Panareda Busto & Gall, 2017). Various criteria based on instance-level predictions have been proposed, including entropy-based (Feng et al., 2021; Saito et al., 2020) and confidence-based (Fu et al., 2020; Saito & Saenko, 2021) methods.

Additionally, privacy and legal considerations increasingly limit access to labeled source data for adaptation purposes. To address this, source-free adaptation methods (Fang et al., 2024) have emerged, enabling adaptation without reliance on labeled source data (Y. Kim et al., 2021; Kundu, Venkat, Babu, et al., 2020; R. Li et al., 2020). In this paper, we focus on Source-free Open-set Domain Adaptation (SF-OSDA), where only a pre-trained source model is available for knowledge transfer, without access to labeled source data. While some Source-free Domain Adaptation (SF-DA) methods have demonstrated effectiveness in addressing SF-OSDA for classification tasks (Liang et al., 2020; Wan et al., 2024; S. Yang et al., 2022), semantic segmentation (Choe et al., 2024), and graph applications (Y. Wang et al., 2024), they primarily focus on the semantics of known classes in the source domain, often overlooking the crucial aspect of novel-class semantics. These methods focus on segregating target samples with low entropy, categorizing them as known classes, and subsequently optimizing specific objectives such as entropy minimization or clustering. In this process, data points associated with known classes are prioritized, while those with high entropy are typically excluded from training, leading to a semantic disparity between the known and unknown classes.

To effectively adapt a pre-trained source model to a target domain facing both category and distribution shifts, we propose **R**ecall and **R**efine for **D**omain **A**daptation (**RRDA**)

for robust SF-OSDA. RRDA employs a two-step strategy. First, we propose to leverage the semantics of the *unknown* classes by introducing a novel target classifier with $K + K'$ decision boundaries. These boundaries extend the K classes from the source domain with K' additional classes for the *unknown* categories. To achieve this, synthetic samples are generated in the feature space from target domain features. These synthetic points are optimized to exhibit low entropy for *known* classes and high entropy for *unknown* classes, which are then clustered into K' categories. The synthetic data are used to refine the decision boundaries of the source classifier, enabling the target classifier to accommodate the unknown classes. In the second step, any off-the-shelf source-free domain adaptation method (e.g., SHOT (Liang et al., 2020), AaD (S. Yang et al., 2022)) can be integrated into our framework to adapt the entire model to the target domain. RRDA directly learns to classify target unknown classes. The framework introduces K' as a hyper-parameter, which we set to $K' = K$ for simplicity. Sensitivity analysis shows that performance improves with higher values of K', though results remain robust across a range of settings. Extensive experiments on three SF-OSDA benchmark datasets demonstrate the effectiveness of our approach, significantly outperforming existing methods.

6.2 Related Work

Unsupervised Domain Adaptation (UDA) aims to adapt a model originally trained on a labeled source domain to perform effectively in an unlabeled target domain. This adaptation process assumes access to data from both the source and target domains during training (Oza et al., 2023). UDA strategies often align feature distributions between domains using metric learning techniques (Kang et al., 2019; Long et al., 2015; Nejjar, Wang, & Fink, 2023b) or adversarial training across various spaces, including image input space (Murez et al., 2018; Pizzati et al., 2020), feature space (Ganin & Lempitsky, 2015), and output space (Y. Luo et al., 2019; Vu et al., 2019b). Additionally, various techniques incorporate pseudo-labeling or self-training algorithms (Dong et al., 2023; T. Sun et al., 2022; Yue et al., 2024), which generate pseudo-labels for unlabeled samples in the target domain. However, existing approaches assume that label spaces are identical across both domains, limiting their applicability in real-world scenarios.

Open-set Domain Adaptation (OSDA) addresses scenarios where the target domain may contain classes not present in the source domain (Dong, Chatzi, & Fink, 2024; Dong, Zhao, et al., 2024; G. Li et al., 2021; Panareda Busto & Gall, 2017). Various approaches have been proposed to tackle this challenge, including assigning target domain images to source categories while discarding unrelated target domain images (Panareda Busto & Gall, 2017), and using adversarial training to separate unknown target samples (Jang et al., 2022; Saito, Yamamoto, et al., 2018). Separate to Adapt (STA) approach (H. Liu et al., 2019) progressively separates unknown and known class samples using a coarse-to-fine weighting mechanism and proposes evaluating OSDA on diverse levels of openness. Rotation-based Open Set (ROS) (Bucci et al., 2020) explores the use of

self-supervised tasks such as rotation recognition for unknown class detection. (M. Jing et al., 2021) project features to a hyperspherical latent space to reject known samples based on angular distance. Adjustment and Alignment for Unbiased Open Set Domain Adaptation (ANNA) (W. Li et al., 2023) addresses semantic-level bias in OSDA by designing Front-Door Adjustment and Decoupled Causal Alignment modules. However, these approaches all assume the availability of labeled source data, which can pose challenges due to privacy concerns in real applications.

Source-free Domain Adaptation (SFDA) leverages only a source-trained model and unlabeled target data for adaptation to the target domain. SFDA approaches can be categorized into data-based and model-based methods (Z. Yu et al., 2023). One of the data-driven methods, SHOT, was introduced by Liang et al. (Liang et al., 2020). It adapts a pre-trained source model via information maximization with self-supervised pseudo-labeling to implicitly align target domain representations to the source hypothesis. Building on this approach, subsequent works(Chu et al., 2022; Lee et al., 2022; Qu et al., 2022) refine the adaptation through self-training techniques. Other works explore different training procedures. For example, historical Contrastive Learning (HCL) (J. Huang et al., 2021) compensates for the absence of source data by leveraging historical models and contrasting current and historical embeddings of target samples. Some methods (S. Yang et al., 2021, 2022) enforce consistency between local neighbors by considering local feature density, with Attract and Disperse (AaD) (S. Yang et al., 2022) treating SFDA as an unsupervised clustering problem. Additionally, Y. Zhang, Wang, and He, 2023 explore leveraging source model classifier weights as class prototypes to embed class relationships into a similarity measure for a target sample.

Source-free Open-set Domain Adaptation (SF-OSDA) extends SFDA to scenarios where the target domain contains novel classes not present in the source domain. While methods like SHOT (Liang et al., 2020), AaD (S. Yang et al., 2022), and Uncertainty-guided Source-free Domain Adaptation (U-SFAN) (Roy et al., 2022) have been adapted for SF-OSDA, they primarily focus on the *known* class semantics in the source domain, which can lead to suboptimal handling of target-private unknown classes. Universal Domain Adaptation (UniDA) aims to handle domain shifts and label set differences between source and target domains, encompassing open, partial, and open-partial set scenarios (Liang et al., 2021; Qu et al., 2023, 2024). Recent SF-UniDA methods proposed one-vs-all clustering approaches (Qu et al., 2023) and subspace decomposition (Qu et al., 2024) to separate and identify common and private target classes in a source-free setup.

Similarly, Progressive Graph Learning (Y. Luo et al., 2023) decomposes the target hypothesis space into shared and unknown subspaces for SF-OSDA. However, current methods either require specific training for the source model to incorporate the unknown classes (Kundu, Venkat, M V, & Babu, 2020), which is usually impractical, or rely on thresholding a metric to distinguish known classes from unknown ones during training and inference, making the prediction sensitive to different thresholds.

6.3 Methodology

6.3.1 Preliminary

For SF-OSDA, we are given a source pre-trained model f_θ^s and an unlabeled target domain with n_t samples, denoted as $\mathcal{D}_t = \{(x_i^t)\}_{i=1}^{n_t}$, where $x_i^t \in \mathcal{X} \subset \mathbb{R}^X$. The target domain follows a distinct data distribution ($P^t \neq P^s$) from the source domain, reflecting both distribution and label shifts.

Let $\mathcal{C}^s$ and $\mathcal{C}^t \subset \mathcal{Y}$ represent the label sets for the source and target domains, respectively, where $\mathcal{C}^s \subset \mathcal{C}^t$. Both domains share K common classes referred to as *known* classes ($\mathcal{C}_k^t = \mathcal{C}^s$). Additionally, the target domain includes target-private novel classes, jointly considered as a single *unknown* class ($\mathcal{C}_{unk}^t = \mathcal{C}^t \setminus \mathcal{C}^s$).

The primary objective of SF-OSDA is to classify both *unknown* and *known* classes, relying exclusively on the target domain data and a pre-trained source model. The pre-trained model can be decomposed as $f_\theta^s = h_\theta^s \circ g_\theta^s$, where $h_\theta^s : \mathbb{R}^X \to \mathbb{R}^D$ is a feature extractor and $g_\theta^s : \mathbb{R}^D \to \mathbb{R}^K$ is the source classifier. Unlike previous works, which freeze the source classifier (e.g., SHOT) during adaptation, we propose training a new target classifier g_θ^t to explicitly account for target-private unknown classes.

One of the challenges in open-set scenarios is the ability to distinguish known from unknown classes in the target domain. Different approaches have been proposed for distinguishing between known and unknown classes, including hand-crafted thresholding criteria and clustering strategies. However, paradigms such as vendor-to-client (Kundu, Venkat, Revanur, Babu, et al., 2020) are more effective, as they incorporate an auxiliary out-of-distribution classifier during source training, enabling better handling of unknown classes in the target domain.

In this paper, we propose a novel approach to address this limitation by adapting the source classifier post hoc to include new decision boundaries for unknown classes. Our method enables the seamless adaptation of any off-the-shelf source pre-trained model to a target domain, even in the presence of novel classes. Motivated by the idea that learning from *unknown* class samples can improve performance in open-set scenarios, *our objective is to simplify adaptation and eliminate the dependency on threshold-based methods during inference.*

6.3.2 RRDA

Our proposed Recall and Refine framework for SF-OSDA consists of three main steps:

1. **Synthetic Data Generation:** Referring to step (b) in Figure 6.1, synthetic feature points are generated for both *known* and *unknown* classes. This involves optimizing target feature representations using entropy objectives.

2. **Target Classifier Training:** Referring to step (c) in Figure 6.1, the synthetic feature points are used to train a new target classifier g_θ^t with extended decision boundaries to accommodate unknown classes.

3. **Target Domain Adaptation:** The entire model is adapted using any off-the-shelf source-free domain adaptation methods (e.g., SHOT, AaD) on target domain data.

This allows the model to (1) learn the semantics of both known and unknown classes in the target domain, (2) treat OSDA as a simple closed-set scenario, and (3) directly output predictions for unknown classes.

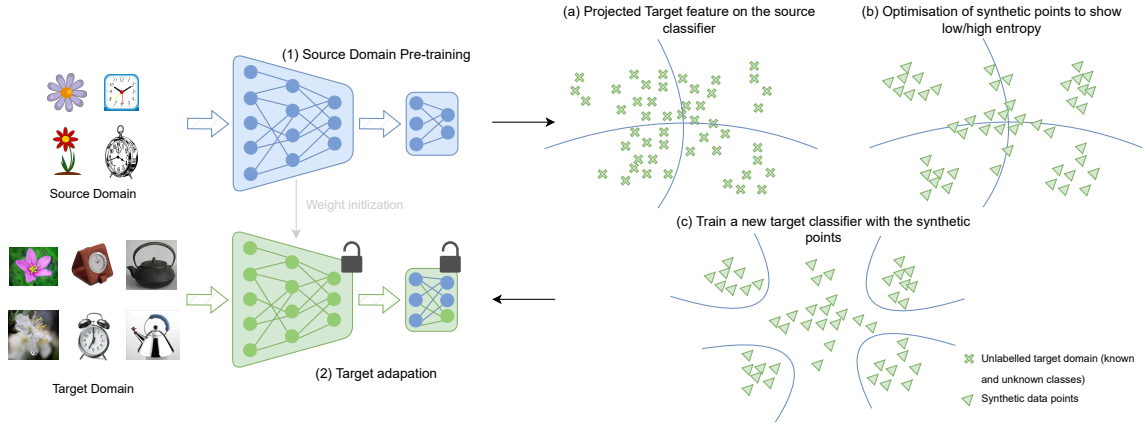


Figure 6.1: Overview of RRDA for Source-free Open-set Adaptation. Unlike conventional methods that overlook unknown class semantics, RRDA explicitly incorporates this information by generating synthetic points for both known and unknown classes from projected target features, enabling the training of a new target classifier that captures the semantics of all classes. The adaptation is then achieved using standard closed-set domain adaptation methods.

Synthetic Data Generation.

The first step of our proposed approach involves generating synthetic features for both *known* and *unknown* classes using the source classifier g_θ^s. Specifically, we optimize the target feature representation $\mathbf{z}^t = h_\theta^s(x^t)$ to generate synthetic samples that exhibit low

entropy for *known* classes and high entropy for the *unknown* class. We denote these optimized synthetic features as $\mathbf{z}_k^{*t}$ and $\mathbf{z}_{unk}^{*t}$. The unknown features are then clustered in K' classes, and a new target classifier g_θ^t is introduced with $K + K'$ classes.

In this section, we describe the process for obtaining feature representations for both *known* and *unknown* classes. We use standard gradient descent optimization to generate the desired feature representations.

Synthetic *Unknown* Classes Generation: To effectively identify points near the source classifier's decision boundary, we aim to find $\mathbf{z}_{unk}^{*t}$ that maximizes entropy while ensuring diverse feature representations, thereby reducing the risk of collapsing to a single-point representation. To prevent feature collapse, we introduce a variance regularization term in the form of a hinge function applied to the standard deviation of features across the batch dimension. Specifically, we initialize the optimization with a noisy version of the original features $\mathbf{z}^t$. This process is formulated as follows:

$$\min_{\mathbf{z}^t} \quad -H(\sigma(g_\theta^s(\mathbf{z}^t))) + \lambda \cdot \max(0, 1 - \sqrt{\mathrm{Var}(\mathbf{z}^t)}), \tag{6.1}$$

where $H(p) = -\sum_{k=1}^{K} p_k \log(p_k)$ represents the entropy, and σ is the softmax activation function, and λ was set to 1 for all the experiments. After optimization, only the points satisfying $H(g_\theta^s(\mathbf{z})) > 0.75 \cdot \log(K)$ (see Ablation section for threshold discussion) are considered as $\mathbf{z}_{unk}^{*t}$. The selected features $\mathbf{z}_{unk}^{*t}$ are then clustered into K' unknown classes using K-means. Each cluster is assigned a pseudo-label corresponding to a new class index, $\hat{y}_{unk}^{*t} \in \{K+1, ..., K+K'\}$, representing the specific unknown class assigned to the synthetic features. These synthetic features and their associated pseudo-labels ($\mathbf{z}_{unk}^{*t}$, $\hat{y}_{unk}^{*t}$) will be used in the subsequent training of the target classifier.

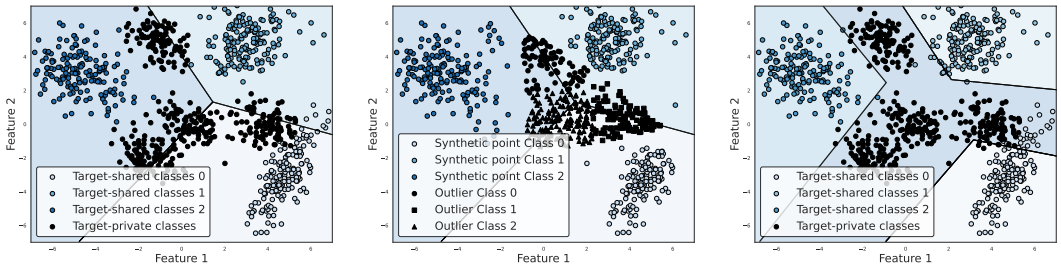
This approach is motivated by the observation in the literature (Lampert et al., 2009) that it is possible to generate meaningful semantics for novel classes using known classes.

Synthetic *Known* Classes Generation: A similar optimization approach is employed to generate synthetic data points for the *known* classes. The optimization is performed iteratively K times, once for each known class k (where $k \in [1, ..., K]$). The objective is to minimize the cross-entropy for each class directly from $\mathbf{z}^t$. The optimization problem for generating a sample for class k is defined as:

$$\min_{\mathbf{z}^t} \quad \mathcal{L}_{CE}(g_\theta^s(\mathbf{z}^t), I_k) + \lambda \cdot \max(0, 1 - \sqrt{\mathrm{Var}(\mathbf{z}^t)}), \tag{6.2}$$

where I_k is the identity function for the k-th class (i.e., a one-hot vector), and λ controls the regularization term, set to 1 in all experiments. After optimization, only the points satisfying $\mathcal{L}_{CE}(g_\theta^s(\mathbf{z}), I_k) < 0.25 \cdot \log(K)$ (see Ablation section for threshold discussion)

are considered as $\mathbf{z}_k^{*t}$. Each selected synthetic feature $\mathbf{z}_k^{*t}$ is assigned the pseudo-label $\hat{y}_k^{*t} = k$, forming the pairs $(\mathbf{z}_k^{*t}, \hat{y}_k^{*t})$. These synthetic data points and their corresponding pseudo-labels are then used to train the target classifier. By iteratively generating feature points for each known class, our method enhances the decision boundaries without requiring access to the original source data or labels.



(a) Unlabeled target domain projected onto the decision boundary of the source classifier.

(b) Optimized synthetic points for known and unknown classes.

(c) Unlabeled target domain projected onto the new decision boundary of the target classifier.

Figure 6.2: Visualization of the synthetic data generation process and the resulting target classifier boundary on a toy example with $K = K' = 3$ classes.

Target Classifier Training.

In the second step, we introduce a new target classifier g_θ^t with $K + K'$ classes, where K is the number of known classes and K' is the number of unknown classes. The weights for the known classes $g_{\theta_{[1:K]}}^t$ are initialized using the source classifier's weights g_θ^s, while the weights for the unknown classes $g_{\theta_{[K+1:K+K']}}^t$ are randomly initialized.

The target classifier g_θ^t is trained using the synthetic feature-label pairs for the *known* classes $(\mathbf{z}_k^{*t}, \hat{y}_k^{*t})$ for $k \in \{1, ..., K\}$, and the *unknown* classes $(\mathbf{z}_{unk}^{*t}, \hat{y}_{unk}^{*t})$. The supervised training objective is defined as:

$$\min_\theta \quad \mathcal{L}_{CE}(g_\theta^t(\mathbf{z}^{*t}), \hat{y^{*t}}), \tag{6.3}$$

where $\mathbf{z}^{*t}$ represents the combined synthetic features for both known and unknown classes, and $\hat{y}^{*t}$ represents their corresponding labels. The results of the previous steps, including the refined decision boundaries achieved by RRDA, are illustrated in Figure 6.2.

Target Domain Adaptation.

Any source-free unsupervised domain adaptation method (originally designed for closed-set scenarios) can be integrated into our approach to address open-set scenarios, provided it

incorporates a diversity loss or a similar mechanism to facilitate self-learning of unknown classes. To empirically validate this hypothesis, we consider SHOT (Liang et al., 2020) and AaD (S. Yang et al., 2022), using their respective training objectives for adaptation. SHOT (Liang et al., 2020) employs information maximization and self-supervised pseudo-labeling to adapt the source model to the target domain. Its objective function can be expressed as:

$$\mathcal{L}_{\text{shot}} = -\frac{\lambda_{ent}}{n_t} \sum_{i=1}^{n_t} \sum_{k=1}^{K+K'} p_{k,i} \log p_{k,i} + \lambda_{\text{div}} \cdot \sum_{k=1}^{K+K'} \bar{p}_k \log \bar{p}_k + \lambda_{\text{ps}} \cdot \mathcal{L}_{\text{pseudo}}, \tag{6.4}$$

where $\bar{p}_k = \frac{1}{n_t} \sum_{i=1}^{n_t} p_k(x_i; \theta)$, and $\mathcal{L}_{\text{pseudo}}$ is the pseudo-labeling loss function from Liang et al., 2020. During adaptation, only the feature encoder is updated while the classifier remains frozen. AaD (S. Yang et al., 2022) leverages local consistency and global dispersion. The objective function for feature i is formulated as:

$$\mathcal{L}_{\text{AaD},i} = -\sum_{j \in \mathcal{C}_i} p_i^T p_j + \lambda \sum_{m \in \mathcal{B}_i} p_i^T p_m, \tag{6.5}$$

where $\mathcal{C}_i$ represents the local neighborhood of feature i and $\mathcal{B}_i$ is the mini-batch feature not in $\mathcal{C}_i$. Unlike SHOT, AaD updates the entire model weights during adaptation.

6.4 Experiments

6.4.1 Experimental Setup

Datasets. Office-Home (Venkateswara et al., 2017) comprises 65 labeled image categories from four distinct domains: Art (Ar), Clipart (Cl), Product (Pr), and Real World (Rw). We designate the first 25 alphabetically ordered categories as known classes, with the remaining 40 as unknown. **Office-31** (Saenko et al., 2010) consists of 31 classes across three domains: Amazon (A), Dslr (D), and Webcam (W). We assign the first 10 as known and the last 10 classes as unknown. **VisDA** (Peng et al., 2017) have 12 categories across two domains: Real (R) and Synthetic (S). The first 6 classes are categorize as known and the remaining 6 as unknown.

Evaluation Metrics. To assess model performance, we adopt standard evaluation metrics widely used in previous OSDA studies (Bucci et al., 2020; W. Li et al., 2023; H. Liu et al., 2019). The Harmonic Open-set (HOS) accuracy balances performance on known and unknown classes and can be calculated as $HOS = \frac{2 \times OS^* \times UNK}{OS^* + UNK}$, where OS^* represents the accuracy of known classes, and UNK denotes the accuracy of unknown classes. The HOS metric provides a comprehensive measure, by equally weighting the model's ability to classify known classes and detect unknown classes.

Implementation Details. All experiments are conducted on a single A100 GPU using PyTorch. For synthetic data generation, we employ the Adam optimizer with a learning rate of 0.001 for 1000 steps, for both known ($\mathbf{z}_k^{*t}$) and unknown ($\mathbf{z}_{unk}^{*t}$) classes. In all main experiments, we set $K' = K$. To maintain class balance, we cap the sample size at 1000 for known classes in Office-Home and Office-31, and 10,000 for VisDA. The target classifier is trained for 50 epochs using SGD with a learning rate of 0.01, momentum of 0.9, weight decay of 0.001, and a fixed batch size of 128. During target model adaptation, we use SGD with momentum 0.9, a batch size of 64, and train for 50 epochs. The learning rate is set to 0.001 for Office-31 and Office-Home, and 0.0001 for VisDA, when using Resnet-50 (K. He et al., 2016) as backbone. When using ViT-B (B. Wu et al., 2020) as the backbone, we set the learning rate to 0.0001 for all experiments. For SHOT, we freeze the target classifier and train only the feature extractor and backbone. For AaD, all model parameters are trained, with the feature extractor's learning rate set to 10 times lower. During inference, samples belonging to the new K' classes are considered unknown target samples. λ_{ps} was set to 0.1, 0.3, and 0.4 for Office-Home, Office-31, and VisDA respectively. The hyper-parameters λ_{div} and λ were set to 1 and λ_{ent} was set to 0.5 in all our experiments.

Baselines. We compare our method against open-set domain adaptation approaches, including both non-source-free and source-free methods. The non-source-free methods include OSBP (Saito, Yamamoto, et al., 2018), CMU (Fu et al., 2020), STA (H. Liu et al., 2019), DANCE (Saito et al., 2020), GATE (L. Chen, Lou, et al., 2022), ANNA (W. Li et al., 2023), and OSLPP (Q. Wang, Meng, & Breckon, 2024). For source-free methods, we consider UMAD (Liang et al., 2021), GLC (Qu et al., 2023), SF-PGL (Y. Luo et al., 2023), and LEAD (Qu et al., 2024).

6.4.2 Experimental Results

From Table 6.1 to 6.3, we compare our method against state-of-the-art (SOTA) OSDA methods in both source-free and non-source-free setups. We include non-SF methods to provide a comprehensive performance benchmark, despite our focus on SF scenarios. We use RRDA alongside AaD and SHOT (vanilla methods for closed-set scenarios), as well as their open-set variants denoted as AaD-O and SHOT-O that rely on entropy-thresholding during training and inference. Results for comparison methods are sourced from (W. Li et al., 2023; Qu et al., 2024), and the mean HOS is reported.

Office-31. Table 6.1 presents results on the Office-31 dataset, where RRDA demonstrates significant improvements over threshold-based methods. AaD+RRDA achieves an average HOS of 94.4%, which is an 8.0% increase over AaD-O. Similarly, SHOT+RRDA reaches 93.8%, representing a 5.6% improvement over SHOT-O. These results surpass all compared source-free and non-source-free SOTA methods.

Table 6.1: HOS (%) results on Office-31 (ResNet-50). SF denotes source-free methods. AaD-O and SHOT-O are the adapted open-set methods of AaD and SHOT. RRDA uses standard AaD and SHOT versions (closed-set scenario).

Methods	SF	Office-31						
		A2D	A2W	D2A	D2W	W2A	W2D	Avg
CMU	×	52.6	55.7	76.5	75.9	65.8	64.7	65.2
DANCE	×	84.9	78.8	79.1	78.8	68.3	78.8	79.8
OSLPP	×	91.5	89.0	79.3	92.3	78.7	9..6	87.4
GATE	×	88.4	86.5	84.2	95.0	86.1	96.7	89.5
ANNA	×	83.8	85.5	82.5	99.5	81.6	98.4	88.6
Source-only	✓	78.2	72.1	44.2	82.2	52.1	88.8	69.6
UMAD	✓	88.5	84.4	86.8	95.0	88.2	95.9	89.8
LEAD	✓	84.9	85.1	90.9	94.8	90.3	96.5	90.3
GLC	✓	82.6	74.6	92.6	96.0	91.8	96.1	89.0
AaD-O	✓	82.3	79.0	84.3	93.1	84.8	95.0	86.4
AaD + RRDA	✓	91.1	94.3	94.1	96.6	94.0	96.2	**94.4**
		+8.8	+15.3	+9.8	+3.5	+9.2	+1.2	**+8.0**
SHOT-O	✓	89.5	83.0	85.9	91.4	84.0	95.2	88.2
SHOT+ RRDA	✓	90.0	92.2	92.6	98.2	91.6	98.2	**93.8**
		+0.5	+9.2	+6.7	+6.8	+7.6	+3.0	**+5.6**

Office-Home. On the Office-Home dataset (Table 6.2), RRDA consistently enhances the performance of both AaD-O and SHOT-O across most domain adaptation tasks. AaD+RRDA and SHOT+RRDA show average HOS improvements of +2.8% and +8.1% respectively. SHOT+RRDA achieves a competitive 70.0% average HOS, outperforming most methods, including both source-free and non-source-free approaches, while falling just slightly short of ANNA a non-source-free adaptation method. A similar observation can be made, when using ViT as backbone, where RRDA consistently improve previous methods AaD+RRDA and SHOT+RRDA show average HOS improvements of +7.3% and +2.7% respectively, and surpass the other baselines.

VisDA. On the challenging VisDA dataset (Table 6.3), RRDA continues to demonstrate its effectiveness. AaD+RRDA improves upon AaD-O by +6.0% in HOS, significantly improving unknown sample recognition and overall class accuracy. Significant improvements are observed in classes such as "Bus" (+15.9%) and "Truck" (+20.4%). Similar improvements can be are observed when applying RRDA to SHOT with an improvement of +22.7% in HOS. We observe that both methods improve over the same class and degrade the performances of the "car" and "motorcycles" classes.

These results demonstrate RRDA's consistent superiority across various domain adaptation scenarios. Our method significantly improves existing SF-OSDA techniques, as evidenced by the consistent performance gains across all datasets. The key advantage of RRDA lies in its novel approach to handling unknown classes. Unlike previous methods that rely on thresholding and discard unknown class data during adaptation, RRDA actively learns the semantics of unknown classes through our adaptive target classifier, which evolves to accommodate the unknown class distribution. Furthermore, the consistent performance gains with both AaD and SHOT demonstrate RRDA's versatility. These

Table 6.2: HOS (%) results on Office-Home (ResNet-50 and ViT). $|C_s| = 25$, $|C_t| = 65$. SF denotes source-free methods.

Methods	SF	Office-Home												
		Ar2Cl	Ar2Pr	Ar2Rw	Cl2Ar	Cl2Pr	Cl2Rw	Pr2Ar	Pr2Cl	Pr2Rw	Rw2Ar	Rw2Cl	Rw2Pr	**Avg**
CMU	×	55.0	57.0	59.0	59.3	58.2	60.6	59.2	51.3	61.2	61.9	53.5	55.3	57.6
DANCE	×	6.5	9.0	9.9	20.4	10.1	9.2	28.1	15.8	12.6	14.2	7.9	13.7	12.9
OSLPP	×	61.0	72.8	74.3	60.9	66.9	70.4	63.6	59.3	74.0	67.2	59.0	74.4	67.0
GATE	×	63.8	70.5	75.8	66.4	67.9	71.7	67.3	61.3	76.0	70.4	61.8	75.4	69.0
ANNA	×	69.0	73.7	76.8	64.7	68.6	73.0	66.5	63.1	76.6	71.3	65.7	78.7	70.7
Source-only	✓	46.1	63.3	72.9	42.8	54.0	58.7	47.8	36.1	66.2	60.8	45.3	68.2	55.2
UMAD	✓	59.2	71.8	76.6	63.5	69.0	71.9	62.5	54.6	72.8	66.5	57.9	70.7	66.4
LEAD	✓	60.7	70.8	76.5	61.0	68.6	70.8	65.3	59.8	74.2	64.8	57.7	75.6	67.2
GLC	✓	65.3	74.2	79.0	60.4	71.6	74.7	63.7	63.2	75.8	67.1	64.3	77.8	69.8
AaD-O	✓	58.0	68.2	75.4	58.8	65.7	69.0	54.6	52.9	72.3	65.8	56.3	72.2	64.1
AaD + RRDA	✓	61.7	72.8	73.5	59.0	74.9	69.9	59.5	58.3	71.2	64.5	64.8	73.2	66.9
		+3.7	+4.6	-1.9	+0.2	+9.2	+0.9	+4.9	+5.4	-1.1	-1.3	+7.7	+1.0	**+2.8**
SHOT-O	✓	57.2	65.4	69.9	58.1	62.6	64.3	60.5	52.8	71.1	64.4	53.5	40.6	61.9
SHOT + RRDA	✓	64.6	74.2	77.2	63.1	71.4	71.3	67.7	59.1	76.7	70.2	67.4	76.7	**70.0**
		+7.4	+8.8	+7.3	+5.0	+8.8	+7.0	+7.2	+6.3	+5.6	+5.8	+13.9	+36.1	**+8.1**
Source-only	✓	57.1	69.5	79.9	50.2	62.5	66.0	52.2	45.7	75.1	69.3	56.4	73.7	63.1
LEAD	✓	58.6	74.7	82.7	58.9	74.6	74.3	59.0	47.1	78.3	71.9	58.7	77.4	68.0
AaD-O	✓	57.8	74.9	82.7	53.9	68.6	70.8	52.5	45.8	76.8	70.6	58.2	77.7	65.9
AaD + RRDA	✓	67.4	77.2	81.2	71.4	71.5	76.1	73.9	63.8	78.5	74.8	67.5	75.0	**73.2**
		+9.6	+1.5	-1.5	+17.5	+2.9	+5.3	+21.4	+18.0	+1.7	+4.2	+9.3	-2.7	+7.3
SHOT-O	✓	63.6	73.5	81.7	66.7	69.8	75.5	66.5	56.2	79.0	73.5	62.6	74.6	70.3
SHOT + RRDA	✓	68.9	75.0	81.4	71.2	73.8	73.6	71.9	60.4	79.2	76.5	66.2	77.5	**73.0**
		+5.3	+1.5	-0.3	+4.5	+4.0	-2.1	+5.4	+4.2	+0.2	+3.0	+3.6	+2.9	**+2.7**

ResNet-50 (rows 1–10) / ViT (rows 11–16)

Table 6.3: Accuracy for each class (%) and HOS (%) results on VisDA (ResNet-50 and ViT), with $|C_s| = 6$, $|C_t| = 12$.

Methods	VisDA							
	Bic	Bus	Car	Mot	Tra	Tru	UNK	HOS
OSBP	35.6	59.8	48.3	76.8	55.5	29.8	81.7	62.7
STA	50.1	69.1	59.7	85.7	84.7	25.1	82.4	71.0
Source-only	16.3	7.9	24.9	48.0	6.1	0.0	72.7	27.9
LEAD	83.5	65.2	57.7	35.7	82.1	79.5	82.7	74.2
SF-PGL	91.5	90.1	74.1	90.3	81.9	74.8	72.0	77.4
AaD-O	86.8	69.8	51.5	38.7	84.3	26.0	65.5	62.4
AaD + RRDA	96.0	85.7	34.5	37.6	92.2	46.4	71.7	68.4
	+9.2	+15.9	-17.0	-1.1	+7.9	+20.4	+6.2	**+6.0**
Shot-O	82.1	67.0	78.6	57.3	72.2	17.9	50.7	56.0
Shot + RRDA	88.6	82.2	66.8	47.3	87.2	74.3	**83.5**	**78.7**
	+6.5	+15.2	-11.8	-10.0	+15.0	+56.4	+32.8	**+22.7**
Source-only	62.3	17.9	17.7	50.7	0.0	0.6	90.8	39.1
LEAD	87.6	65.3	49.8	30.5	70.9	54.4	98.2	74.3
AaD-O	87.9	77.6	47.7	36.8	61.2	16.6	67.6	60.4
AaD + RRDA	98.2	91.1	84.8	39.4	94.2	97.5	81.7	**82.9**
	+10.3	+13.5	+37.1	+2.6	+33.0	+80.9	+14.1	+22.5
Shot-O	96.7	77.0	80.4	75.9	3.0	4.7	80.1	66.1
Shot + RRDA	95.4	84.8	74.1	48.7	85.1	82.3	79.3	**78.9**
	-1.3	+7.8	-6.3	-27.2	+82.1	+77.6	-0.8	+ 12.8

ResNet-50 (rows 1–10) / ViT (rows 11–16)

results underscore the importance of explicitly modeling unknown classes in open-set domain adaptation, rather than treating them as outliers to be discarded.

6.4.3 Ablation Study and Sensitivity Analysis

Optimization Process. We conducted ablation studies on Office-31 with three different settings to train the new classifier. The results are shown in Table 6.4. We compare the following scenarios: (1) selecting target features based on entropy threshold without optimization, (2) optimizing entropy without hinge loss for diversity, and (3) the full proposed method optimizing feature points based on entropy and diversity. Our findings are as follows: (1) Using target features based on entropy directly to train the target classifier leads to the worst results in terms of HOS. SHOT-O achieves an HOS of 88.2 %, while using features directly without optimization achieves an HOS of 86.5%. (2) Optimizing the points significantly improves performance. There is a slight additional improvement when using hinge loss during optimization to promote diversity. (3) The full proposed method, which optimizes feature points based on both entropy and diversity, yields the best performance. We used SHOT for adaptation in the experiment as it keeps the classifier frozen, allowing for a direct performance comparison with the new classifier.

Table 6.4: Ablation on the optimization objective to generate synthetic points. Results using SHOT+RRDA on Office-31.

Entropy	Diversity	OS*	UNK	HOS
		95.6	79.8	86.5
✓		95.0	91.8	93.4
✓	✓	**95.8**	**91.9**	**93.8**

Threshold Sensitivity Analysis. To further analyze the hyperparameter sensitivity and its impact on performance, we examined the effect of varying the entropy threshold used for feature selection during the optimization process. The thresholds were evaluated on the A2D task (ref Table 6.5).

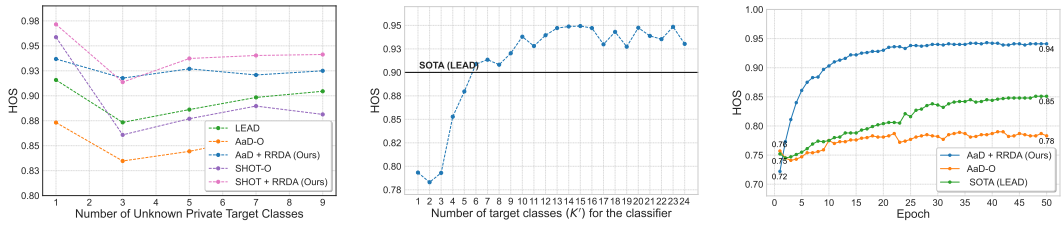
Table 6.5: Ablation on the optimization objective to generate synthetic points. Results using SHOT+RRDA on Office-31.

Methods	0.1/0.9	0.2/0.8	**0.25/0.75**	0.3/0.7	0.4/0.6	0.5/0.5
SHOT	93.3	91.3	91.1	91.1	91.1	91.9
AaD	90.9	89.5	90.0	89.7	90.1	90.0

We observe that the best-performing threshold on this task is $T = 0.1$. However, the HOS score remains consistent across different thresholds. For larger datasets, such as VisDA, where the domain shifts are more significant, lower thresholds (e.g., $T = 0.1$) can result in highly imbalanced datasets, with some classes being excluded entirely. For example, under such thresholds, a subset of classes may not meet the selection criteria. To ensure consistency across all datasets while maintaining a balanced feature distribution,

we report results using $T = 0.25$ throughout the experiments. This threshold provides a balance between maintaining sufficient class representation and achieving competitive performance, particularly in scenarios with large domain shifts.

Varying Unknown Classes. We investigated the robustness of our framework against an increased number of unknown private classes, which complicates the distinction between known and unknown classes. We compared our method to LEAD, SHOT-O, and AaD-O on the Office-31 dataset. As shown in Figure 6.3a, our RRDA method in combination with SHOT and AaD achieves stable results and consistently outperforms existing approaches. For consistency with our main results, we kept K' fixed at 10.

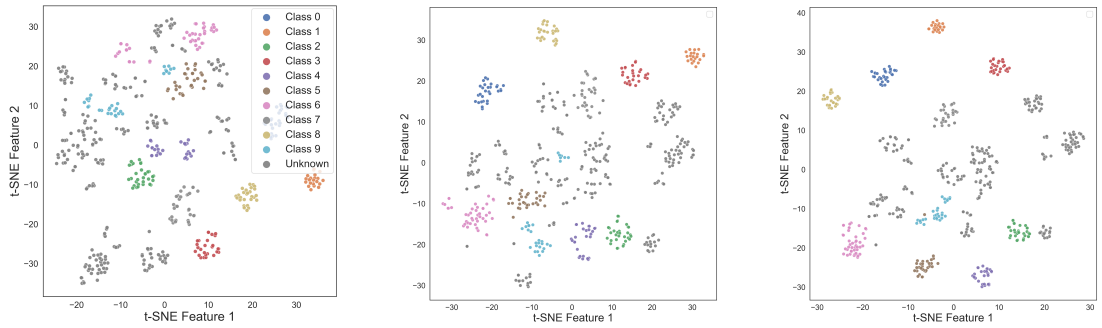


(a) Various openness settings (b) Sensitivity to K' (c) HOS convergence

Figure 6.3: Sensitivity analysis on Office-31. (a) Adaptation performance across different openness levels (average across all transfer tasks). (b) Sensitivity to K' target classifier classes (average across all transfer tasks). (c) HOS curves for the A2W task.

Sensitivity to K′. Figure 6.3b shows adaptation performance for different K' values of the target classifier on Office-31 dataset. The performance improves as K' increases, validating the benefit of inheriting class separability knowledge, before eventually reaching a plateau. In fact, $K' = 15$ yields the best results. For the main experiments, we reported Office-31 results using $K' = K = 10$.

Training Stability. Figure 6.3c illustrates the training curves for the A2W task on the Office-31 dataset. Our method shows consistent HOS improvement on the test set, with steadily increasing before plateauing. In contrast, AaD-O exhibits unstable training, with noticeable performance fluctuations throughout the training process.



(a) Source Only (b) AaD-O (c) AaD + RRDA (Ours)

Figure 6.4: T-SNE visualization of the pre-classifier feature space for the A2W task on Office-31 dataset.

6.4.4 Feature Space Visualization

Figure 6.4 shows t-SNE embeddings of pre-classifier features for the source-only model, AaD-O, and our method on the A2W task on the Office-31 dataset. The source-only model (Figure 6.4a) exhibits well-separated known class clusters but mixes unknown samples with known classes. AaD-O (Figure 6.4b) slightly improves known-unknown separation, but class overlap remains. Our method (Figure 6.4c) achieves superior separation of known and unknown classes, maintaining tight, well-defined known class clusters while isolating unknown samples. This demonstrates our method's effectiveness in inheriting class separability during adaptation.

6.5 Conclusion

In this work, we introduce **R**ecall and **R**efine for **D**omain **A**daptation, a simple but effective framework for SF-OSDA. RRDA enables the successful adaptation of off-the-shelf source pre-trained models to target domains, effectively addressing both distribution and category shift problems. RRDA achieves this by introducing a new target classifier that aids in classifying and learning the semantics of both known and unknown classes. This approach enables the direct use of source-free adaptation methods designed for closed-set scenarios in open-set contexts. Extensive experiments on three challenging benchmarks demonstrate that RRDA significantly outperforms existing SF-OSDA methods and even surpasses OSDA methods that have access to the source domain. Future work could explore its potential for continuous adaptation in the setup where new classes appear over time.

7 Discussions

Detailed discussions for the individual studies have been presented in their respective chapters. This chapter discusses the key findings across these studies, examining their interactions, and potential integrations, particularly in the context of Prognostics and Health Management applications. We discuss how these different approaches contribute to the broader goal of building robust regression models that can handle real-world distribution shifts and data imbalance challenges.

7.1 Module 1: Operating-Phase Aware Distribution Alignment

In Chapter 2, we introduced a phase-aware domain adaptation method for Remaining Useful Life prediction in turbofan engines, which independently adapts each operational phase, such as take-off, cruise, and landing. This approach emphasizes the importance of aligning these operating phases separately rather than treating them as a single uniform distribution, as this mitigates issues of phase underrepresentation or overrepresentation (e.g., take-off vs. cruise phases), which can otherwise affect the adaptation process.

Performance Our experiments demonstrated that phase-aware domain adaptation is particularly effective for applications where varying operational conditions substantially influence degradation patterns. By aligning distributions within each operational phase separately, this approach improved adaptability across various RUL prediction scenarios. In the most challenging adaptation setting (short-to-long flights, S→L), this method achieved up to a 45% reduction in RMSE compared to the baseline (from 7.34 to 3.90). Traditional domain adaptation methods, which treat the operational cycle as a single distribution, proved less effective, reinforcing the importance of phase-specific alignment. Among the evaluated methods, OPS-DANN (hard) delivered the best performance, achieving an additional 9% RMSE reduction over standard DANN (from 4.38 to 3.90) and improving the NASA score from 1.43 to 1.39. While this approach successfully addressed

phase-specific alignment, it did not explicitly consider regression-specific challenges such as feature scale preservation. This aspect was later addressed through the regression-specific domain adaptation approach introduced in Chapter 3.

Complexity While phase-aware alignment improves performance, it introduces additional complexity by requiring separate discriminators for each phase. This requirement, particularly in OPS-DANN (hard), increases computational demands compared to simpler methods like Multi-Class OPS-DANN, which employs a shared structure across all phases. The additional computational complexity of OPS-DANN (hard) may present challenges in resource-constrained environments, although it consistently achieves the highest accuracy across adaptation scenarios.

Domain Gap Size The effectiveness of phase-specific alignment showed strong dependence on the magnitude of the domain gap. For large domain gaps, such as S→L, phase-aware alignment substantially reduced prediction errors. For medium gaps (S→M), improvements were more modest (RMSE reduction from 4.59 to 3.71). In small-gap scenarios (M→L), while traditional domain adaptation methods showed negative transfer, actually worsening the performance compared to the baseline, our method maintained stable performance (RMSE from 2.63 to 2.66). This demonstrates that our phase-aware approach can effectively mitigate negative transfer even in scenarios with small domain gaps. Proxy A-distance measurements confirmed these findings, showing the largest domain discrepancy reductions in cases with large gaps. This observation motivated our investigation of uncertainty in domain adaptation, explored in Chapter 4, where we demonstrated that prediction uncertainty correlates with domain gap size, providing an additional metric for guiding adaptation in variable environments.

7.2 Module 2: Regression-Specific Domain Adaptation

In Chapters 3 and 4, we introduced regression-specific approaches for domain adaptation to address unique challenges posed by continuous output spaces in regression tasks. Two methods were developed: DARE-GRAM, which focuses on aligning inverse Gram matrices for feature scale preservation, and Uncertainty-Guided Alignment, which leverages predictive uncertainty to guide the adaptation process by enriching the feature space.

Comparison on Computer Vision Tasks

DARE-GRAM and UGA both address the limitations of traditional feature alignment in regression, although with different objectives. DARE-GRAM emphasizes preserving the scale and relational consistency of features by aligning the inverse Gram matrices of the source and target domains. This approach ensures that feature correlations remain intact, leading to more reliable predictions under domain shift. In contrast, UGA challenges the assumption of constant variance in MSE loss by introducing uncertainty quantification,

which accounts for variability in prediction confidence across the feature space. By adapting the loss function to account for target data that can be considered out-of-distribution of the source domain, UGA guides the model to prioritize high-confidence predictions, effectively mitigating the impact of domain shift.

On computer vision tasks, both methods achieved state-of-the-art performance with similar results for most benchmarks. However, UGA showed improved performance in head position estimation task, which is the smallest dataset in our evaluation. This improved performance under data-scarce regime, which is commonly encountered in PHM applications, can be attributed to the evidential loss term in UGA which provides additional regularization by enforcing consistency between predictions and their associated uncertainties, helping prevent overfitting when training data is limited.

Application to PHM Tasks

We evaluated both approaches on two PHM case studies: battery State-of-Charge prediction and turbofan engine degradation using the N-CMAPSS dataset. Both DARE-GRAM and UGA demonstrated superior performance compared to traditional domain adaptation techniques across these tasks. For state-of-charge prediction, we investigated two transfer scenarios with different data availability. In the limited data scenario, UGA's uncertainty acted as an implicit regularizer, improving generalization and reducing prediction error, similar to our observations in computer vision tasks. In the data-abundant scenario, DARE-GRAM performed comparably to UGA, demonstrating that aligning feature relationships through the MSE loss can effectively handle domain shifts.

These observations suggest a practical guideline: UGA excels in low-data regimes by leveraging uncertainty as a form of implicit regularization while quantifying the uncertainty. This is particularly valuable in prognostics applications where reliable uncertainty estimates are often required for deployment, enabling informed decision-making in maintenance planning and risk assessment.

Potential Future Research: Combining Regression-Specific Adaptations with Phase Awareness

A promising direction for future work is integrating DARE-GRAM or UGA into a Phase-Aware Distribution Alignment framework. Replacing DANN with DARE-GRAM would involve aligning the inverse Gram matrices within each operational phase independently. This would ensure that the scale and relational consistency of phase-specific features are maintained, effectively addressing biases introduced by overrepresented phases, such as steady-state operations.

Similarly, incorporating UGA into this framework would require redefining the adaptation strategy to include uncertainty-weighted phase alignments. Each phase could be modeled with its own uncertainty distribution, guiding the adaptation process to focus on confident

predictions within each phase. This would not only improve the robustness of phase-specific alignments but also help handle transition phases where prediction confidence varies.

In both cases, leveraging these regression-specific methods within a phase-aware framework could significantly improve model adaptability and generalization across complex operational scenarios of PHM tasks.

7.3 Module 3: In-Context Learning for Imbalanced Regression

In Chapter 5, we introduced in-context learning as a promising approach to address the challenges of imbalanced regression. Unlike traditional in-weight learning, which relies on gradient updates, in-context learning allows a model to generate predictions by conditioning on a sequence of input-label pairs (prompts) during inference. This paradigm shift has demonstrated significant potential for tackling label imbalance without requiring extensive retraining.

Our study of in-context learning for imbalanced regression has uncovered several insights that challenge prevailing assumptions about the use of context. Through both theoretical analysis and empirical validation, we demonstrated that increasing the context size can unexpectedly degrade performance in minority regions. This behavior manifests as a characteristic U-shaped error curve in minority regions: performance initially improves with additional context but declines when too many examples are added. We demonstrated that the effectiveness of ICL in regression strongly depends on retrieving semantically relevant examples, a particularly challenging task in imbalanced datasets where majority sample dominate. To overcome this, we proposed an augmented neighbor retrieval strategy that combines samples from both original and inverse-density datasets. This approach ensures balanced representation while maintaining diversity. Across multiple datasets, this method consistently improved performance in minority regions without significantly compromising accuracy in majority regions.

Training and Inference Trade-offs In-context learning introduces a fundamental shift in computational requirements compared to in-weight learning. While in-weight learning requires extensive computational resources for model training or fine-tuning on each new task, in-context learning leverages pre-trained models without requiring task-specific fine-tuning, effectively reducing training time to zero. However, this shift increases computational demand during inference, as the model must process context examples in real-time. The attention mechanism, in particular, becomes more computationally expensive with longer context sequences. This trade-off enables more flexible deployment but necessitates careful management of inference-time efficiency.

In-context learning for PHM In-context learning holds great promise for Prognostics and Health Management tasks, particularly where labeled data is scarce. For Remaining Useful Life prediction, temporal dependencies play a critical role, as RUL tasks typically involve sequential time-series data capturing progressive degradation patterns. Future work could explore leveraging models similar to the meta-model used in this study, such as decoder-based transformers, to handle sequential inputs. These models could condition on complete degradation trajectories as context, alongside current measurements, to better capture long-term trends. Foundation models like Chronos (Ansari et al., 2024), TimeGPT-1 (Garza et al., 2024) offer a practical starting point for embedding time-series data and leveraging pre-trained in-context learning models for RUL prediction. Another significant challenge lies in handling long sequences effectively, as transformers tend to over-allocate attention to irrelevant context. Alternative approaches, such as the differential transformer architecture proposed in (Ye et al., 2024), offer promising solutions for more efficient attention allocation in temporal sequences.

7.4 Module 4: Source-Free Open-Set Domain Adaptation

While Chapter 5 addresses data scarcity, Chapter 6 introduces another critical real-world constraint – adapting to new classes not present in the source domain. In industrial settings, such as fault diagnosis, new fault types may arise during system operation, presenting a significant challenge for adaptation. These unknown classes can hurt domain alignment and degrade model performance. The proposed Recall and Refine Domain Adaptation (RRDA) framework overcomes the limitations of traditional open-set, source-free methods that rely on thresholding to discard unknown samples, which often results in suboptimal adaptation.

Open-set DA for Discrete Degradation States and Fault Diagnostics

In many industrial applications, continuous monitoring of system degradation is not always feasible due to limited sensor measurements or incomplete degradation trajectories. While Module 2 assumes continuous measurement of system condition over time, real applications often face constraints that make this assumption impractical. Consequently, prognostics tasks can be formulated as classification problems with discrete degradation states. In safety-critical systems, assets are typically removed from operation when they reach certain degradation levels, before actual end of life occurs, such as with brake pads or railway wheels. This preventive maintenance practice means that severe degradation states may be entirely absent in the training data, creating an inherent open-set scenario where some health states are unobserved. Similarly, in fault diagnostics, new fault types may emerge during system operation that were not present in the initial training data. Both scenarios - unobserved degradation states in prognostics and novel fault types in diagnostics - can benefit from the same open-set methodology, as they share the fundamental challenge of adapting to previously unseen classes in the target domain.

Impact of Missing Classes on Model Performance Our experiments demonstrate that methods relying on discarding high-entropy outliers can perform worse than the source-only baseline in source-free settings. However, when integrated into the proposed RRDA framework, these same methods outperform the source-only baseline by a large margin and prove as competitive as non-source-free open-set domain adaptation methods. This underscores the importance of learning from the semantics of unknown classes.

While the current approach uses synthetic points solely to train the classifier and discards them during adaptation, an interesting direction for future work would be to explore how these synthetic features could be retained and integrated into the adaptation process to address class imbalance. Specifically, these features could be leveraged to compensate for the scarcity of certain unknown classes.

Robustness to Domain Shifts The RRDA framework demonstrated to be resilient under significant domain shifts. In the challenging VisDA synthetic-to-real adaptation task, RRDA maintained robust performance despite the pronounced domain gap. This robustness extended to scenarios with varying proportions of unknown classes, where sensitivity analyses showed consistent improvements in adaptation quality, regardless of the number of unknown samples in the target domain. While the proposed method has been extensively validated on computer vision tasks, its application to PHM represents a promising direction.

7.5 Comprehensive Framework Analysis

The framework developed in this thesis offers a progressive and integrated approach to addressing the challenges of regression tasks under domain shifts and data scarcity in industrial applications, particularly within PHM. Achieving reliable performance in such contexts requires not only individual methods for specific challenges but also a cohesive strategy that adapts to varying data availability and distribution shifts encountered in real-world deployments.

Early Deployment Phase: Addressing Data Scarcity The initial deployment of a condition monitoring system often involves limited data that may not represent future operating conditions or degradation patterns. To address this, the framework employs *in-context learning*, which uses a few semantically relevant examples — such as a single run-to-failure trajectory — to focus on critical aspects of the data. This method does not require training or fine-tuning, allowing for efficient use of limited data in early deployment phases. However, further research is needed to refine the selection of context examples, especially for temporally dependent sequences in PHM.

Intermediate Deployment Phase: Leveraging Operational Knowledge As operational data accumulates, the framework incorporates *phase-aware distribution alignment*,

which utilizes expert knowledge to align individual operational phases between domains. When complete degradation trajectories are unavailable, this approach could be integrated with *discrete health state information* to improve alignment and generalization. Combining these methods leverages both continuous operational variations and discrete health states. Joint training of regression models and auxiliary classification tasks has been shown to improve generalization (Pintea et al., 2023).

Mature Deployment Phase: Ensuring Robust Adaptation In mature deployments, where diverse operating conditions and larger domain shifts are more likely, robustness becomes crucial. This phase focuses on *uncertainty-guided adaptation*, which could help enrich the embedding space with confidence estimates, improving reliability under domain shifts. The *inverse Gram matrix alignment* method complements this by preserving feature relationships across domains. These methods can be combined with *phase-aware distribution alignment* to improve adaptability further. Furthermore, Uncertainty estimates can be used to identify instances where the current operating conditions deviate significantly from those seen during training, thereby helping to detect new or previously unseen domains.

Dataset Considerations and Practical Implementation The N-CMAPSS dataset, collected using a simulator, provides controlled conditions for evaluating turbofan engine degradation. While it has been a standard benchmark for PHM research, it does not fully capture real-world complexities such as sensor noise, maintenance interventions, and varying environmental factors. For the battery State-of-Charge prediction task, domain shifts due to temperature and manufacturer differences were explored, demonstrating the effectiveness of regression-specific methods. However, since only a few trajectories were collected, this dataset is not widely used. The limited diversity of available datasets highlights the need for more realistic industrial datasets to evaluate model performance under domain shifts and data imbalance. A specific challenge lies in end-of-life definitions, where expert opinions may vary significantly. Future research should focus on creating standardized benchmarks that capture real-world complexities, including noise, maintenance effects, and varying expert assessments.

8 Conclusions

This chapter revisits the main objectives of this dissertation, highlighting the contributions made to domain adaptation for regression under real-world constraints in industrial applications, and imbalance regression. Key conclusions are drawn to address the primary research questions and objectives. Lastly, the limitations and possible future research directions are discussed.

8.1 Research Objectives Revisited

The aim of this research was to develop a framework for regression tasks in industrial settings, capable of overcoming challenges related to domain shifts and imbalanced data distributions with minimal supervision.

This objective was achieved by focusing on four primary modules: (1) phase-aware adaptation that leverages expert knowledge to align distinct operational phases across domains, (2) regression-specific domain adaptation methods that preserve feature relationships and incorporate uncertainty estimation, (3) in-context learning strategies for generalization in minority regions of imbalanced distributions, and (4) adapting to open-set, source-free scenarios in which the target domain may contain new or unknown states of degradation.

8.2 Summary

Based on the main research aim and objectives, this dissertation introduced a framework with four distinct modules that address the unique challenges of domain adaptation and imbalance in regression tasks. This framework answered the following research questions:

Can phase-aware domain adaptation improve RUL predictions in variable operating conditions? Chapter 2 introduces a novel approach to domain adaptation for Remaining Useful Life prediction, leveraging phase-aware alignment across distinct

operational phases. The proposed methods — OPS-DANN (hard), OPS-DANN (soft), and Multi-Class OPS-DANN — address the complexities introduced by diverse feature distributions and distinct degradation patterns within each operational phase. Unlike traditional domain adaptation methods, which treat the dataset as a single, homogeneous entity, these methods align the marginal distributions for each operational phase separately, enabling the model to capture phase-dependent degradation trends.

Experimental results demonstrate that this approach provides substantial improvements over existing domain adaptation methods, particularly in settings with significant operational variability. Specifically, OPS-DANN (soft) and OPS-DANN (hard) preserved distinctions between phases in the feature space, effectively capturing unique degradation patterns associated with each operational phase. In scenarios with large domain shifts, the proposed methods achieved notable gains in RUL prediction accuracy, making them ideal for real-world prognostics with high operational variability. Furthermore, even for smaller domain shifts, our approach outperformed established methods like AdaBN, DANN, and MMD, which often underperformed compared to a source-only baseline. The effectiveness of phase-aware alignment in RUL prediction was validated through improved performance on the N-CMAPSS dataset.

Can regression-specific domain adaptation methods improve performance in regression tasks? In both Chapters 3 and 4 we introduce two methods, to tackle different challenges in regressions:

- In Chapter 3, we introduce a new domain adaptation from a new perspective by aligning a selected subspace of the pseudo-inverse Gram matrix, rather than directly aligning the original feature embedding space. This alignment preserves the critical scale and angle relationships within the feature space, which are essential for accurate regression predictions under domain shift. By incorporating regularization terms that adjust scale and angle, DARE-GRAM ensures that both feature relationships and scale are retained across source and target domains, enhancing the robustness of the model under varying conditions. Experimental results demonstrate that DARE-GRAM achieves substantial improvements in performance on benchmark regression datasets, validating its effectiveness over traditional feature alignment methods. The approach establishes that adapting the pseudo-inverse Gram matrix not only supports more accurate predictions but also ensures stability in training, making it a robust choice for domain adaptation in regression.
- In Chapter 4 Regression models are typically trained using Mean Squared Error loss, which assumes a constant variance in the prediction likelihood. However, this assumption is invalid under domain shift, we observe that the target features are collapsed and that the model predicted value in a smaller range. UGA modifies the alignment process by relaxing the constant variance assumption, allowing predictions to incorporate uncertainty alongside target values. In practice, we observed that

target domain samples exhibit higher predictive uncertainty compared to the source domain, and we leverage this difference to guide the alignment process. This adaptation enables the model to concentrate on more reliable regions within the target domain.

By using uncertainty as a guiding metric, UGA facilitates a more effective alignment between source and target domains, outperforming traditional MSE-based approaches. Through comprehensive experiments across multiple computer vision benchmarks and on state-of-charge prediction for batteries, UGA consistently demonstrated superior adaptation performance. By calibrating uncertainty within the alignment process, UGA enhances the model's robustness, enabling baseline models to achieve more accurate adaptation under domain shift conditions.

How can in-context learning counterbalance the impact of data scarcity and imbalance in regression? In Chapter 5, we present in-context learning as a solution to the challenges of imbalanced regression, allowing the model to leverage context examples during inference without updating model weights. Unlike traditional regression models, which require specific training to handle different tasks and often struggle in minority regions, in-context learning adapts dynamically by learning directly from selected context samples. This approach improves performance in minority regions, providing more reliable predictions in regions with limited representation than in-weight learning approaches.

Our analyses underscore the importance of localized context: by selecting a small, relevant set of neighboring samples, the model effectively mitigates the bias toward majority regions. Interestingly, performance follows a U-shaped trend relative to context size — initially improving as more context is provided, but then declining as excessive context introduces bias toward the majority. By carefully choosing a smaller, relevant set of context samples, the model achieves robust predictions in both high-imbalance and data-sparse regions.

In-context learning consistently outperforms conventional regression methods in data-scarce regions, adapting effectively across diverse label distributions, on computer vision benchmarks, and engineering datasets. By improving generalization in underrepresented areas without task-specific training, in-context learning provides an alternative solution for imbalanced regression tasks.

Can we learn target unknown class semantics in source-free open-set settings? The source-free open-set adaptation module in Chapter 6 addresses the challenge of adapting to a target domain that includes new classes not present in the source domain, alongside domain shifts, all while operating without access to source domain data due to privacy or data-sharing constraints. Traditional approaches often rely on entropy thresholding to identify unknown classes, but this method does not explicitly capture the distinguishing features of these new classes. To overcome this limitation, we propose a two-stage approach.

In the first stage, synthetic samples are generated within the target feature space to represent both known and unknown classes. These samples are then used to train a new classifier with expanded decision boundaries, maintaining the known classes from the source while accommodating the novel unknown classes in the target domain.

In the second stage, we treat the open-set problem as a closed-set problem by leveraging standard source-free adaptation techniques. Using self-training, we adapt the model to the target domain and directly learn the semantics of unknown classes, rather than simply discarding them as outliers.

Extensive experiments on benchmark datasets such as Office-31, Office-Home, and VisDA demonstrate the efficacy of our method. By effectively capturing the semantics of unknown classes, RRDA improves harmonic open-set accuracy, showcasing robust performance in source-free open-set adaptation.

8.3 Limitations and Outlook

Although this thesis has extended the applicability of deep regression learning in data-scarce settings, some potential limitations remain, and further improvements could strengthen the proposed framework. This section outlines these limitations and explores promising directions for future work.

Phase-Aware Distribution Alignment In Chapter 2, the proposed framework introduces a phase-aware alignment approach that distinguishes degradation patterns across different operational phases, addressing overrepresented and underrepresented phases using N-CMAPSS as datasets. While our proposed method assumes that RUL prediction could be improved by accounting for degradation variations across operational phases, in the N-CMAPSS dataset the degradation is assumed to be constant during one flight cycle. Therefore, the dataset does not enable evaluation of the immediate impact of operating conditions on degradation, as this impact is only observable after the end of the cycle. Although our approach improves robustness by balancing phases in scenarios with over- and under-represented regions, future research could explore alternatives to relying solely on expert knowledge. Such definitions could be data-driven, derived from the observed impact of operating conditions on degradation, or informed by simulators and system models where such knowledge may already be integrated. This flexibility would expand the applicability of phase-aware methods to other domains where operational conditions vary dynamically.

Uncertainty Alignment for Continuous Domain Shifts Chapter 4 introduces uncertainty alignment to quantify and guide adaptation under domain shifts, but its potential for addressing continuous domain shifts remains underexplored. While we demonstrated that domain shifts can be detected by comparing model confidence levels

and aligning uncertainty directly, future work could extend this approach to handle continuous streaming data from evolving domains. This extension would be particularly valuable for scenarios involving continuous system evolution, such as machines undergoing maintenance or experiencing changes in operational profiles over time.

In-Context Learning for Time Series Data in RUL Prediction: In Chapter 5 the in-context learning approach demonstrated effectiveness across tabular data and in extracting features from images and text through foundation models. However, we did not explore its applicability to time series data for RUL prediction. A key challenge lies in capturing long-range dependencies in temporal data, where the relationship between the current system state and future degradation may span extensive time periods. While the current method performs well in localized contexts, it would be valuable to investigate its performance over extended time horizons, analyzing how effectively it can leverage temporal dependencies without requiring model parameter updates. This extension could be particularly valuable for real-world applications, allowing for more flexible adaptation to changing conditions in predictive maintenance.

Domain Generalization for Regression Tasks: Chapters 3 and 4 show how regression models are particularly sensitive to domain shifts, with notable performance drops in new, unseen domains. A fundamental challenge lies in preserving feature scaling across domains while maintaining the relative relationships between features crucial for regression accuracy. This sensitivity is more pronounced in regression tasks than in classification, as the model must learn not only the decision boundaries but also maintain a continuous mapping between input features and predicted values. Future research could focus on domain generalization approaches that use synthetic or auxiliary datasets to develop domain-invariant representations, while carefully preserving the scale-sensitive nature of regression tasks.

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Publications

Journals

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Nejjar, I., F. Ahmed, & O. Fink: IM-Context: In-Context Learning for Imbalanced Regression Tasks, Transactions on Machine Learning Research, 2024

Frusque, G., I. Nejjar, M. Nabavi, & O. Fink: Semi-Supervised Health Index Monitoring with Feature Generation and Fusion, IEEE Transactions on Reliability, 2024

Bahari, M., I.Nejjar, & A.Alahi: Injecting knowledge in data-driven vehicle trajectory predictors Transportation research part C: emerging technologies 128

Conferences

Nejjar, I., Q. Wang, & O. Fink: DARE-GRAM: Unsupervised Domain Adaptation Regression by Aligning Inverse Gram Matrices. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023

Han, S., R.Gong, I.Nejjar, & O. Fink: DynAlign: Unsupervised Dynamic Taxonomy Alignment for Cross-Domain Segmentation.International Conference on Learning Representations (ICLR), 2025

Dong, H., I.Nejjar, H. Sun, E. Chatzi, & O. Fink: SimMMDG: A Simple and Effective Framework for Multi-modal Domain Generalization, NeurIPS 2023

Workshop

Nejjar, I., J. Meunier-Pion, G. Frusque, & O. Fink: DG-Mix: Domain Generalization for Anomalous Sound Detection Based on Self-Supervised Learning, DCASE, 2022

Work Experience

AI Research Intern — 09/2020 – 03/2021
Swisscom AG, Digital Lab — Lausanne, Switzerland

- Create a first working Multi-Task Learning (MTL) system for customer understanding behavior, increasing by 6.3% the overall performance of the classifiers in comparison to the single-task baselines.
- Creation of an extensible code base for MTL, where 7 state-of-the-art approaches have been implemented.
- Improved MTL with a new deep learning approach leveraging metric-based features against commonly used soft-max-based features.

Data Analytics and A.I Engineer Intern 02/2020 – 07/2020
Omnisens SA Morges, Switzerland

- Collected, analyzed and interpreted raw big data from the Distributed Acoustic Sensing (DAS) sensor.
- Developed of real-time Machine Learning algorithm for threat detection and led its implementation in a smart city: Berne -Switerzland. Covering a distance 22 Km inside the city and resulting in at most 1 false alarm a weak.

Professional Service

Journal Reviewer

Reliability Engineering & System Safety (RESS), IEEE Transactions on Automation Science and Engineering, Engineering Applications of Artificial Intelligence, IEEE Sensors

Conference Reviewer

International Conference on Learning Representations (ICLR), Conference on Neural Information Processing Systems (NeurIPS), International Conference on Artificial Intelligence and Statistics (AISTATS), International Conference on Machine Learning (ICML)